Electrochemistry

Question1

A solution of aluminium chloride is electrolysed for 30 minutes using a current of 2 A . The amount of the aluminium deposited at the cathode is _____ .

[Given : molar mass of aluminium and chlorine are $27\ \mathrm{g\ mol^{-1}}$ and $35.5\ \mathrm{g\ mol^{-1}}$ respectively. Faraday constant $= 96500\mathrm{C\ mol^{-1}}$]

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Options:

A. 1.660 g

B. 1.007 g

C. 0.336 g

D. 0.441 g

Answer: C

Solution:

$\mathrm{Al}^{3+} + 3e^- \rightarrow \mathrm{Al}$

The mass of aluminium deposited can be calculated using Faraday's laws of electrolysis. The steps are as follows:

Compute the total charge passed:

$Q = I \cdot t = 2\,\mathrm{A} \cdot (30 \times 60\,\mathrm{s}) = 3600\,\mathrm{C}$

Determine the moles of aluminium deposited. Since three electrons are required to deposit one mole of aluminium, the number of moles is given by:

$n(\mathrm{Al}) = \frac{Q}{3F} = \frac{3600}{3 \times 96500} \approx 0.01242\,\mathrm{mol}$

Finally, calculate the mass of aluminium using its molar mass ($M = 27\,\mathrm{g/mol}$):

$m(\mathrm{Al}) = n(\mathrm{Al}) \times M = 0.01242 \times 27 \approx 0.335\,\mathrm{g}$

Thus, the amount of aluminium deposited at the cathode is approximately

$0.336\ \mathrm{g}$

Question2

Which of the following electrolyte can be used to obtain $\mathrm{H_2\ S_2O_8}$ by the process of electrolysis ?

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Options:

A. Concentrated solution of sulphuric acid

B. Dilute solution of sodium sulphate.

C. Acidified dilute solution of sodium sulphate.

D. Dilute solution of sulphuric acid

Answer: A

Solution:

$\mathrm{H_2\ S_2O_8}$ is obtained by electrolysis of concentrated solution of sulphuric acid.

At anode :

$2\mathrm{HSO_4^-} \rightarrow \mathrm{H_2\ S_2O_8} + 2\mathrm{e^-}$

Question3

Given below are two statements :

Statement (I) : Corrosion is an electrochemical phenomenon in which pure metal acts as an anode and impure metal as a cathode.

Statement (II) : The rate of corrosion is more in alkaline medium than in acidic medium.

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A. Both Statement I and Statement II are true

B. Both Statement I and Statement II are false

C. Statement I is true but Statement II is false

D. Statement I is false but Statement II is true

Answer: C

Solution:

Let's examine both statements one by one and see whether they hold true for typical corrosion processes (e.g., iron rusting):

Statement (I)

> "Corrosion is an electrochemical phenomenon in which pure metal acts as an anode and impure metal as a cathode."

Corrosion is electrochemical

True. Corrosion, especially rusting of iron, involves oxidation at one region (the anode) and reduction at another region (the cathode) on the metal's surface.

Pure metal as anode, impurity as cathode

In many practical cases (e.g., iron containing small amounts of carbon or other impurities), the *relatively pure* region of the iron is more active (less noble) and tends to oxidize (lose electrons) — that is, it serves as the anode.

The impurity (e.g., carbon-rich region) often behaves more nobly and thus becomes the cathode region where reduction (e.g., oxygen reduction) occurs.

This difference in electrode potentials between the pure region and the impurity region drives the corrosion cell.

Hence, **Statement (I)** is generally considered **true** in the usual context of corrosion (like rusting of iron).

Statement (II)

> "The rate of corrosion is more in alkaline medium than in acidic medium."

For most common metals (like iron), **acidic** media typically enhance corrosion because abundant H^+ ions (and possibly other acidic species) can accelerate the oxidation/dissolution of the metal.

In *mildly* alkaline or neutral environments, metals often form protective oxide or hydroxide layers that can slow down further corrosion.

While certain strong bases can attack specific metals (e.g., Al in strong NaOH), **in general** for iron and many other metals, corrosion is more severe in acidic environments than in alkaline ones.

Thus, **Statement (II)** is **false** under normal corrosion scenarios (e.g., rusting of iron).

Conclusion

Statement (I): **True**

Statement (II): **False**

Therefore, the correct choice (matching these truth values) is:

(C) Statement I is true but Statement II is false.

Question4

$$\mathrm{FeO_4^{2-}} \xrightarrow{+2.0\ \mathrm{V}} \mathrm{Fe^{3+}} \xrightarrow{0.8\ \mathrm{V}} \mathrm{Fe^{2+}} \xrightarrow{-0.5\ \mathrm{V}} \mathrm{Fe^0}$$

In the above diagram, the standard electrode potentials are given in volts (over the arrow).

The value of $\mathrm{E}_{\mathrm{FeO_4^{2-}/Fe^{2+}}}$ is :

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Options:

A. 1.2 V

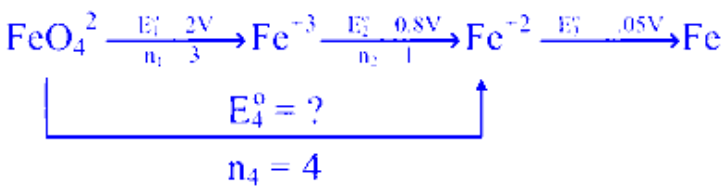
B. 2.1 V

C. 1.4 V

D. 1.7 V

Answer: D

Solution:



$$\Delta G_4^o = \Delta G_1^o + \Delta G_2^o$$
$$\Rightarrow -n_4 F E_4^o = -n_1 F E_1^0 - n_2 F E_2^o$$
$$\Rightarrow +4E_4^o = 3 \times 2 + (1 \times 0.8)$$
$$\Rightarrow E_4^o = \frac{6.8}{4}\ V$$
$$\Rightarrow E_4^o = 1.7\ V$$

Question5

Standard electrode potentials for a few half cells are mentioned below :

$$E^{\circ}_{Cu^{2+}/Cu} = 0.34\ V, E^{\circ}_{Zn^{2+}/Zn} = -0.76\ V$$
$$E^{\circ}_{Ag^{+}/Ag} = 0.80\ V, E^{\circ}_{Mg^{2+}/Mg} = -2.37\ V$$

Which one of the following cells gives the most negative value of ΔG° ?

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Options:

A. $Zn\left|Zn^{2+}(1M)\right\|Ag^{+}(1M)\left|Ag\right.$

B. $Ag\left|Ag^{+}(1M)\right|\left|Mg^{2+}(1M)\right|Mg$

C. $Zn\left|Zn^{2+}(1M)\right|\left|Mg^{2+}(1M)\right|Mg$

D. $Cu\left|Cu^{2+}(1M)\right|\left|Ag^{+}(1M)\right|Ag$

Answer: A

Solution:

$$
\begin{aligned}
\because \Delta G^{\circ} &= -nFE^{\circ} \\
\text{Option } (1) E^{\circ} &= 0.8 + 0.76 \\
&= 1.56\,V \\
\therefore \Delta G^{\circ} &= -2 \times F \times 1.56 \\
&= -3.12\,V \\
\text{Option (2) } E^{\circ} &= -2.37 + 0.76 \\
&= -1.61\,V \\
\therefore \Delta G^{\circ} &= -2 \times F \times (-1.61) \\
&= +3.22\,V \\
\text{Option (3) } E^{\circ} &= -2.37 - 0.8 \\
&= -3.17\,V \\
\therefore \Delta G^{\circ} &= -2 \\
&= F \times (-3.17) \\
&= +6.34 \\
\text{Option } (4) E^{\circ} &= 0.8 - 0.34 \\
&= 0.46\,V \\
\Delta G^{\circ} &= -2 \times F \times 0.46 \\
&= -0.92\,V
\end{aligned}
$$

Question6

For the given cell

$$Fe^{2+}(aq) + Ag^{+}_{(aq)} \rightarrow Fe^{3+}(aq) + Ag_{(s)}$$

The standard cell potential of the above reaction is Given:

$$
\begin{aligned}
Ag^{+} + e^{-} &\rightarrow Ag \quad E^{\theta} = xV \\
Fe^{2+} + 2e^{-} &\rightarrow Fe \quad E^{\theta} = yV \\
Fe^{3+} + 3e^{-} &\rightarrow Fe \quad E^{\theta} = zV
\end{aligned}
$$

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Options:

A. $x + 2y - 3z$

B. $x + 2y$

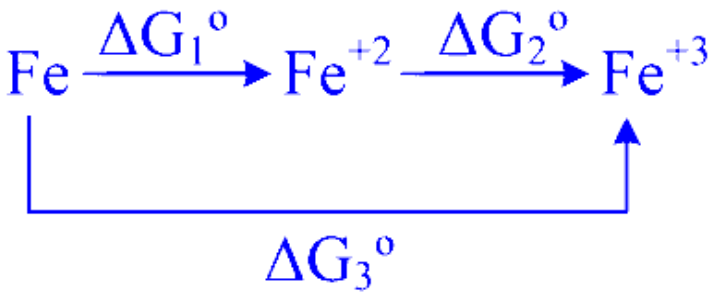
C. $y - 2x$

D. $x + y - z$

Answer: A

Solution:

$Fe^{2+}(aq) + Ag^{+}(aq) \rightarrow Fe^{3+}(aq) + Ag(s)$



$\Delta G_3^0 = \Delta G_1^0 + \Delta G_2^0$
$-3\,F(-z) = -2\,F(-y) + \Delta G_2^0$
$\Delta G_2^0 = 3Fz - 2Fy$

Also $\Delta G_2^0 = -nFE^0_{Fe^{+2}/Fe^{+3}}$

$3Fz - 2Fy = -1\,F\left(E^0_{Fe^{+2}/Fe^{+3}}\right)$
$E^0_{Fe^{+2}/Fe^{+3}} = 2y - 3z$

E^0_{Cell} for reaction will be

$E^0_{Ag^{+}/Ag^0} + E^0_{Fe^{+2}/Fe^{+3}}$
$= x + 2y - 3z$

Option (2)

Question7

Based on the data given below :

$E^{\circ}_{Cr_2O_7^{2-}/Cr^{3+}} = 1.33\ V \quad E^{\circ}_{Cl_2/Cl^{(-)}} = 1.36\ V$

$E^0_{MnO_4^-/Mn^{2+}} = 1.51\ V \quad E^{\circ}_{Cr^{3+}/Cr} = -0.74\ V$

the strongest reducing agent is :

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Options:

A. Cl^-

B. MnO_4^-

C. Cr

D. Mn^{2+}

Answer: C

Solution:

To determine the strongest reducing agent among the options, we need to consider which species most readily donates electrons (i.e., is most easily oxidized). One standard method is to compare the standard reduction potentials of the corresponding half-reactions. A lower (more negative) standard reduction potential indicates that the reduced species is less stable and more prone to oxidation.

Here's the data provided:

$\mathrm{Cr_2O_7^{2-}} + 14H^+ + 6e^- \rightarrow 2\mathrm{Cr^{3+}} + 7H_2O, \quad E^\circ = +1.33\,V$

$\mathrm{Cl_2} + 2e^- \rightarrow 2\mathrm{Cl^-}, \quad E^\circ = +1.36\,V$

$\mathrm{MnO_4^-} + 8H^+ + 5e^- \rightarrow \mathrm{Mn^{2+}} + 4H_2O, \quad E^\circ = +1.51\,V$

$\mathrm{Cr^{3+}} + 3e^- \rightarrow \mathrm{Cr}, \quad E^\circ = -0.74\,V$

Now, let's consider the species from the Options:

Option A: Cl^-

The relevant half-reaction is:

$\mathrm{Cl_2} + 2e^- \rightarrow 2\mathrm{Cl^-}, \quad E^\circ = +1.36\,V.$

The reduced form here is Cl^-. Its corresponding oxidation (the reverse reaction) would have an electrode potential of

$E^\circ_{\text{oxidation}} \quad = -1.36\,V.$

Option B: MnO_4^-

In the given half-reaction, MnO_4^- acts as the oxidizing agent (it is reduced to Mn^{2+}). Therefore, MnO_4^- is not a good reducing agent.

Option C: Cr (metallic chromium)

The relevant half-reaction is:

$\mathrm{Cr^{3+}} + 3e^- \rightarrow \mathrm{Cr}, \quad E^\circ = -0.74\,V.$

Here, Cr is the reduced species. Reversing this reaction gives:

$\mathrm{Cr} \rightarrow \mathrm{Cr^{3+}} + 3e^-,$

with an effective oxidation potential of

$E^\circ_{\text{oxidation}} \quad = +0.74\,V.$

Option D: Mn^{2+}

In the MnO_4^-/Mn^{2+} couple, Mn^{2+} is the reduced form. Its reverse (oxidation) potential would be:

$E^\circ_{\text{oxidation}} = -1.51\,V.$

When comparing the oxidation potentials (remember, a higher oxidation potential means the species is more willing to lose electrons), we have:

Cl^-: $E^\circ_{ox} = -1.36\,V$

Cr: $E^\circ_{ox} = +0.74\,V$

Mn^{2+}: $E^\circ_{ox} = -1.51\,V$

Among these, metallic Cr stands out because its corresponding reduction half-reaction has the most negative potential ($-0.74\,V$). This negative value indicates that Cr (in its elemental form) is unstable relative to its oxidized form (Cr^{3+}) and, therefore, is the most eager to lose electrons. In other words, Cr is the strongest reducing agent.

Thus, the answer is:

Option C: Cr

Question8

The standard reduction potential values of some of the p-block ions are given below. Predict the one with the strongest oxidising capacity.

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Options:

A.

$E^o_{Sn^{4+}/Sn^{2+}} = +1.15\ V$

B.

$E^o_{Al^{3+}/Al} = -1.66\ V$

C.

$E^o_{Pb^{4+}/Pb^{2+}} = +1.67\ V$

D.

$E^o_{Tl^{3+}/Tl} = +1.26\ V$

Answer: C

Solution:

Oxidizing capacity is an element's tendency to donate electrons, or the atmosphere's ability to oxidize compounds.

A higher standard reduction potential indicates a stronger oxidizing capacity. A substance with a higher standard reduction potential is move likely to gain electrons and act as an oxidizing agent in a redon reaction. The more positive the reduction potential, the stronger the oxidizing power of a species.

From the given options, the higher standard reduction potential is for lead (Pb). $(+1.67)$.

so, the p-block ion with strongest oxidizing capacity is Pb (lead).

Answer: Option 3) $E^0_{Pb^{4+}/Pb^{2+}} = +1.67\ \mathrm{V}$

Question9

The molar conductivity of a weak electrolyte when plotted against the square root of its concentration, which of the following is expected to be observed?

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Options:

A.

Molar conductivity increases sharply with increase in concentration.

B.

Molar conductivity decreases sharply with increase in concentration.

C.

A small increase in molar conductivity is observed at infinite dilution.

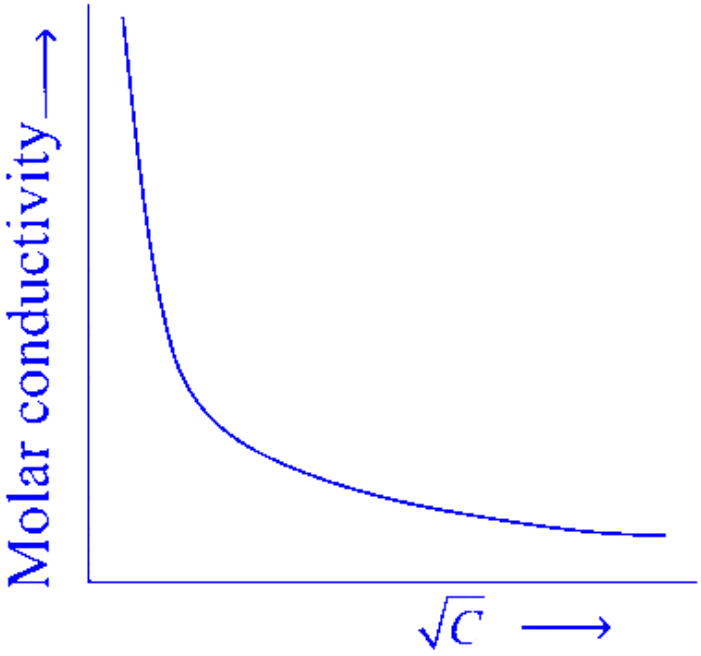
D.

A small decrease in molar conductivity is observed at infinite dilution.

Answer: B

Solution:

The molar conductivity of a weak electrolyte is plotted against the square root of its concentration:



c-concentration of weak electrolyte

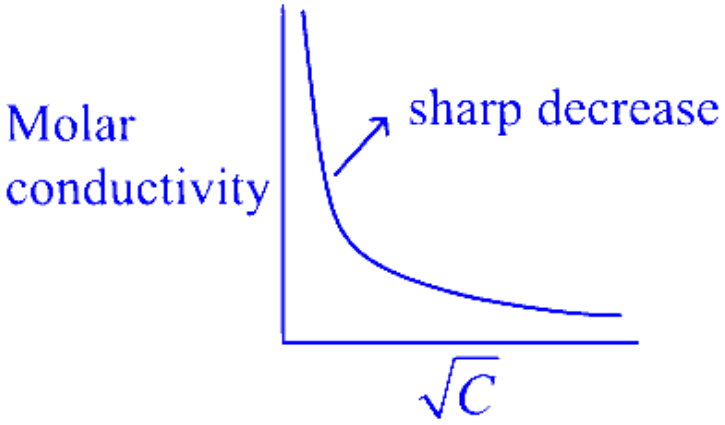
Weak electrolytes give curved decreasing graph.

1) Molar conductivity increases sharply with increase in concentration.

Incorrect.

With increase in concentration, molar conductivity decreases and it is lean from the graph of molar conductivity vs $\sqrt{c}$.

2) Molar conductivity decreases sharply with increase in concentration.



Correct

From the graph it is clean that, molar conductivity decreases sharply with increase in concentration.

3) A small increase in molar conductivity is observed at infinite dilution.

Incorrect

There is no small increase in molas conductivity with increase in concentration for a weak electrolyte. And there is no small increase in mola conductivity. Significant increase occurs.

4) A small decrease in molar conductivity is observed at infinite dilution.

Incorrect

For a weak electrolyte, the molar conductivity significantly increases with dilution, especially at infinite dilution.

Correct answer:

Option 2) Molen conductivity decreases sharply with increase in concentration.

Question10

For a Mg | Mg^{2+} (aq) || Ag^{+} (aq) | Ag the correct Nernst Equation is :

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Options:

A. $\mathrm{E_{cell}} = \mathrm{E^{o}_{cell}} - \frac{\mathrm{RT}}{2\ \mathrm{F}} \ln \frac{\left[\mathrm{Ag^{+}}\right]^{2}}{\left[\mathrm{Mg^{2+}}\right]}$

B. $\mathrm{E_{cell}} = \mathrm{E^{o}_{cell}} - \frac{\mathrm{RT}}{2\ \mathrm{F}} \ln \frac{\left[\mathrm{Mg^{2+}}\right]}{\left[\mathrm{Ag^{+}}\right]}$

C. $\mathrm{E_{cell}} = \mathrm{E^{o}_{cell}} - \frac{\mathrm{RT}}{2\ \mathrm{F}} \ln \frac{\left[\mathrm{Ag^{+}}\right]}{\left[\mathrm{Mg^{2+}}\right]}$

D. $\mathrm{E_{cell}} = \mathrm{E^{o}_{cell}} + \frac{\mathrm{RT}}{2\ \mathrm{F}} \ln \frac{\left[\mathrm{Ag^{+}}\right]^{2}}{\left[\mathrm{Mg^{2+}}\right]}$

Answer: D

Solution:

$Mg\left|Mg^{2+}(aq)\right|\left|Ag^{+}(aq)\right|Ag$
$\underset{(Anode)}{}\qquad\qquad\qquad\qquad\underset{(Cathode)}{}$

Nernst equation (general)

$E_{Cell} = E^{0}_{Cell} - \frac{RT}{nF} \ln Q$

For the cell, the chemical equation can be written as

$\mathrm{Mg_{(s)} + 2Ag^{+}_{(aq\ )} \longrightarrow Mg^{2+}_{(aq\ )} + 2Ag(\ s)}$

n (number of electrons transferred) $= 2$

$E_{cell} = E^{o}_{cell} - \frac{RT}{2F} \ln \frac{[Mg^{2+}][Ag]^{2}}{[Mg][Ag^{+}]^{2}}$

$E_{cell} = E^{o}_{cell} - \frac{RT}{2F} \ln \frac{[Mg^{2+}]}{[Ag^{+}]^{2}}$

$E_{cell} = E^{o}_{cell} + \frac{RT}{2F} \ln \frac{[Ag^{+}]^{2}}{[Mg^{2+}]}$

Concentration of solid species

$[Mg] = [Ag] = 1$

Correct answer :

Option D) $E_{cell} = E^o_{cell} + \frac{RT}{2F} \ln \frac{[Ag^+]^2}{[Mg^{2+}]}$

Question11

O_2 gas will be evolved as a product of electrolysis of :

(A) an aqueous solution of $AgNO_3$ using silver electrodes.

(B) an aqueous solution of $AgNO_3$ using platinum electrodes.

(C) a dilute solution of H_2SO_4 using platinum electrodes.

(D) a high concentration solution of H_2SO_4 using platinum electrodes.

Choose the correct answer from the options given below :

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Options:

A. (B) and (C) only

B. (B) and (D) only

C. (A) and (D) only

D. (A) and (C) only

Answer: A

Solution:

(A) An aqueous solution of $AgNO_3$ using silver electrodes.

Cathode - Reduction - $Ag^+_{(aq)} + e^- \rightarrow Ag(s)$

Anode - Oxidation - $Ag(s) \rightarrow Ag^+_{(aq)} + e^-$

Solid silver will be deposited at the cathode. Solid anode (silver) will dissolve, releasing silver ions into the solution.

So, there is no formation of O_2 gas in this electrolysis.

(B) An aqueous solution of $AgNO_3$ using platinum electrodes.

Cathode - Reduction - $Ag^{+}_{(aq)} + e^{-} \rightarrow Ag(s)$

Anode - Oxidation - $2H_2O \rightarrow 4H^{+} + O_2 + 4e^{-}$

When platinum electrodes are used, Ag^{+} from solution is reduced and deposited at cathode whereas O_2 is produced at the anode.

(C) A dilute soution of H_2SO_4 using platinum electrodes.

Cathode - Reduction - $2H^{+} + 2e^{-} \rightarrow H_2$

Anode - Oxidation - $2H_2O \rightarrow O_2 + 4H^{+} + 4e^{-}$

H_2 gas is produded at the cathode and O_2 gas is produced at the anode.

(D) a high concentration solution of H_2SO_4 using platinum electrodes.

O_2 gas is not formed in this case.

Cathode - Reduction - The substance formed is H_2 gas.

Anode - Oxidation - The substance formed is not O_2 gas.

So, statements (B) and (C) are correct.

Question12

Match List - I with List - II :

List - I (Applications)	List - II (Batteries/Cell)
(A) Transistors	(I) Anode - Zn/Hg; Cathode - HgO + C
(B) Hearing aids	(II) Hydrogen fuel cell
(C) Inverters	(III) Anode - Zn; Cathode - Carbon
(D) Apollo space ship	(IV) Anode - Pb; Cathode - Pb | PbO_2

Choose the correct answer from the options given below :

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Options:

A.

(A)-(III), (B)-(I), (C)-(IV), (D)-(II)

B.

(A)-(II), (B)-(III), (C)-(IV), (D)-(I)

C.

(A)-(IV), (B)-(III), (C)-(II), (D)-(I)

D.

(A)-(III), (B)-(II), (C)-(IV), (D)-(I)

Answer: A

Solution:

For each application listed in List-I, match it with the appropriate battery or cell type from List-II based on their characteristics:

(A) Transistors: Generally use a dry cell or Leclanché cell, where the anode is zinc (Zn) and the cathode is a carbon rod surrounded by manganese dioxide (MnO_2).

Therefore, (A) Transistors match with (III) Anode - Zn; Cathode - Carbon.

(B) Hearing aids: Operate using a mercury cell. The anode in this cell is a zinc-mercury amalgam, and the cathode is composed of mercury oxide (HgO) with carbon.

Thus, (B) Hearing aids match with (I) Anode - Zn/Hg; Cathode - HgO + C.

(C) Inverters: Use a lead-acid storage cell. In this type of cell, the anode consists of lead (Pb), and the cathode is a lead dioxide (PbO_2) grid.

As a result, (C) Inverters match with (IV) Anode - Pb; Cathode - Pb | PbO_2.

(D) Apollo space ship: Utilizes a hydrogen-oxygen fuel cell. The hydrogen is oxidized at the anode, while oxygen is reduced at the cathode.

Therefore, (D) Apollo space ship matches with (II) Hydrogen fuel cell.

Summary of matches:

A - III

B - I

C - IV

D - II

This corresponds with Answer Option A.

Question13

Correct order of limiting molar conductivity for cations in water at 298 K is :

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Options:

A.

$H^+ > Na^+ > K^+ > Ca^{2+} > Mg^{2+}$

B.

$H^+ > Ca^{2+} > Mg^{2+} > K^+ > Na^+$

C.

$Mg^{2+} > H^+ > Ca^{2+} > K^+ > Na^+$

D.

$H^+ > Na^+ > Ca^{2+} > Mg^{2+} > K^+$

Answer: B

Solution:

The limiting molar conductivities of ions are important in determining how well they conduct electricity in solution. Here are the values for some common cations at 298 K:

H^+: 349.8 S cm² mol⁻¹

Na^+: 50.11 S cm² mol⁻¹

K^+: 73.52 S cm² mol⁻¹

Ca^{2+}: 119 S cm² mol⁻¹

Mg^{2+}: 106.12 S cm² mol⁻¹

Based on these values, the correct order of limiting molar conductivity for these cations is:

$\overset{\oplus}{H}^+ > Ca^{2+} > Mg^{2+} > K^+ > Na^+$

This order reflects the decreasing ability of each ion to conduct electricity in water under the given conditions.

Question14

The standard cell potential $\left(E^{\ominus}_{cell}\right)$ of a fuel cell based on the oxidation of methanol in air that has been used to power television relay station is measured as 1.21 V . The standard half cell reduction potential for O_2 $\left(E^{\circ}_{O_2/H_2O}\right)$ is 1.229 V .

Choose the correct statement :

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Options:

A. Reactants are fed at one go to each electrode.

B. Oxygen is formed at the anode.

C. The standard half cell reduction potential for the reduction of CO_2 $\left(E^{\circ}_{CO_2/CH_3OH}\right)$ is 19 mV

D. Reduction of methanol takes place at the cathode.

Answer: C

Solution:

To determine the reaction potentials at the anode and cathode in the methanol fuel cell, we start with the given standard cell potential:

$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}}$

Given:

$E^{\circ}_{\text{cell}} = 1.21\,\text{V}$

$E^{\circ}_{\text{O}_2/\text{H}_2\text{O}} = 1.229\,\text{V}$

Substituting the values,

$1.21 = 1.229 - E^{\circ}_{\text{anode}}$

Solving for E°_{anode} :

$E^{\circ}_{\text{anode}} = 1.229 - 1.21 = 0.019\,\text{V}$

Since this is a fuel cell based on the oxidation of methanol, methanol undergoes oxidation at the anode, while oxygen is reduced at the cathode. Therefore:

Anode Reaction: Oxidation of methanol occurs here, contributing to the fuel cell's electrical output.

Cathode Reaction: Reduction of oxygen occurs at the cathode, where the $E^{\circ}_{\text{O}_2/\text{H}_2\text{O}}$ potential is utilized.

Thus, we verify that the half-cell reduction potential for oxidation of methanol by CO_2 is indeed 19 mV, aligning with the calculated E°_{anode} . The processes correctly identify the roles of anode and cathode in this fuel cell.

Question15

On charging the lead storage battery, the oxidation state of lead changes from x_1 to y_1 at the anode and from x_2 to y_2 at the cathode. The values of x_1, y_1, x_2, y_2 are respectively :

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Options:

A. $0, +2, +4, +2$

B. $+2, 0, 0, +4$

C. $+4, +2, 0, +2$

D. $+2, 0, +2, +4$

Answer: D

Solution:

For charging of lead storage battery cell reaction is

$2PbSO_4(s) + 2H_2O(l) \rightarrow Pb(s) + PbO_2(s) + 2H_2SO_4(aq)$

At anode $PbSO_4$ reduced back to Pb and at cathode $PbSO_4$ oxidised back to PbO_2.

$\because \ x_1 = +2, y_1 = 0$
$x_2 = +2, y_2 = 4$

Question16

Given below are two statements :

1 M aqueous solutions of each of $Cu(NO_3)_2$, $AgNO_3$, $Hg_2(NO_3)_2$, $Mg(NO_3)_2$ are electrolysed using inert electrodes. Given: $E^0_{Ag^+/Ag} = 0.80$ V, $E^0_{Hg_2^{2+}/Hg} = 0.79$ V, $E^0_{Cu^{2+}/Cu} = 0.24$ V and $E^0_{Mg^{2+}/Mg} = -2.37$ V.

Statement (I) : With increasing voltage, the sequence of deposition of metals on the cathode will be Ag, Hg and Cu.

Statement (II) : Magnesium will not be deposited at the cathode instead oxygen gas will be evolved at the cathode.

In the light of the above statements, choose the most appropriate answer from the options given below :

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Options:

A.

Both **Statement I** and **Statement II** are incorrect

B.

Statement I is incorrect but **Statement II** is correct

C.

Statement I is correct but **Statement II** is incorrect

D.

Both **Statement I** and **Statement II** are correct

Answer: C

Solution:

Let's analyze the situation step by step.

We have 1 M aqueous solutions of:

$AgNO_3$ with $Ag^+/Ag \quad E^0 = +0.80\,V$

$Hg_2(NO_3)_2$ with $Hg_2^{2+}/Hg \quad E^0 = +0.79\,V$

$Cu(NO_3)_2$ with $Cu^{2+}/Cu \quad E^0 = +0.24\,V$

$Mg(NO_3)_2$ with $Mg^{2+}/Mg \quad E^0 = -2.37\,V$

In an electrolytic cell with inert electrodes, reduction occurs at the cathode. The ion with the highest (most positive) reduction potential will be reduced (or deposited) first. Therefore, as we increase the applied voltage (making the cathode sufficiently negative), the metals will begin to deposit in the order of their reduction potentials.

The order of reduction (deposition) will be:

First, silver (+0.80V)

Next, mercury (+0.79V)

Then, copper (+0.24V)

So, Statement (I) which says "With increasing voltage, the sequence of deposition of metals on the cathode will be Ag, Hg and Cu" is correct.

For magnesium, the reduction potential is very negative (−2.37V). In aqueous solution, before reaching such a negative potential, water will be reduced. The cathodic reaction for water typically is:

$2H_2O+2e^- \rightarrow H_2+2OH^-$

which leads to the evolution of hydrogen gas, not oxygen. (Oxygen is produced at the anode during the oxidation of water.)

Hence, Statement (II) erroneously states that “oxygen gas will be evolved at the cathode” when in fact, if magnesium were to be reduced (which it won’t be), water reduction would yield hydrogen gas. Thus, Statement (II) is incorrect.

Summary:

Statement (I) is correct.

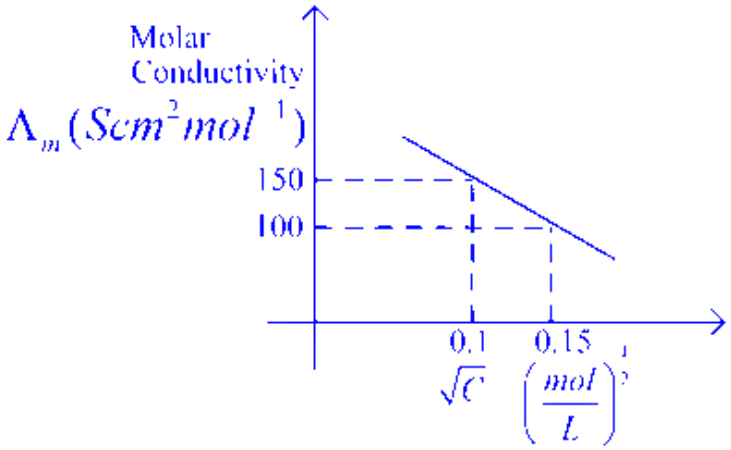
Statement (II) is incorrect.

Therefore, the most appropriate answer is:

Option C: Statement I is correct but Statement II is incorrect.

Question17

Given below is the plot of the molar conductivity vs √concentration for KCl in aqueous solution.



If, for the higher concentration of KCl solution, the resistance of the conductivity cell is , then the resistance of the same cell with the dilute solution is ' x '

The value of is _____ (Nearest integer)

JEE Main 2025 (Online) 28th January Morning Shift

Answer: 150

Solution:

Resistance and Resistivity:

$R = \rho \frac{\ell}{A}$

Where R is resistance, ρ is resistivity, ℓ is the length, and A is the cross-sectional area of the cell. The term $\frac{\ell}{A}$ is known as the cell constant, G^*.

Conductivity and Conductance:

$\kappa = G \cdot G^*$

$G = \frac{1}{R} \quad ; \quad \kappa = \frac{1}{\rho}$

Where G is conductance and κ is conductivity.

Molar Conductivity:

$\lambda_m = \frac{\kappa \times 1000}{C}$

Where λ_m is molar conductivity and C is concentration.

Given Information:

The resistance of the cell with the concentrated solution $R_c = 100\,\Omega$.

We need to find the resistance R_d for the dilute solution.

Steps to Solution:

The conductivity ratio for concentrated (κ_c) and dilute (κ_d) solutions relates to resistance:

$\frac{\kappa_c}{\kappa_d} = \frac{R_d}{R_c}$

From the plot or given values (assuming values for molar conductivity):

$\lambda_{m,c}$ and $\lambda_{m,d}$ are the molar conductivities for the concentrated and dilute solutions.

If from the plot or data, $\lambda_{m,c} = 150\,\text{S cm}^2\text{mol}^{-1}$ and $\lambda_{m,d} = 100\,\text{S cm}^2\text{mol}^{-1}$.

Calculate the ratio of conductivities:

$\frac{\kappa_c}{\kappa_d} = \frac{(\lambda_{m,c} \cdot C_c)}{(\lambda_{m,d} \cdot C_d)}$

Substituting values into the equation:

$\frac{100 \cdot (0.15)^2}{150 \cdot (0.1)^2} = \frac{R_d}{100}$

Solving the equation gives:

$R_d = 150\,\Omega$

Thus, the resistance of the conductivity cell with the dilute solution is $R_d = 150\,\Omega$.

Question18

Electrolysis of 600 mL aqueous solution of NaCl for 5 min changes the pH of the solution to 12 .

The current in Amperes used for the given electrolysis is _____ . (Nearest integer).

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Answer: 2

Solution:

Electrolysis of NaCl is

$\mathrm{NaCl}+\mathrm{H_2O(aq)} \rightarrow \mathrm{NaOH(aq)}+\frac{1}{2}\mathrm{Cl_2(\ g)}+\frac{1}{2}\mathrm{H_2(\ g)}$

Since during electrolysis pH changes to 12

So $\left[\mathrm{OH}^{\ominus}\right]=10^{-2}$ and $\left[\mathrm{H}^{+}\right]=10^{-12}$

So by Faraday law

Gram amount of substance deposited =Amount of electricity passed

$$10^{-2} \times \frac{600}{1000} \times 96500=\mathrm{I} \times \mathrm{t}$$

$$\frac{10^{-2} \times 600}{1000} \times 96500=\mathrm{I} \times 5 \times 60$$

$$\mathrm{I}=\frac{10^{-2} \times 600 \times 96500}{1000 \times 5 \times 60}$$

$\mathrm{I}=1.93$ ampere

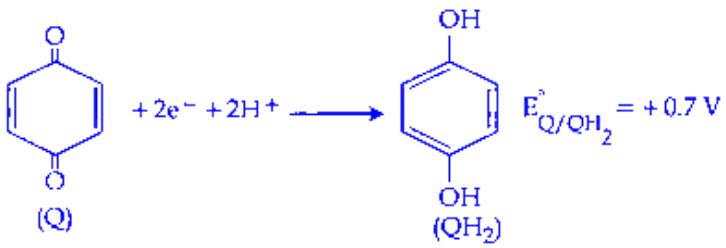
So, $\mathrm{I}=2$ ampere (nearest integer)

Question19

Consider the following electrochemical cell at standard condition.

$$\mathrm{Au(\ s)\left|QH_2, Q\right|NH_4X(0.01M)||Ag^{+}(1M)\ |\ Ag(\ s)} \mathrm{E_{cell}}\ =+0.4\,\mathrm{V}$$

The couple $\mathrm{QH_2/Q}$ represents quinhydrone electrode, the half cell reaction is given below:



$\left[\text{Given : } E^0_{Ag^+/Ag} = +0.8\,V \text{ and } \frac{2.303RT}{F} = 0.06\,V\right]$

The pK_b value of the ammonium halide salt (NH_4X) used here is _____ . (nearest integer)

Answer: 6

Solution:

The cell reaction is:

$QH_2 + 2Ag^+ \rightarrow 2Ag + Q + 2H^+$

The Nernst equation for the reaction can be expressed as:

$E = E^\circ - \frac{0.06}{2}\log[H^+]^2$

Simplifying, we obtain:

$E = E^\circ - 0.06\log[H^+]$

Given data includes:

$E_{cell} = +0.4\,V$

Standard potential for Ag^+/Ag, $E^0_{Ag^+/Ag} = +0.8\,V$

Using the cell potential and the standard potential, we calculate:

$pH = -\log(H^+) = \frac{E-E^\circ}{0.06} = \frac{0.4-0.1}{0.06}$

$pH = \frac{0.3}{0.06} = 5$

This established pH must now relate to the buffer equation involving NH_4X, at

$pH + NH_4X = 7 - \frac{1}{2}pK_b - \frac{1}{2}\log C$

Given the concentration $C = 0.01\,M = 10^{-2}$, we substitute into the equation:

$5 = 7 - \frac{1}{2} \times pK_b - \frac{1}{2}\log(10^{-2})$

Simplifying further, the equation resolves to:

$pK_b = 6$

Thus, the calculated pK_b value of the ammonium halide salt NH_4X is 6.

Question20

$0.2\%(w/v)$ solution of NaOH is measured to have resistivity $870.0\ m\Omega\ m$. The molar conductivity of the solution will be_____ $\times 10^2 mSdm^2\ mol^{-1}$. (Nearest integer)

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Answer: 23

Solution:

Determine the conductivity from the resistivity.

The resistivity is given as

$\rho = 870.0\ \text{m}\Omega \cdot \text{m}.$

Convert this into ohm·meters:

$870.0\ \text{m}\Omega \cdot \text{m} = 870.0 \times 10^{-3}\ \Omega \cdot \text{m} = 0.87\ \Omega \cdot \text{m}.$

Conductivity is the reciprocal of resistivity:

$\kappa = \frac{1}{\rho} = \frac{1}{0.87} \approx 1.15\ \text{S/m}.$

Find the concentration of NaOH in mol/m³.

A $0.2\%\ (\text{w/v})$ solution means there are 0.2 grams of NaOH per 100 mL of solution.

In 1 liter (1000 mL) there are:

$\frac{0.2\ \text{g}}{100\ \text{mL}} \times 1000\ \text{mL} = 2\ \text{g/L}.$

The molar mass of NaOH is approximately 40 g/mol. Thus, the molarity is:

$$\text{Molarity} = \frac{2 \text{ g/L}}{40 \text{ g/mol}} = 0.05 \text{ mol/L}.$$

Since $1 \text{ L} = 0.001 \text{ m}^3$, converting to SI units (mol/m³):

$$c = 0.05 \text{ mol/L} \times \frac{1}{0.001 \text{ m}^3/\text{L}} = 50 \text{ mol/m}^3.$$

Calculate the molar conductivity in SI units.

Molar conductivity Λ_m is given by:

$$\Lambda_m = \frac{\kappa}{c}.$$

Substituting the values:

$$\Lambda_m = \frac{1.15 \text{ S/m}}{50 \text{ mol/m}^3} = 0.023 \text{ S} \cdot \text{m}^2/\text{mol}.$$

Convert the molar conductivity to the desired units (mS dm² mol⁻¹).

First, convert the conductivity:

$$0.023 \text{ S} \cdot \text{m}^2/\text{mol} \times 1000 \frac{\text{mS}}{\text{S}} = 23 \text{ mS} \cdot \text{m}^2/\text{mol}.$$

Now, convert the area units. Recall that:

$$1 \text{ m}^2 = 100 \text{ dm}^2.$$

So,

$$23 \text{ mS} \cdot \text{m}^2/\text{mol} = 23 \times 100 \text{ mS} \cdot \text{dm}^2/\text{mol} = 2300 \text{ mS} \cdot \text{dm}^2/\text{mol}.$$

The problem asks for the answer in the form

$$[\text{blank}] \times 10^2 \text{ mS} \cdot \text{dm}^2/\text{mol}.$$

Expressing 2300 mS dm²/mol in this form:

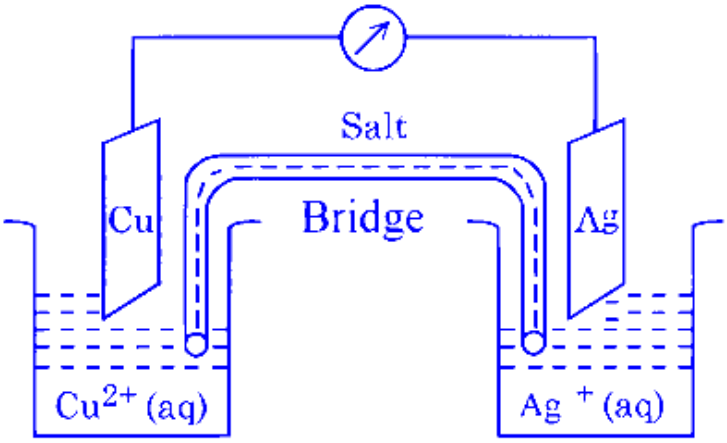
$$2300 \text{ mS} \cdot \text{dm}^2/\text{mol} = 23 \times 10^2 \text{ mS} \cdot \text{dm}^2/\text{mol}.$$

Thus, the molar conductivity is

$$\boxed{23 \times 10^2 \text{ mS} \cdot \text{dm}^2/\text{mol}}.$$

Question21

1 Faraday electricity was passed through $Cu^{2+}(1.5M, 1\ L)/Cu$ and 0.1 Faraday was passed through $Ag^{+}(0.2M, 1\ L)/Ag$ electrolytic cells. After this the two cells were connected as shown below to make an electrochemical cell. The emf of the cell thus formed at 298 K is_____ mV (nearest integer)



$$\text{Given : } E^{\circ}Cu^{2+}/Cu = 0.34\ V$$

$$E^{\circ}Ag^{+}/Ag = 0.8\ V$$

$$\frac{2 \cdot 303RT}{F} = 0.06\ V$$

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Answer: 400

Solution:

t = 0 1.5 1 mole

t = t 1 – 0.5 mole

$[Cu^{+2}]$ = 1 M after electrolysis

* $Ag\oplus + e^- \longrightarrow Ag$

t = 0 0.2 0.1 mole

t = t 0.1 – –

$[Ag^+]$ = 0.1 M after electrolysis

Cell $Cu_{(s)} + 2Ag^+_{(aq)} \rightarrow Cu^{+2}_{(aq)} + 2Ag_{(s)}$

reaction

$$E = E^\circ - \frac{0.06}{n}\log\frac{[Cu^{+2}]}{[Ag^+]^2}$$

$$E = (0.8 - 0.34) - \frac{0.06}{2}\log\frac{1}{(0.1)^2} = 0.4\ V$$

Correct answer = 400mV

Question22

Consider the following half cell reaction

$$Cr_2O_7^{2-}(aq) + 6e^- + 14H^+(aq) \rightarrow 2Cr^{3+}(aq) + 7H_2O(\ell)$$

The reaction was conducted with the ratio of $\frac{[Cr^{3+}]^2}{[Cr_2O_7^{2-}]} = 10^{-6}$. The pH value at which the EMF of the half cell will become zero is _____.

(nearest integer value)

[Given: standard half cell reduction potential $E^\circ_{Cr_2O_7^{2-},H^+/Cr^{3+}} = 1.33\ V$, $\frac{2.303RT}{F} = 0.059\ V$.]

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Answer: 10

Solution:

$$Cr_2O_{7(aq)}^{-2} + 14H_{(aq)}^{+} + 6e^{-} \rightarrow 2Cr_{(aq)}^{+3} + 7H_2O_{(\ell)}$$

$$E_R = E_R^0 - \frac{0.059}{6}\log\frac{\left[Cr^{+3}\right]^2}{\left[Cr_2O_7^{-2}\right]\left[H^+\right]^{14}}$$

$$0 = 1.33 - \frac{0.059}{6}\log\frac{10^{-6}}{\left[H^+\right]^{14}}$$

$$\frac{1.33 \times 6}{0.059} = \log\frac{10^{-6}}{[H]^{14}}$$

$$135.254 = -6 - 14\log\left[H^+\right]$$

$$141.254 = 14pH$$

$$pH = \frac{141.254}{14} = 10.08$$

Question23

The mass of silver (Molar mass of Ag : 108gmol^{-1}) displaced by a quantity of electricity which displaces 5600mL of O_2 at S.T.P. will be _____g.

[27-Jan-2024 Shift 1]

Answer: 108

Solution:

Eq. of Ag = Eq. of O_2

Let x gm silver displaced,

$$\frac{x \times 1}{108} = \frac{5.6}{22.7} \times 4$$

(Molar volume of gas at STP = 22.7 lit)

$x = 106.57$ gm

Ans. 107

OR,

as per old STP data, molar volume = 22.4 lit

$$\frac{x \times 1}{108} = \frac{5.6}{22.4} \times 4, x = 108 \text{ gm}.$$

Ans. 108

Question24

Which of the following statements is not correct about rusting of iron?

[27-Jan-2024 Shift 2]

Options:

A.

Coating of iron surface by tin prevents rusting, even if the tin coating is peeling off.

B.

When pH lies above 9 or 10 , rusting of iron does not take place.

C.

Dissolved acidic oxides SO_2, NO_2 in water act as catalyst in the process of rusting.

D.

Rusting of iron is envisaged as setting up of electrochemical cell on the surface of iron object.

Answer: A

Solution:

As tin coating is peeled off, then iron is exposed to atmosphere.

Question25

The hydrogen electrode is dipped in a solution of pH = 3 at 25°C. The potential of the electrode will be____ $\times 10^{-2}$V.

$\left(\frac{2.303\,RT}{F} = 0.059V\right)$

[27-Jan-2024 Shift 2]

Answer: 18

Solution:

$$2H^{+}_{(aq.)} + 2e^{-} \rightarrow H_2(g)$$

$$E_{cell} = E_{cell}^{0} - \frac{0.059}{2}\log\frac{P_{H_2}}{[H^+]^2}$$

$$= 0 - 0.059 \times 3 = -0.177 \text{ volts.} \quad = -17.7 \times 10^{-2} V.$$

Question26

The mass of zinc produced by the electrolysis of zinc sulphate solution with a steady current of 0.015 A for 15 minutes is ____ × 10^{-4}g.

(Atomic mass of zinc = 65.4amu)

[29-Jan-2024 Shift 1]

Answer: 45.75

Solution:

$$Zn^{+2} + 2e^{-} \rightarrow Zn$$

$$W = Z \times i \times t$$

$$= \frac{65.4}{2 \times 96500} \times 0.015 \times 15 \times 60$$

$$= 45.75 \times 10^{-4} \text{ gm}$$

Question27

A constant current was passed through a solution of $AuCl_4^-$ion between gold electrodes. After a period of 10.0 minutes, the increase in mass of cathode was 1.314g. The total charge passed through the solution is _____ × 10^{-2}F.

(Given atomic mass of $\overline{Au}$ = 197)

[29-Jan-2024 Shift 2]

Answer: 2

Solution:

$$\frac{W}{E} = \frac{\text{charge}}{1F}$$

$$\frac{\frac{1.314}{197}}{3} = \frac{Q}{1F}$$

$$Q = 2 \times 10^{-2} F$$

Question28

Reduction potential of ions are given below:

ClO_4^-	IO_4^-	BrO_4^-
$E^\circ = 1.19V$	$E^\circ = 1.65V$	$E^\circ = 1.74V$

The correct order of their oxidising power is:

[30-Jan-2024 Shift 2]

Options:

A.

$ClO_4^- > IO_4^- > BrO_4^-$

B.

$BrO_4^- > IO_4^- > ClO_4^-$

C.

$BrO_4^- > ClO_4^- > IO_4^-$

D.

$IO_4^- > BrO_4^- > ClO_4^-$

Answer: B

Solution:

Higher the value of $\oplus$ ve SRP (Std. reduction potential) more is tendency to undergo reduction, so better is oxidising power of reactant.

Hence, ox. Power:- $BrO_4^- > IO_4^- > ClO_4^-$

Question29

Identify the factor from the following that does not affect electrolytic conductance of a solution.

[31-Jan-2024 Shift 1]

Options:

A.

The nature of the electrolyte added.

B.

The nature of the electrode used.

C.

Concentration of the electrolyte.

D.

The nature of solvent used.

Answer: B

Solution:

Conductivity of electrolytic cell is affected by concentration of electrolyte, nature of electrolyte and nature of solvent.

Question30

The metals that are employed in the battery industries are
A. Fe
B. Mn
C. Ni
D. Cr
E. Cd

Choose the correct answer from the options given below:

[31-Jan-2024 Shift 1]

Options:

A.

B, C and E only

B.

A, B, C, D and E

C.

A, B, C and D only

D.

B, D and E only

Answer: A

Solution:

Mn, Ni and Cd metals used in battery industries.

Question31

One Faraday of electricity liberates $x \times 10^{-1}$ gram atom of copper from copper sulphate, x is_____

[31-Jan-2024 Shift 1]

Answer: 5

Solution:

$Cu^{2+} + 2e^- \longrightarrow Cu$

2 Faraday $\longrightarrow$ 1 mol Cu

1 Faraday $\longrightarrow$ 0.5 mol Cu deposit

$0.5\,\text{mol} = 0.5\text{g atom} = 5 \times 10^{-1}$

$x = 5$

Question32

The values of conductivity of some materials at $298.15K^{-1}Sm^{-1}$ are 2.1×10^3, 1.0×10^{-16}, 1.2×10, 3.91, 1.5×10^{-2}, 1×10^{-7}, 1.0×10^3. The number of conductors among the materials is____

[31-Jan-2024 Shift 2]

Answer: 4

Solution:

Conductivity (Sm^{-1})

2.1×10^3
1.2×10
3.91
1×10^3
} conductors at 298.15K

1×10^{-16} Insulator at 298.15K

1.5×10^{-2}
1×10^{-7}
} Semiconductor at 298.15K

Therefore number of conductors is 4 .

Question33

The potential for the given half cell at 298K is × 10^{-2}V.

$2H^+_{(aq)} + 2e^- \longrightarrow H_2(g)$

$[H^+] = 1M, P_{H_2} = 2\,atm$

(Given: $2.303\,RT/F = 0.06V, \log 2 = 0.3$)

[1-Feb-2024 Shift 1]

Answer: 1

Solution:

$$E = E_{H^+/H_2}{}^\circ - \frac{0.06}{2}\log\frac{P_{H_2}}{[H^+]^2}$$

$$E = 0.00 - \frac{0.06}{2}\log\frac{2}{[1]^2}$$

$$E = -0.03 \times 0.3 = -0.9 \times 10^{-2}V$$

Question34

The amount of electricity in Coulomb required for the oxidation of 1mol of H_2O to O_2 is _____ × 10^5C.

[1-Feb-2024 Shift 2]

Answer: 2

Solution:

$$2H_2O \rightarrow O_2 + 4H^+ + 4e^-$$

$$\frac{W}{E} = \frac{Q}{96500}$$

$$\text{mole} \times \text{n-factor} = \frac{Q}{96500}$$

$$1 \times 2 = \frac{Q}{96500}$$

$$Q = 2 \times 96500C$$

$$= 1.93 \times 10^5 C$$

Question35

At 298K, a 1 litre solution containing 10 mmol of $Cr_2O_7^{2-}$ and 100 mmol of Cr^{3+} shows a pH of 3.0.

Given : $Cr_2O_7^{2-} \rightarrow Cr^{3+}$; $E^0 = 1.330V$ and $\frac{2.303\,RT}{F} = 0.059V$

The potential for the half cell reaction is $x \times 10^{-3}$ V. The value of x is____
[24-Jan-2023 Shift 1]

Answer: 917

Solution:

Solution:

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$$

$$E = 1.33 - \frac{0.059}{6} \log \frac{(0.1)^2}{(10^{-2})(10^{-3})^{14}}$$

$$E = 1.33 - \frac{0.059}{6} \times 42 = 0.917$$

$$E = 917 \times 10^{-3}$$

$$x = 917$$

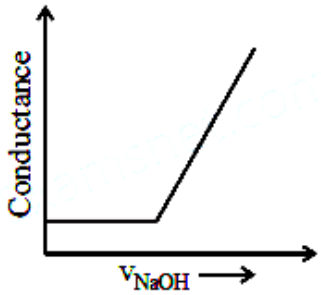
Question36

Choose the correct representation of conductometric titration of benzoic acid vs sodium hydroxide.

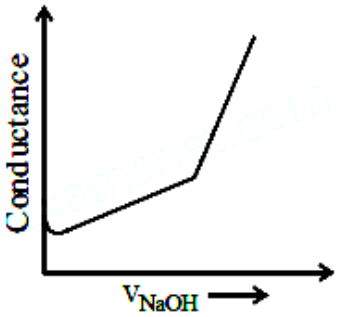
[24-Jan-2023 Shift 2]

Options:

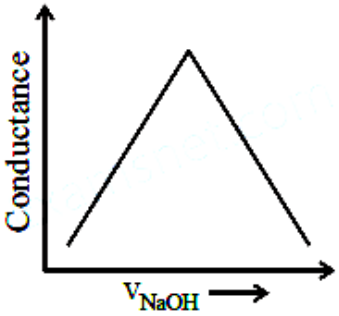
A.



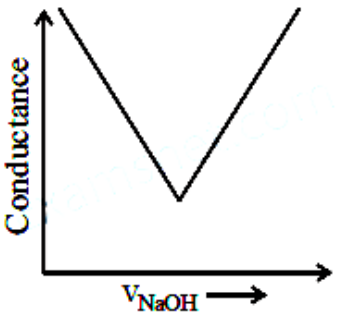
B.



C.



D.

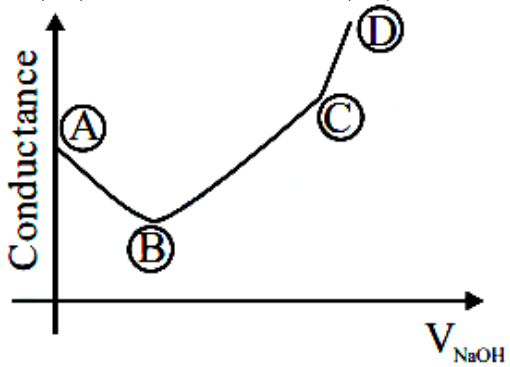


Answer: B

Solution:

Solution:

$$\underset{\text{(WA)}}{C_6H_5COOH} + \underset{\text{(SB)}}{NaOH} \rightarrow \underset{\text{(Salt)}}{C_6H_5COONa} + H_2O$$



(A) → (B) Free H^+ions are replaced by $Na^{\oplus}$ which decreases conductance.
(B) → (C) Un-dissociated benzoic acid reacts with $NaOH$ and forms salt which increases ions & conductance increases.
(C) → (D) After equivalence point at (3), $NaOH$ added further increases $Na^{\oplus}$&$OH^{\odot}$ ions which further increases the conductance.

Question37

Consider the cell

$Pt(s) \mid H_2(s)(1atm)H^+(aq, [H^+] = 1)$

$Fe^{3+}(aq), Fe^{2+}(aq) \mid Pt(s)$

Given: $E^{\circ}_{Fe^{3+}//Fe^{2+}} = 0.771V$ and $E^{\circ}_{H^+/\frac{1}{2}H_2} = 0V$, T = 298K

If the potential of the cell is 0.712V the ratio of concentration of Fe^{2+} to Fe^{2+} is _____ (Nearest integer)

[25-Jan-2023 Shift 1]

Answer: 10

Solution:

Solution:

$$\frac{1}{2}H_2(g) + Fe^{3+}(aq.) \longrightarrow H^+(aq) + Fe^{2+}(aq.)$$

$$E = E^o - \frac{0.059}{1}\log\frac{[Fe^{2+}]}{[Fe^{3+}]}$$

$$\Rightarrow \quad 0.712 = (0.771 - 0) - \frac{0.059}{1}\log\frac{[Fe^{2+}]}{[Fe^{3+}]}$$

$$\Rightarrow \log \frac{[Fe^{2+}]}{[Fe^{3+}]} = \frac{(0.771 - 0712)}{0.059} = 1$$

$$\Rightarrow \frac{[Fe^{2+}]}{[Fe^{3+}]} = 10$$

Question38

Pt(s) H_2(g)(1 bar) | H^+(aq)(1M)| | M^{3+}(aq), M^+(aq) | Pt(s)

The E_{cell} for the given cell is 0.1115V at 298K when $\frac{[M^+(aq)]}{[M^{3+}(aq)]} = 10^a$

The value of a is _____

Given : $E^0 M^{3+}/M^+ = 0.2V$

$\frac{2.303\,RT}{F}$ **= 0.059V**

[25-Jan-2023 Shift 2]

Answer: 3

Solution:

Solution:
Overall reaction :-

$$H_{2(g)} + M_{(aq)}^{3+} \longrightarrow M_{(aq)}^{+} + 2H_{(aq)}^{+}$$

$$E_{Cell} = E_{Cathode}^{0} - E_{anode}^{0} - \frac{0.059}{2} \log \frac{[M^+] \times 1^2}{[M^{+3}]1}$$

$$0.1115 = 0.2 - \frac{0.059}{2} \log \frac{[M^+]}{[M^{+3}]}$$

$$3 = \log \frac{[M^+]}{[M^{+3}]}$$

$$\therefore a = 3$$

Question39

The standard electrode potential (M^{3+}/M^{2+}) for V, Cr, Mn& Co are −0.26V, −0.41V, +1.57V and +1.97V, respectively. The metal ions which can liberate H_2 from a dilute acid are

[29-Jan-2023 Shift 1]

Options:

A. V^{2+} and Mn^{2+}

B. Cr^{2+} and CO^{2+}

C. V^{2+} and Cr^{2+}

D. Mn^{2+} and Co^{2+}
Official Ans. by NTA (3)

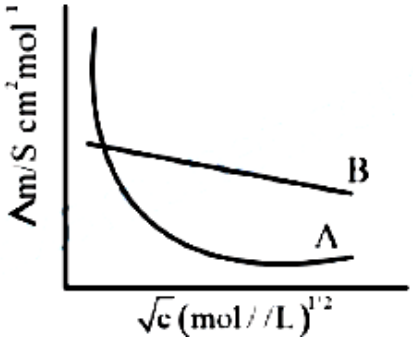
Answer: C

Solution:

Solution:
Metal cation with (−) value of reduction potential (M^{+3}/M^{+2}) or with (+) value of oxidation potential (M^{+2}/M^{+3}) will liberate H_2
Therefore they will reduce H^+
i.eV eV^{+2} and Cr^{+2}

Question40

Following figure shows dependence of molar conductance of two electrolytes on concentration.
Λ_m^0 is the limiting molar conductivity



The number of Incorrect statement(s) from the following is _____
(A) Λm for electrolyte A is obtained by extrapolation
(B) For electrolyte B, vx Λm vs $\sqrt{c}$ graph is a straight line with intercept equal to Λ_m^0
(C) At infinite dilution, the value of degree of dissociation approach zero for electrolyte B.
(D) Λm for any electrolyte A or B can be calculated using λ^0 for individual ions.
[29-Jan-2023 Shift 1]

Answer: 2

Solution:

Solution:
Statement (A) and Statement (C) are incorrect

Question41

The equilibrium constant for the reaction
$Zn(s) + Sn^{2+}(aq) \rightleftharpoons Zn^{2+}(aq) + Sn(s)$ is 1×10^{20}
at 298K. The magnitude of standard electrode potential of Sn/Sn^{2+} if $E_{Zn^{2+}/Zn}^{0} = -0.76V$ is _____ $\times 10^{-2}$V. (Nearest integer)
[29-Jan-2023 Shift 2]

Answer: 17

Solution:

Solution:

Given : $\frac{2.303\,RT}{F} = 0.059V$

$Zn(s) + Sn^{2+}(aq) \rightleftharpoons Zn^{2+}(aq) + Sn(s)$

$\Delta G^{\circ} = -2.303\,RT \log_{10} Keq$

$-nF(E_{cell}^{0}) = -2.303RT \log_{10} Keq$

$0.76 + E_{SQ^{2+}/Su}^{0} = \frac{0.059}{2}\log_{10}10^{20}$

$0.76 + E_{Sn^{2+}/Sn}^{0} = \frac{0.059 \times 20}{2}$

$E_{Sn^{2+}/Sa}^{0} = 0.59 - 0.76 = -0.17$

$E_{Su/Sn^{2+}}^{0} = 17 \times 10^{-2}V$

Ans. $= 17$

Question42

Consider the cell

$Pt_{(s)} \mid H_2(g, 1\,atm) \mid H^+(aq, 1M)$

$Fe^{3+}(aq), Fe^{2+}(aq) \mid Pt(s)$

When the potential of the cell is 0.712V at 298K, the ratio $[Fe^{2+}]/[Fe^{3+}]$ is _____.

(Nearest integer)

Given: $Fe^{3+} + e^- = Fe^{2+}$, $E^\circ_{Fe^{3+}, Fe^{2+} \mid Pt} = 0.771$

$\frac{2.303\,RT}{F} = 0.06V$

[30-Jan-2023 Shift 1]

Answer: 10

Solution:

Solution:

$Pt_{(s)} \mid H_2(g, 1\,latm) \mid H^+(aq, 1M) \mid Fe^{3+}(aq), Fe^{2+}(aq) \mid Pt(s)$

at anode $H_2 \rightarrow 2H^+ + 2e^-$

At cathode $Fe_{aq}^{3+} + e^- \rightarrow Fe_{aq}^{2+}$

$$E^\circ = E^\circ_{H_2 \mid H^+} + E^\circ_{Fe^{3+} \mid Fe^{2+}} = 0 \cdot 771V$$

$$E = E^\circ - \frac{0 \cdot 06}{1} \log \frac{Fe^{2+}}{Fe^{3+}}$$

$$0 \cdot 712 = (0 + 0 \cdot 771) - \frac{0 \cdot 06}{1} \log \frac{Fe^{2+}}{Fe^{3+}}$$

$$\log \frac{Fe^{2+}}{Fe^{3+}} = \frac{0 \cdot 059}{0 \cdot 06} \approx 1$$

$$\frac{Fe^{2+}}{Fe^{3+}} = 10$$

Question43

The electrode potential of the following half cell at 298K .

$X \mid X^{2+}(0.001M) \mid Y^{2+}(0.01M) \mid Y$

is _____ $\times 10^{-2}$V (Nearest integer).

Given: $E^0_{x^{2+} \mid x} = -2.36V$

$$E^{0}_{Y^{2+}/Y} = +0.36V$$

$$\frac{2.303\,RT}{F} = 0.06V$$

[30-Jan-2023 Shift 2]

Answer: 275

Solution:

Solution:

$X + Y^{2+} \rightarrow Y + X^{2+}$

$E_{Cell}^{0} = 0.36 - (-2.36) = 2.72V$

$E_{Cell} = 2.72 - \frac{0.06}{2} \log \frac{0.001}{0.01}$

$= 2.72 + 0.03 = 2.75V$

$= 275 \times 10^{-2}V$

Question44

Which one of the following statements is correct for electrolysis of brine solution?

[31-Jan-2023 Shift 1]

Options:

A. Cl_2 is formed at cathode

B. O_2 is formed at cathode

C. H_2 is formed at anode

D. OH^- is formed at cathode

Answer: D

Solution:

Solution:

Electrolysis of brine solution

$NaCl\ (aq.) \rightarrow Na^{+}_{(aq)} + Cl^{+}_{(aq)}$

At anode : $2Cl^{+}_{(aq.)} \rightarrow \underset{\text{Major}}{Cl_2(g)} + 2e^{-}$

$2H_2O_{(\ell)} \rightarrow \underset{\text{Minor}}{O_{2(g)}} + 4H^+_{(aq)} + 4e^-$

At Cathode : $2H_2O_{(\ell)} + 2e^- \rightarrow H_{2(g)} \uparrow + 2OH^-_{(aq)}$

$2Na^+ + 2OH^- \rightarrow 2\,NaOH$

Question45

The resistivity of a 0.8M solution of an electrolyte is _____ $5 \times 10^{-3}\Omega$ cm. Its molar conductivity is $\times 10^4 \Omega^{-1}cm^2mol^{-1}$. (Nearest integer)
[31-Jan-2023 Shift 2]

Answer: 25

Solution:

Solution:

$$\Lambda_m = \frac{\kappa \times 1000}{M}$$

$$\Lambda_m = \frac{1}{\rho} \times \frac{1000}{M}$$

$$\frac{1}{5 \times 10^{-3}} \times \frac{1000}{0.8}$$

Ans. $25 \times 10^4 \Omega^{-1}cm^{-2}mol^{-1}$

Question46

At what pH, given half cell MnO_4^-(0.1M) | Mn^{2+} (0.001 M) will have electrode potential of 1.282 V ? _____ (Nearest Integer)
Given $E^{o}_{MnO_4^-/Mn^{2+}} = 1.54V$, $\frac{2.303RT}{F} = 0.059V$
[1-Feb-2023 Shift 1]

Answer: 3

Solution:

Solution:

$MnO_4^- + 8H^+ + 5e^- \rightleftharpoons Mn^{2+} + 4H_2O$

$E = E^\circ - \frac{0.059}{5} \log \frac{[Mn^{2+}]}{[MnO_4^-][H^+]^8}$

$1.282 = 1.54 - \frac{0.059}{5} \log \frac{10^{-3}}{10^{-1} \times [H^+]^8}$

$\frac{0.258 \times 5}{0.059} = \log \frac{10^{-2}}{[H^+]^8}$

$\Rightarrow \ 21.86 = -2 + 8\,pH$

$\therefore \ pH = 2.98$

$\simeq 3$

Question47

1×10^{-5}M AgNO 3 is added to 1L of saturated solution of AgBr. The conductivity of this solution at 298K is _____ $\times 10^{-8} Sm^{-1}$.

[Given : $K_{sp}(AgBr) = 4.9 \times 10^{-13}$ at 298K

$\lambda_{Ag^+}^0 = 6 \times 10^{-3} Sm^2 mol^{-1}$

$\lambda_{Br^-}^0 = 8 \times 10^{-3} Sm^2 mol^{-1}$

$\lambda_{NO_3^-}^0 = 7 \times 10^{-3} Sm^2 mol^{-1}$]

[1-Feb-2023 Shift 2]

Answer: 14

Solution:

Solution:

$[Ag^+] = 10^{-5}$

$[NO_3^-] = 10^{-5}$

$[Br^-] = \frac{Ksp}{[Ag^+]} = 4.9 \times 10^{-8}$

$\Lambda_m = \frac{k}{1000 \times M}$

For Ag^+

$6 \times 10^{-3} = \frac{K_{Ag^+}}{1000 \times 10^{-5}}$

$K_{Ag+} = 6 \times 10^{-5}$

$\Rightarrow 6000 \times 10^{-8}$

for Br^-

$$8 \times 10^{-3} = \frac{K_{Br}}{1000 \times 4.9 \times 10^{-8}}$$

$$K_{Br^-} = 39.2 \times 10^{-8}$$

for NO_3^-

$$7 \times 10^{-3} = \frac{K_{NO_3^-}}{1000 \times 10^{-5}}$$

$$K_{NO_3^-} = 7 \times 10^{-5}$$

$$= 7000 \times 10^{-8}$$

Conductivity of solution

$$\Rightarrow (6000 + 7000 + 39.2) \times 10^{-8}$$

$$\Rightarrow 13039.2 \times 10^{-8} Sm^{-1}$$

Question48

The standard electrode potential of M^+/M in aqueous solution does not depend on
[6-Apr-2023 shift 1]

Options:

A. Ionisation of a solid metal atom

B. Sublimation of a solid metal

C. Ionisation of a gaseous metal atom

D. Hydration of a gaseous metal ion

Answer: A

Solution:

Solution:

Question49

The product, which is not obtained during the electrolysis of brine solution is
[6-Apr-2023 shift 2]

Options:

A. NaOH

B. Cl_2

C. H_2

D. HCl

Answer: D

Solution:

Solution:
Brine is aq. Solution of NaCl
$NaCl_{(aq)} \rightarrow Na^+ + Cl^-$
Cathode reaction
$2H_2O + 2e^- \xrightarrow{\text{reduction}} H_{2(g)} + 2OH^-$
Anode reaction
$2Cl^- \xrightarrow{\text{oxidation}} Cl_{2(g)} + 2e^-$
So HCl will not form during electrolysis.

Question50

The standard reduction potential at 298K for the following half cells are given below :-

$NO_3^- + 4H^+ + 3e^- \rightarrow NO(g) + 2H_2O$	$E^\circ = 0.97V$
$V^{2+}(aq) + 2e^- \rightarrow V(s)$	$E^\circ = -1.19V$
$Fe^{3+}(aq) + 3e^- \rightarrow Fe(s)$	$E^\circ = -0.04V$
$Ag^+(aq) + e^- \rightarrow Ag(s)$	$E^\circ = 0.80V$
$Au^{3+}(aq) + 3e^- \rightarrow Au(s)$	$E^\circ = 1.40V$

The number of metal(s) which will be oxidized by NO_3^- in aqueous solution is
[6-Apr-2023 shift 2]

Answer: 3

Solution:

Solution:
Metal having lower SRP than $0.97V$ will be oxidised by NO_3^-.

Question51

The reaction
$$\frac{1}{2}H_2(g) + Ag(Cl)(s) \rightleftharpoons H^+(aq) + Cl^-(aq) + Ag(s)$$
occurs in which of the given galvanic cell.
[8-Apr-2023 shift 1]

Options:

A. $Pt \mid H_2(g) \mid HCl(sol^n) \mid AgNO_3(sol^n) \mid Ag$

B. $Pt \mid H_2(g) \mid HCl(sol^n) \mid AgCl(s) \mid Ag$

C. $Pt \mid H_2(g) \mid KCl(sol^n) \mid AgCl(s) \mid Ag$

D. $Ag \mid AgCl(s) \mid KCl(sol^n) \mid AgNO_3 \mid Ag$

Answer: B

Solution:

Solution:
Anode $\rightarrow H_2 \rightarrow 2H^+ + 2e^-$
Cathode $\rightarrow AgCl + e^- \rightarrow Ag + Cl^-$

Question52

The specific conductance of 0.0025M acetic acid is $5 \times 10^{-5} Scm^{-1}$ at a certain temperature. The dissociation constant of acetic acid is _____ $\times 10^{-7}$. (Nearest integer)
Consider limiting molar conductivity of CH_3COOH as $400 Scm^2 mol^{-1}$.
[10-Apr-2023 shift 2]

Answer: 66

Solution:

Solution:

$\Lambda_m = \frac{k}{C} \times 1000$

Given $k = 5 \times 10^{-5} Scm^{-1}$

$C = 0.0025M$

$\Lambda_m = \frac{5 \times 10^{-5} \times 10^3}{0.0025} = \frac{5 \times 10^{-2}}{2.5 \times 10^{-3}}$

$= 20 Scm^2 mol^{-1}$

$\alpha = \frac{20}{400} = \frac{1}{20}$

$K_a = \frac{C\alpha^2}{1-\alpha} = \frac{0.0025 \times \frac{1}{20} \times \frac{1}{20}}{\frac{19}{20}}$

$= \frac{0.0025}{19 \times 20} = 6.6 \times 10^{-6}$

$= 66 \times 10^{-7}$

Question53

In an electrochemical reaction of lead, at standard temperature, if $E_{(pb^{2+}/pb)}^{\circ}$ = m volt and $E_{(pb^{+}/Pb_b)}^{\circ}$ = n volt, then the value of $E_{(Pb^{2+}/Pb^{+})}^{\circ}$ is given by m − xn. The value of x is _____ (Nearest integer)
[11-Apr-2023 shift 1]

Answer: 2

Solution:

Solution:

$Pb^{2+} + 2e^- \rightarrow Pb$	$E^\circ = m$	$\Delta G_1^\circ = -2Fm$
$Pb^{4+} + 4e^- \rightarrow Pb$	$E^\circ = n$	$\Delta G_2^\circ = -4Fn$
$Pb^{2+} \rightarrow Pb^{4+} + 2e^-$	$\Delta G_3^\circ = \Delta G_1^\circ - \Delta G_2^\circ$	
	$-2FE^\circ = -2Fm + 4Fn$	
	$\boxed{E^\circ = m - 2n}$	
	$\boxed{x = 2}$	

Question54

The number of correct statements from the following is _____
A. E_{cell} is an intensive parameter

B. A negative $E^{\ominus}$ means that the redox couple is a stronger reducing agent than the H^{+}/H_2 couple.

C. The amount of electricity required for oxidation or reduction depends on the stoichiometry of the electrode reaction.
D. The amount of chemical reaction which occurs at any electrode during electrolysis by a current is proportional to the quantity of electricity passed through the electrolyte.
[11-Apr-2023 shift 2]

Answer: 4

Solution:

Solution:
Given statements A, B, C and D are correct.

Question55

For lead storage battery pick the correct statements
A. During charging of battery, $PbSO_4$ on anode is converted into PbO_2

B. During charging of battery, $PbSO_4$ on cathode is converted into PbO_2

C. Lead storage battery consists of grid of lead packed with PbO_2 as anode

D. Lead storage battery has ~38% solution of sulphuric acid as an electrolyte Choose the correct answer from the options given below:
[12-Apr-2023 shift 1]

Options:

A. B, D only

B. B, C only

C. B, C, D only

D. A,B,D only

Answer: A

Solution:

Solution:
Lead storage battery consists of lead anode and a grid of lead packed with lead oxide (PbO_2) as cathode, a 38% solution of H_2SO_4 is used as an electrolyte.
On charging the battery the reaction is reversed and $PbSO_4(s)$ on anode and cathode is converted into Pb and PbO_2 respectively.

Question56

A metal surface of $100cm^2$ area has to be coated with nickel layer of thickness 0.001 mm. A current of 2A was passed through a solution of $Ni(NO_3)_2$ for ' x ' seconds to coat the desired layer. The value of x is (Nearest integer)
(ρ_{Ni}(.. density of Nickel) is $10gmL^{-1}$, Molar mass of Nickel is _____ $60gmol^{-1}F = 96500Cmol^{-1}$)
[13-Apr-2023 shift 1]

Answer: 161

Solution:

Solution:

$$\text{Volume of nickel required} = 100 \times 0.001 \times 10^{-3} \times 100$$
$$= 0.01 cm^3$$
$$\text{Mass of Nickel required} = 0.01 \times 10$$
$$= 0.1 \text{ gm}$$
$$\text{Moles} = \frac{0.1}{60} = \frac{1}{600} \text{ mol}$$
$$Ni^{2+} + 2e^- \rightarrow Ni(s)$$
$$\text{for coating of } 1 \text{ mol Ni, charge required} = 2 \times 96500C$$
$$\text{for coating of } \frac{1}{600} \text{ mol, charge required} = 2 \times 96500 \times \frac{1}{600} C$$
$$= \frac{965}{3} C$$
$$I = \frac{q}{t}$$

$t = \frac{965/3}{2} = 160.83 \text{ sec} \approx 161$

Question57

At 298K, the standard reduction potential for Cu^{2+}/Cu electrode is 0.034V.

Given : $K_{sp}\ Cu(OH)_2 = 1 \times 10^{-20}$

Take $\frac{2.303RT}{F} = 0.059V$

The reduction potential at pH = 14 for the above couple is $(-)x \times 10^{-2}V$. The value of x is _____.

[13-Apr-2023 shift 2]

Answer: 25

Solution:

Solution:

$Cu(OH)_2(s) \rightleftharpoons Cu^{2+}(aq) + 2OH^-(aq)$

$Ksp = [Cu^{2+}][OH^-]^2$

$pH = 14;\ pOH = 0;\ [OH^-] = 1M$

$\therefore [Cu^{2+}] = \frac{Ksp}{[1]^2} = 10^{-20}M$

$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

$E = E^\circ - \frac{0.059}{2}\log_{10}\frac{1}{[Cu^{2+}]}$

$= 0.34 - \frac{0.059}{2}\log_{10}\frac{1}{10^{-20}}$

$= -0.25 = -25 \times 10^{-2}$

Question58

The number of correct statements from the following is _____.
(A) Conductivity always decreases with decrease in concentration for both strong and weak electrolysis.
(B) The number of ions per unit volume that carry current in a solution increases on dilution.
(C) Molar conductivity increases with decrease in concentration.

(D) The variation in molar conductivity is different for strong and weak electrolysis.
(E) For weak electrolysis, the change in molar conductivity with dilution is due to decrease in degree of dissociation.
[15-Apr-2023 shift 1]

Answer: 3

Solution:

Solution:
(A) Conductivity decreases with dilution for strong electrolyte as well as weak electrolyte.
(B) On dilution, The number of ions per unit volume that carry current in a solution decreases.
(C) Molar conductivity increases with dilution.
(D) Molar conductivity of strong electrolyte follows DHO equation but it is not applicable for weak electrolyte.
(E) On dilution degree of dissociation of weak electrolyte increases.
So answer is (A), (C) & (D)

Question59

The cell potential for the following cell
Pt | $H_2(g)$ | $H^+(aq)$ | $Cu^{2+}(0.01M)$ | Cu(s)
is 0.576V at 298K. The pH of the solution is _____(Nearest integer)
(Given : $E^{o}_{Cu^{2+}/Cu} = 0.34V$ and $\frac{2.303RT}{F} = 0.06V$)
[24-Jun-2022-Shift-1]

Answer: 5

Solution:

Solution:

$$E_{cell} = E_{cell}^{0} - \frac{0.06}{2}\log\frac{[H^{\oplus}]^2}{[Cu^{+2}]}$$

$$0.576 = 0.34 - 0.03\log\frac{[H^{\oplus}]^2}{[0.01]}$$

$$0.576 - 0.34 = -0.03\log[H^{\oplus}]^2 + 0.03\log(0.01)$$

$$= 0.06\,pH - 0.06$$

$$pH \simeq 4.93 \simeq 5$$

Question60

The resistance of a conductivity cell containing 0.01M KCl solution at 298K is 1750Ω. If the conductivity of 0.01M KCl solution at 298K is $0.152 \times 10^{-3} Scm^{-1}$, then the cell constant of the conductivity cell is _____ $\times 10^{-3} cm^{-1}$
[24-Jun-2022-Shift-2]

Answer: 266

Solution:

Solution:
Molarity of KCl solution $= 0.1M$
Resistance $= 1750\,ohm$
Conductivity $= 0.152 \times 10^{-3} Scm^{-1}$
Conductivity $= \dfrac{\text{Cell constant}}{\text{Resistance}}$
$\therefore$ Cell constant $= 0.152 \times 10^{-3} \times 1750$
$= 266 \times 10^{-3} cm^{-1}$

Question61

In a cell, the following reactions take place

$$Fe^{2+} \rightarrow Fe^{3+} + e^- \quad E^o_{Fe^{3+}/Fe^{2+}} = 0.77V$$

$$2I^- \rightarrow I_2 + 2e^- \quad E^o_{I_2/I^-} = 0.54V$$

The standard electrode potential for the spontaneous reaction in the cell is $x \times 10^{-2} V298K$. The value of x is ___(Nearest Integer)
[25-Jun-2022-Shift-1]

Answer: 23

Solution:

Solution:

$$\underset{\text{Cathode}}{Fe^{+3}} + \underset{\text{anode}}{I^{-}} \longrightarrow I_2 + Fe^{+2}$$

$$E_{cell}^{\circ} = E_{cathode}^{\circ} - E_{anode}^{\circ}$$
$$= 0.77 - 0.54$$
$$= 0.23V$$
$$= 23 \times 10^{-2}V$$

Question62

The correct order of reduction potentials of the following pairs is

A. Cl_2/Cl^-

B. $1^2/1^-$

C. Ag^+/Ag

D. Na^+/Na

E. Li^+/Li

Choose the correct answer from the options given below.
[25-Jun-2022-Shift-2]

Options:

A. A > C > B > D > E

B. A > B > C > D > E

C. A > C > B > E > D

D. A > B > C > E > D

Answer: A

Solution:

Solution:

$E_{C_2/CI}^{\circ} = +1.36V$

$E_{I_2/I^-}^{0} = +0.54V$

$E_{Ag^+/Ag}^{\circ} = +0.80V$

$E_{Na^+/Na}^{\circ} = -2.71V$

$E_{L^+/Li} = -3.05V$

Question63

A solution of $Fe_2(SO_4)_3$ is electrolyzed for ' x ' min with a current of 1.5A to deposit 0.3482g of Fe. The value of x is [nearest integer]
Given : 1F = 96500Cmol^{-1}
Atomic mass of Fe = 56gmol^{-1}
[25-Jun-2022-Shift-2]

Answer: 20

Solution:

Solution:
$Fe^{3+} + 3e^- \longrightarrow Fe$
$3F \longrightarrow 1$ mole Fe is deposited
For $56g \longrightarrow 3 \times 96500$ (required charge)
For $1g \longrightarrow \frac{3 \times 96500}{56}$ (required charge)
For $0.3482g \longrightarrow \frac{3 \times 96500}{56} \times 0.3482$
$= 1800.06$
$Q = it$
$1800.06 = 1.5t$
$t = 20$ min

Question64

The $\left(\frac{\partial E}{\partial T}\right)_P$ of different types of half cells are as follows:

A	B	C	D
1×10^{-4}	2×10^{-4}	0.1×10^{-4}	0.2×10^{-4}

(Where E is the electromotive force)
Which of the above half cells would be preferred to be used as reference electrode?
[26-Jun-2022-Shift-1]

Options:

A. A

B. B

C. C

D. D

Answer: C

Solution:

Solution:
A cell with less variation in EMF with temperature is preferred as a reference electrode because it can be used for a wider range of temperatures without much derivation from standard value so a cell with less $\left(\frac{\partial E}{\partial T}\right)_P$ is preferred.

Question65

$Cu(s) + Sn^{2+}(0.001M) \rightarrow Cu^{2+}(0.01M) + Sn(s)$
The Gibbs free energy change for the above reaction at 298K is $x \times 10^{-1}$ kJ mol^{-1}. The value of x is _____ [nearest integer]
[Given : $E_{Cu^{2+}/Cu}^{\Theta} = 0.34V$; $E_{Sn^{2+}/Sn}^{\Theta} = -0.14V$; $F = 96500 C mol^{-1}$]
[26-Jun-2022-Shift-2]

Answer: 983

Solution:

Solution:
$Cu_{(s)} + Sn^{2+}(0.001M) \rightarrow Cu^{2+}(0.01M) + Sn_{(s)}$

$E_{cell}^{\circ} = E_{cathode}^{\circ} - E_{anode}^{\circ}$

$= -0.14 - (0.34)$

$= -0.48V$

$E_{cell} = E_{cell}^{\circ} - \frac{0.059}{2} \log \frac{[Cu^{2+}]}{[Sn^{2+}]}$

$= -0.48 - \frac{0.059}{2} \log \frac{0.01}{0.001}$

$= -0.509$

$\Delta G = -nF E_{cell}$

$= -2 \times 96500 \times (-0.5095)$

$= 98333.5 J/mol$

$= 98.335 kJ/mol$

$= 983.35 \times 10^{-1} kJ/mol$

Question66

In 3d series, the metal having the highest M^{2+}/M standard electrode potential is
[27-Jun-2022-Shift-2]

Options:

A. Cr

B. Fe

C. Cu

D. Zn

Answer: C

Solution:

Solution:
$Cr^{+2}/Cr \rightarrow -0.90V$
$Fe^{+2}/Fe \rightarrow -0.44V$
$Cu^{+2}/Cu \rightarrow +0.34V$
$Zn^{+2}/Zn \rightarrow -0.76V$
So Ans. is Cu^{+2}/Cu

Question67

For the reaction taking place in the cell :
$Pt(s) \mid H_2(g) \mid H^+(aq) \mid Ag^+(aq) \mid Ag(s)\ E_{cell}^{o} = +0.5332V.$
The value of $\Delta_f G^\circ$ is ____kJ mol^{-1}. (in nearestinteger)
[27-Jun-2022-Shift-2]

Answer: 103

Solution:

Solution:

$H_2 \left| H^+ \right| \left| Ag^+ \right| Ag$

Anode Cathode

At anode, oxidation occur

$H_2 \rightarrow 2H^+ + 2e^-$

\# At cathode, reduction occur

$2Ag^+ + 2e^- \rightarrow 2\,Ag$

Adding equation (1) and (2), we get $n = 2$, where $n =$ cancelled out electron

Now,

$\Delta G^\circ = -nF\,E_{cell}^{o}$

$= -2 \times 96500 \times 0.5332$

$= -102907.6$

$= -102.9\,kJ/mol$

$= -103\,kJ/mol$

Question68

The solubility product of a sparingly soluble salt A_2X_3 is 1.1×10^{-23}. If specific conductance of the solution is $3 \times 10^{-5} Sm^-1$, the limiting molar conductivity of the solution is $x \times 10^{-3} Sm^2 mol^{-1}$. The value of x is____ [28-Jun-2022-Shift-1]

Answer: 3

Solution:

Solution:

$A_2X_3 \rightleftharpoons 2\underset{2S}{A} + \underset{3S}{3X}$

$K_{sp} = (2\,s)^2(3s)^3 = 1.1 \times 10^{-23}$

$S \approx 10^{-5}$

For sparingly soluble salts

$\Lambda_m = \Lambda_m^0$

$\Lambda_m = \dfrac{k}{S \times 10^3}$

$= \dfrac{3 \times 10^{-5}}{10^{-5}} \times 10^{-3}$

$= 3 \times 10^{-3} Sm^2 mol^{-1}$

Question69

The quantity of electricity in Faraday needed to reduce 1 mol of $Cr_2O_7^{2-}$ to Cr^{3+} is _____
[28-Jun-2022-Shift-1]

Answer: 6

Solution:

Solution:
$Cr_2O_7^{-2} + 6e^- \longrightarrow 2Cr^{+3}$
1 mol 6 mol
⇒ number of faradays = moles of electrons = 6

Question70

For the given reactions
$Sn^{2+} + 2e^- \rightarrow Sn$
$Sn^{4+} + 4e^- \rightarrow Sn$
the electrode potentials are ; $E_{Sn^{2+}/Sn}^{o} = -0.140V$ and $E_{Sn^{4+}/Sn}^{o} = -0.010V$.
The magnitude of standard electrode potential for Sn^{4+}/Sn^{2+} i.e. $E_{Sn^{4+}/Sn^{2+}}^{o}$ is _____$\times 10^{-2}$V. (Nearest integer)
[28-Jun-2022-Shift-2]

Answer: 16

Solution:

Solution:
$Sn \longrightarrow Sn^{2+} + 2e^- \quad E_1^0 = 0.140V$
$Sn^{4+} + 4e^- \longrightarrow Sn \quad E_2^0 = 0.010V$
$Sn^{4+} + 2e^- \longrightarrow Sn^{2+} \quad E_{cell}^0$
$E_{cell}^{o} = \frac{n_2E_2^{o} + n_1E_1^0}{n} = \frac{4(0.010) + 2(0.140)}{2}$
$E_{cell}^0 = 0.16V = 16 \times 10^{-2}V$

Question71

A dilute solution of sulphuric acid is electrolysed using a current of 0.10A for 2 hours to produce hydrogen and oxygen gas. The total volume of gases produced a STP is_____cm^3. (Nearest integer)
[Given : Faraday constant F = 96500$Cmol^{-1}$ at STP, molar volume of an ideal gas is 22.7$Lmol^{-1}$]
[29-Jun-2022-Shift-1]

Answer: 127

Solution:

Solution:
At anode

$$2H_2O \rightarrow O_2(g) + 4H^+ + 4e^-$$

At cathode

$$2H^+ + 2e^- \rightarrow H_2(g)$$

Now number of gm eq. $= \frac{i \times t}{96500}$

$$= \frac{0.1 \times 2 \times 60 \times 60}{96500}$$

$$= 0.00746$$

$$V_{O_2} = \frac{0.00746}{4} \times 22.7 = 0.0423$$

$$V_{H_2} = \frac{0.00746}{2} \times 22.7 = 0.0846$$

$$V_{Total} \approx 127 \text{ ml or cc}$$

Question72

The cell potential for the given cell at 298K
Pt | H_2(g, 1 bar) | H^+(aq) | Cu^{2+}(aq) | Cu(s)
is 0.31V. The pH of the acidic solution is found to be 3 , whereas the concentration of Cu^{2+} is 10^{-x}M. The value of x is_____
(Given : $E^{\Theta}_{Cu^{2+}/Cu}$ = 0.34V and $\frac{2.303RT}{F}$ = 0.06V)
[29-Jun-2022-Shift-2]

Answer: 7

Solution:

Solution:

$$Q = \frac{[H^+]^2}{[Cu^2]pH_2} = \frac{10^{-6}}{C} \quad pH_2 = 1$$

$$E = E_{cell}^{\circ} - \frac{0.06}{n}\log Q$$

$$0.31 = 0.34 - \frac{0.06}{2}\log\frac{10^{-6}}{C}$$

$$\log\frac{10^{-6}}{C} = 1$$

$$C = 10^{-7}M$$

$$x = 7$$

Question73

The cell potential for $Zn \mid Zn^{2+}(aq) \mid\mid Sn^{x+} \mid Sn$ is 0.801V at 298K . The reaction quotient for the above reaction is 10^{-2}. The number of electrons involved in the given electrochemical cell reaction is
(Given : $E^{o}_{Zn^{2+} \mid Zn} = -0.763V$, $E^{o}_{Sn^{x+} \mid Sn} = +0.008V$ and $\frac{2.303RT}{F} = 0.06V$)
[25-Jul-2022-Shift-1]

Answer: 4

Solution:

Solution:

$$A : Zn \rightarrow Zn^{2+} + 2e^-$$

$$C : Sn^{+x} + xe^- \rightarrow Sn$$

$$E_{Cell}^{\circ} = E_{Zn \mid Zn^{2+}}^{\circ} + E_{Sn^{+x} \mid Sn}^{\circ}$$

$$\Rightarrow 0.763 + 0.008 = 0.771V$$

From the Nernst equation,

$$E_{Cell} = E_{Cell}^{\circ} \frac{-2.303\,RT}{nF}\log Q$$

$$0.801 = 0.771 - \frac{0.06}{n}\log 10^{-2}$$

$0.03 = \frac{0.06}{n} \times 2$

$n = 4$

Question74

The molar conductivity of a conductivity cell filled with 10 moles of 20mLN aCl solution is Λ_{m1} and that of 20 moles another identical cell heaving 80mLN aCl solution is Λ_{m2}. The conductivities exhibited by these two cells are same. The relationship between Λ_{m2} and Λ_{m1} is [25-Jul-2022-Shift-2]

Options:

A. $\Lambda_{m2} = 2\Lambda_{m1}$

B. $\Lambda_{m2} = \Lambda_{m1}/2$

C. $\Lambda_{m.2} = \Lambda_{m1}$

D. $\Lambda_{m2} = 4\Lambda_{m1}$

Answer: A

Solution:

Solution:

$$\Lambda_{m_1} = \frac{k_1 \times 1000}{M_1} = \frac{k \times 1000}{\frac{10}{0.02}}$$

$$\Lambda_{m_2} = \frac{k_2 \times 1000}{\frac{20}{0.08}}$$

It is given that $k_1 = k_2$

$$k_1 = \frac{\Lambda_{m_1}}{2} \quad k_2 = \frac{\Lambda_{m_2}}{4}$$

Applying the given condition on conductivity.

$$\frac{\Lambda_{m_1}}{2} = \frac{\Lambda_{m_2}}{4}$$

$$\Lambda_{m_2} = 2\Lambda_{m_1}$$

Question75

The spin-only magnetic moment value of M^{3+} ion (in gaseous state) from the pairs Cr^{3+}/Cr^{2+}, Mn^{3+}/Mn^{2+}, Fe^{3+}/Fe^{2+} and Co^{3+}/Co^{2+} that has negative standard electrode potential, is _____ B.M. [Nearest integer]
[25-Jul-2022-Shift-2]

Answer: 4

Solution:

Solution:
Among the pairs given, Cr^{3+}/Cr^{2+} has negative reduction potential which is $-0.41V$.
$Cr(III) \Rightarrow d^3$
Number of unpaired electrons $= 3$
$\mu = \sqrt{3(3+2)} = \sqrt{15} \simeq 4$ B.M.

Question76

The amount of charge in F (Faraday) required to obtain one mole of iron from Fe_3O_4 is _____. (Nearest Integer)
[26-Jul-2022-Shift-1]

Answer: 3

Solution:

Solution:
For Fe_3O_4,
$x = \frac{+8}{3}$
where x is oxidation state of Fe.
$Fe_3O_4 + 8H^+ + 8e^- \longrightarrow 3Fe + 4H_2O$
Charge required $= \frac{8}{3} \times F = \frac{8F}{3} \simeq 3F$

Question77

Given below are two statements :
Statement I : For KI, molar conductivity increases steeply with dilution
Statement II : For carbonic acid, molar conductivity increases slowly with dilution
In the light of the above statements, choose the correct answer from the options given below:
[27-Jul-2022-Shift-2]

Options:

A. Both Statement I and Statement II are true

B. Both Statement I and Statement II are false

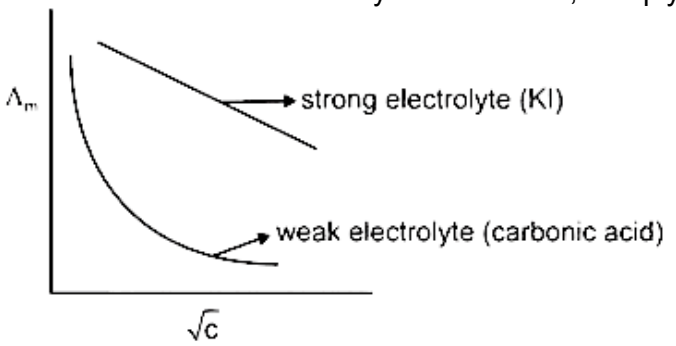
C. Statement I is true but Statement II is false

D. Statement I is false but Statement II is true

Answer: B

Solution:

Solution:
Statement I: KI is strong electrolyte thus almost constant on dilution.
Statement II: In weak electrolyte it increases, sharply.



Question78

Match List - I with List - II.

List - I	List - II
(A) $Cd(s)+2\,Ni(OH)_3(s) \rightarrow CdO(s)+2\,Ni(OH)_2(s)+H_2O(l)$	(I) Primary battery
(B) $Zn(Hg)+HgO(s) \rightarrow ZnO(s)+Hg(l)$	(II) Discharging of secondary battery
(C) $2PbSO_4(s)+2H_2O(l) \rightarrow Pb(s)+PbO_2(s)+2H_2SO_4(aq)$	(III) Fuel cell
(D) $2H_2(g)+O_2(g) \rightarrow 2H_2O(l)$	(IV) Charging of secondary battery

Choose the correct answer from the options given below:
[28-Jul-2022-Shift-1]

Options:

A. (A) − (I), (B) − (II), (C) − (II), (D) − (IV)

B. (A) − (IV), (B) − (I), (C) − (I), (D) − (II)

C. (A) − (II), (B) − (I), (C) − (IV), (D) − (III)

D. (A) − (II) ,(B) − (I), (C) − (II), (D) − (IV)

Answer: C

Solution:

Solution:
(a) $Cd(s) + 2\,Ni(OH)_3(s) \rightarrow CdO(s) + 2\,Ni(OH)_2(s) + H_2O(l)$ Discharge of secondary Battery
(b) $Zn(Hg) + HgO(s) \rightarrow ZnO(s) + Hg(l)$ (Primary Battery Mercury cell)
(c) $2PbSO_4(s) + 2H_2O(l) \rightarrow Pb(s) + PbO_2(s) + 2H_2SO_4(aq)$ (Charging of secondary Battery)
(d) $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$ (Fuel cell)

Question79

Resistance of a conductivity cell (cell constant $129m^{-1}$) filled with 74.5 ppm solution of KCl is 100Ω (labelled as solution 1). When the same cell is filled with KCl solution of 149 ppm, the resistance is 50Ω (labelled as solution 2). The ratio of molar conductivity of solution 1 and solution 2 is i.e. $\frac{\Lambda_1}{\Lambda_2} = x \times 10^{-3}$. The value of x is _____.(Nearest integer)

Given, molar mass of KCl is $74.5gmol^{-1}$.
[29-Jul-2022-Shift-1]

Answer: 1000

Solution:

Solution:
$\frac{l}{A} = 129m^{-1}$
KCl solution 1 ⇒ 74.5 ppm, $R_1 = 100\Omega$

KCl solution 2 ⇒ 149 ppm, $R_2 = 50\Omega$

Here, $\frac{ppm_1}{ppm_2} = \frac{M_1}{M_2} = \left(\frac{w_{1/M_0}}{V} \times \frac{V}{w_{2/M_0}} \right)$

$$\frac{\Lambda_1}{\Lambda_2} = \frac{k_1 \times \frac{1000}{M_1}}{k_2 \times \frac{1000}{M_2}}$$

$$= \frac{k_1}{k_2} \times \frac{M_1}{M_2}$$

$$= \frac{50}{100} \times 2$$

$$= \frac{\Lambda_1}{\Lambda_2} = 1000 \times 10^{-3}$$

=1000

Question80

For a cell, Cu(s) | Cu^{2+}(0.001M) | Ag^{+}(0.01M) | Ag(s) the cell potential is found to be 0.43V at 298K. The magnitude of standard electrode potential for Cu^{2+}/Cu is _____$\times 10^{-2}$V

[Given : $E^{\Theta}_{Ag^{+}/Ag} = 0.80V$ and . $\frac{2.303RRT}{F} = 0.06V$]

[29-Jul-2022-Shift-2]

Answer: 34

Solution:

Solution:

Anode : $Cu(s) \longrightarrow Cu^{2+}(aq) + 2e^{-}$

Cathode: $[Ag^{+} + e^{-} \longrightarrow Ag(s)]2$

Cus(s) $+2Ag^{+}(aq) \longrightarrow Cu^{2+}(aq) + 2\,Ag(s)$

$$E_{cell} = E_{cell}^{0} - \frac{0.06}{2} \log \frac{[Cu^{2+}]}{[Ag^{+}]^2}$$

$$0.43 = E_{cell}^{0} - \frac{0.06}{2} \log \left(\frac{10^{-3}}{(10^{-2})^2} \right)$$

$$0.43 = E_{cell}^{0} - 0.03 \log 10$$

$$E_{cell}^{0} = 0.46V$$

$$E_{cell}^{0} = E_{Ag^{+}/Ag}^{0} - E_{Cu^{2+}/Cu}^{0}$$

$$E_{Cu^{2+}/Cu^{0}} = (0.80 - 0.46) = 0.34V = 34 \times 10^{-2}$$

Question81

Emf of the following cell at 298K in V is
$x \times 10^{-2}$, $Zn \mid Zn^{2+}(0.1M) \mid Ag^{+}(0.01M) \mid Ag$ The value of x is ..... .
(Rounded off to the nearest integer).
[Given, $E_{Zn^{2+}/Zn}^{\circ} = -0.76V$, $E_{Ag^{+}/Ag}^{\circ} = +0.80V$, $\frac{2.303RT}{F} = 0.059$]
[26 Feb 2021 Shift 2]

Answer: 147

Solution:

Solution:
$Zn \mid Zn^{2+}(0.1M) \mid Ag^{+}(0.01M) \mid Ag$
Cell reaction : $Zn(s) + 2Ag^{+} \rightleftharpoons Zn^{2+} + 2Ag(s)$

$$\Rightarrow Q = \frac{[Zn^{2+}][Ag]^2}{[Zn][Ag^{+}]^2} = \frac{0.1M}{1 \times (0.01)^2} = 10^3$$

$n = 2$ and $E_{cell}^{\circ} = E_{Ag^{+}/Ag}^{\circ} - E_{Zn^{\circ}}^{2+}/Zn$

$= 0.8 - (-0.76) = 1.56V$

$$\Rightarrow E_{cell} = E_{cell}^{\circ} - \frac{0.059}{n}\log Q = 1.56 - \frac{0.059}{2}\log 10^3$$

$= 1.4715V$
$= 147.15 \times 10^{-2}V = 147 \times 10^{-2}V$
$= x \times 10^{-2}V$
$x = 147$

Question82

Copper reduces NO_3^- into NO and NO_2 depending upon the concentration of HNO_3 in solution. (Assuming fixed $[Cu^{2+}]$ and $p_{NO} = p_{NO_2}$), the HNO_3 concentration at which the thermodynamic tendency for reduction of NO_3^- into NO and NO_2 by copper is same is $10^x M$. The value of 2x is ..... .(Rounded off to the nearest integer)
[Given, $E_{Cu^{2+}/Cu}^{\circ} = 0.34V$, $E_{NO_3^-/NO}^{\circ} = 0.96V$
$E_{NO_3^{\circ}/NO_2}^{\circ} = 0.79V$ and at 298K, $\frac{RT}{F}$

(2.303) = 0.059]
[25 Feb 2021 Shift 2]

Answer: 4

Solution:

Solution:

Cell-I $(HNO_3 \longrightarrow NO)$ $3Cu + 2NO_3^- + 8H^+ \longrightarrow 3Cu^{2+} + 2NO + 4H_2O$

$Q_1 = \dfrac{[Cu^{2+}]^3 \times (p_{NO})^2}{[NO_3^-]^2 \times [H^+]^8}$

$\because \ E_1^\circ = 0.96 - (-0.34) = 1.3V$

$E_1 = 1.3 - \dfrac{0.059}{6} \log Q_1$

Cell-II $(HNO_3 \rightarrow NO_2)$

$Cu + 2NO_3^- + 4H^+ \longrightarrow Cu^{2+} + 2NO_2 + 2H_2O$

$Q_2 = \dfrac{[Cu^{2+}] \times (p_{NO_2})^2}{[NO_3^-]^2 \times [H^+]^4}$

$E_2^\circ = 0.79 - (-0.34)V = 1.13V$

$E_2 = 1.13 - \dfrac{0.059}{2} \log Q_2$

$\because \ E_2$

Now, $E_1 = E_2$

$1.3 - \dfrac{0.059}{6} \log Q_1 = 1.13 - \dfrac{0.059}{2} \log Q_2$

$0.17 = \dfrac{0.059}{6}[\log Q_1 - 3 \log Q_2] = \dfrac{0.059}{6} \log \dfrac{Q_1}{Q_2}$

$= \dfrac{0.059}{6} \log \dfrac{[Cu^{2+}]^3 \times (p_{NO})^2}{[NO_3^-]^2 \times [H^+]^8} \times \dfrac{[NO_3^-]^6 \times [H^+]^{12}}{[Cu^{2+}]^3 \times (p_{NO_2})^6} = \dfrac{0.059}{6} \log \dfrac{[H^+]^4 \times [NO_3^-]^4}{(p_{NO_2})^4} \quad [\because p_{NO} = p_{NO_2}]$

$= \dfrac{0.059}{6} \log \dfrac{[HNO_3]^4}{(p_{NO_2})^4}$

Now, $p_{NO_2} \equiv [HNO_3]$

So, $0.17 = \dfrac{0.059}{6} \log [HNO_3]^8$

$= \dfrac{0.059}{6} \times 8 \log [HNO_3]$

$\log [HNO_3] = 2.16$

$[HNO_3] = 10^{2.16} M = 10^x M$

$\therefore \ x = 2.16$

$\Rightarrow \ 2x = 2 \times 2.16 = 4.32 \sim eq4$

Question83

The magnitude of the change in oxidising power of the MnO_4^-/Mn^{2+} couple is $x \times 10^{-4}$ V , if the H^+concentration is decreased from 1M to 10^{-4}M at 25°C. (Assume concentration of $MnO_4^-$and Mn^{2+} to be same on change in H^+concentration). The value of x is ..... (Rounded off to the nearest integer). [Given, $\frac{2.303RT}{F} = 0.059$]

[24 Feb 2021 Shift 2]

Answer: 3776

Solution:

Solution:

Reaction, $MnO_4^- + H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$

Applying Nernst equation, $E_{cell} = E_{cell}^\circ - \frac{0.0591}{n} \log \frac{[P]}{[R]}$

or $E_{cell} = E_{cell}^\circ - \frac{0.0591}{5} \log \frac{[Mn^{2+}]}{[MnO_4^-]} \left[\frac{1}{H^+}\right]^8$

(I) Given, $[H^+] = 1M$

$E_1 = E^\circ - \frac{0.0591}{5} \log \frac{[Mn^{2+}]}{[MnO_4^-]}$

(II) Now, $[H^+] = 10^{-4}M$

$E_2 = E^\circ - \frac{0.0591}{5} \log \frac{[Mn^{2+}]}{[MnO_4^-]} \times \frac{1}{(10^{-4})^8}$

$\therefore |E_1 - E_2|$

$|E_1 - E_2| = \frac{0.0591}{5} \times 32 = 0.3776V = 3776 \times 10^{-4}$

$x = 3776$

Question84

Compound A used as a strong oxidising agent is amphoteric in nature. It is the part of lead storage batteries. Compound A is

[26 Feb 2021 Shift 1]

Options:

A. PbO_2

B. PbO

C. $PbSO_4$

D. Pb_3O_4

Answer: A

Solution:

Solution:
In the set of four lead compounds Pb (II) compounds are PbO and $PbSO_4$. PbO_2 is a Pb (IV) compound whereas Pb_3O_4 is a mixed oxide of Pb (II) and Pb (IV) i.e. $2PbO \cdot PbO_2$.

Pb is a member of group 14 and it shows +2 and +4 oxidation states. But due to inert pair effect, Pb^{2+} is more stable than Pb^{4+}. So, Pb (IV) compounds are strong oxidising agents as Pb^{4+} gets easily reduced to more stable Pb^{2+}.

$Pb^{+4} + 2e^- \longrightarrow Pb^{2+}$, $\Delta C^\circ < 0$ (spontaneous)

So, PbO_2 or Pb_3O_4 can be the compound A. But out of these two compounds only PbO_2 is used in lead storage batteries where a grid of lead packed with PbO_2 acts as cathode and also it is amphoteric in nature. It reacts with both acids and alkali.

(i) $PbO_2 + 2HCl \longrightarrow PbCl_2 + Cl_2 + H_2O$

Here, PbO_2 acts as a basic oxide as well as an oxidising agent.

(ii) $PbO_2 + 2NaOH \longrightarrow Na_2PbO_2 + H_2O$

Here, PbO_2 acts as an acidic oxide.

So, the compound A is PbO_2 (option-a).

Question85

Consider the following reaction, $MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$, $E^\circ = 1.51V$. The quantity of electricity required in Faraday to reduce five moles of MnO_4^- is ..... .

[26 Feb 2021 Shift 1]

Answer: 25

Solution:

Solution:
In the reduction half-cell,

$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H$

1mol 5F

5mol 25F

As one mole of MnO_4^- required $5F$ of charge,

5 moles of MnO_4^- will require charge,

$Q = 5 \times 5$ Faraday $= 25$ Faraday

Question86

A 5.0mmol d m^{-3} aqueous solution of K Cl has a conductance of 0.55mS when measured in a cell constant 1.3cm^{-1}.

The molar conductivity of this solution is mSm^2mol^{-1}.

(Round off to the nearest integer)

[16 Mar 2021 Shift 2]

Answer: 14

Solution:

Solution:

$$\Lambda_m = \frac{K}{1000 \times C}$$

where, Λ_m = molar conductivity κ = conductivity, C = molarity

Conductivity $(\kappa) = G\frac{I}{A} = 0.55 \times (1.3 \times 100)$

$= 55 \times 1.3$

$$\Lambda_m = \frac{55 \times 1.3}{1000 \times 5 \times 10^{-3}}$$

$\Lambda_m = 14.3$

Closest integer, $\Lambda_m = 14$

Question87

The molar conductivities at infinite dilution of barium chloride, sulphuric acid and hydrochloric acid are 280,860 and 426Scm^2 mol^{-1} respectively. The molar conductivity at infinite dilution of barium sulphate is ..... Scm^2mol^{-1} (Round off to the nearest Integer).

[18 Mar 2021 Shift 2]

Answer: 288

Solution:

Solution:
Molar conductivity of $BaCl_2 = 280 Scm^2 mol^{-1}$
Molar conductivity of $H_2SO_4 = 860 Scm^2 mol^{-1}$
Molar conductivity of $HCl = 426 Scm^2 mol^{-1}$
Molar conductivity of $BaSO_4 =$?
From Kohlrausch's law,
$\Lambda_m^{\infty}(BaSO_4) = \lambda_m^{\omega}(Ba^{2+}) + \lambda_m^{\omega}(SO_4^{2-})$
$\Lambda_m^{\infty}(BaSO_4) = \Lambda_m^{-}(BaCl_2) + \Lambda_m^{\infty}(H_2SO_4) - 2\Lambda_m^{\infty}(HCl)$
On putting given values we get,
$= 280 + 860 - 2(426)$
$= 288 Scm^2 mol^{-1}$

Question88

For the reaction,
$2Fe^{3+}(aq) + 2l^{-}(aq) \longrightarrow 2Fe^{2+}(aq) + I_2(s)$
The magnitude of the standard molar Gibbs free energy change,
$\Delta, G_m^{\circ} = -.....$ kJ
(Round off to the nearest integer).

$$\left[\begin{array}{ll} E_{Fe^{2+}/Fe(s)}^{\circ} = -0.440V; & E_{Fe^{3+}/Fe(s)}^{\circ} = -0.036V \\ E_{I_2/2I^-} = 0.539V; & F = 96500C \end{array}\right]$$

[18 Mar 2021 Shift 1]

Answer: 45

Solution:

Solution:
Given reaction is
$2Fe^{3+}(aq) + 2I^{-}(aq) \longrightarrow 2Fe^{2+}(aq) + I_2(s)$

Now, $Fe^{3+} \xrightarrow{E_1^{\circ}} Fe^{2+} \xrightarrow{E_2^{\circ}} Fe$

$\xrightarrow{E_3^{\circ}}$

$nE_1^\circ + nE_2^\circ = nE_3^\circ$ [n = number of electron transferred]

$E_1^\circ + 2E_2^\circ = 3E_3^\circ$

$E_{Fe^{3+}/Fe^{2+}}^{0} + 2E_{Fe^{2+}/Fe(s)}^{0} = 3E_{Fe^{3+}/Fe(s)}^{0}$

$E_{Fe^{3+}/Fe^{2+}}^\circ + (-0.440) \times 2 = (-0.036) \times 3$

$E_{Fe^{3+}/Fe^{2+}}^\circ = 0.772V$

$E_{cell}^\circ = E_{cathode}^\circ - E_{anode}^\circ$

$= 0.772 - 0.539 = 0.233V$

Using standard Gibb's free energy,

$\Delta G^\circ = nF E_{cell}^\circ = +2 \times 96500 \times 0.233$

$\Delta G^\circ = 44969J = 44.9kJ = 45kJ$

Question89

Potassium chlorate is prepared by electrolysis of KCl in basic solution as shown by following equation.

$6OH^- + Cl^- \rightarrow ClO_3^- + 3H_2O + 6e^-$

A current of x A has to be passed for 10h to produce 10.0g of potassium chlorate. the value of x is _____. (Nearest integer)

(Molar mass of $KClO_3$ = 122.6g mol^{-1}, F = 96500C)

[20 Jul 2021 Shift 2]

Answer: 1

Solution:

Solution:

$W = \frac{E}{F} \times I \times t$

$10 = \frac{122.6}{96500 \times 6} \times x \times 10 \times 3600$

$X = 1.311$

Question90

For the cell

$Cu(s) \mid Cu^{2+}(aq)(0.1M) \parallel Ag^+(aq)(0.01M) \mid Ag(s)$

the cell potential E_1 = 0.3095V For the cell

$Cu(s) \mid Cu^{2+}(aq)(0.01M) \parallel Ag^{+}(aq)(0.001M) \mid Ag(s)$
the cell potential $= \text{——} 10^{-2}V$. (Round off the Nearest Integer).

$\left[\text{Use : } \frac{2.303RT}{F} = 0.059\right]$

[27 Jul 2021 Shift 2]

Answer: 28

Solution:

Solution:
Cell reaction is :

$$Cu(s) + 2Ag^{+}(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$$

Now, $E_{cell} = E^{\circ}_{Cell} - \frac{0.059}{2} \log \frac{[Cu^{2+}]}{[Ag^{+}]^2}$ .....(1)

$\therefore E_1 = 0.3095 = E^{\circ}_{Cell} - \frac{0.059}{2} .\log \frac{0.01}{(0.001)^2}$

From (1) and (2), $E_2 = 0.28V = 28 \times 10^{-2}V$

Question91

Consider the cell at 25°C
$Zn \mid Zn^{2+}(aq), (1M) \parallel Fe^{3+}(aq), Fe^{2+}(aq)) Pt(s)$
The fraction of total iron present as Fe^{3+} ion at the cell potential of 1.500V is $x \times 10^{-2}$. The value of x is _____ . (Nearest integer)
(Given : $E^{\circ}_{Fe^{3+}/Fe^{2+}} = 0.77V$, $E^{\circ}_{Zn^{2+}/Zn} = -0.76V$)

[25 Jul 2021 Shift 1]

Answer: 24

Solution:

Solution:

$$Zn \rightarrow Zn^{2+} + 2e^{-}$$
$$2Fe^{3+} \rightarrow 2e^{-} + 2e^{2+}$$

$$Zn + 2Fe^{3+} \rightarrow Zn^{2+} + 2Fe^{2+}$$

$$E^\circ_{cell} = 0.77 - (0.76)$$
$$= 1.53V$$
$$1.50 = 1.53 - \frac{0.06}{2}\log\left(\frac{Fe^{2+}}{Fe^{3+}}\right)^2$$
$$\log\left(\frac{Fe^{2+}}{Fe^{3+}}\right) = \frac{0.03}{0.06} = \frac{1}{2}$$
$$\frac{[Fe^{2+}]}{[Fe^{3+}]} = 10^{1/2} = \sqrt{10}$$
$$\frac{[Fe^{3+}]}{[Fe^{2+}]} = \frac{1}{\sqrt{10}}$$
$$\frac{[Fe^{3+}]}{[Fe^{2+}] + [Fe^{3+}]} = \frac{1}{1+\sqrt{10}} = \frac{1}{4.16}$$
$$= 0.2402$$
$$= 24 \times 10^{-2}$$

Question92

Assume a cell with the following reaction

$Cu_{(s)} + 2Ag^{+}(1 \times 10^{-3}M) \rightarrow Cu^{2+}(0.250M) + 2Ag_{(s)}$

$E_{cell}^{\ominus} = 2.97V$

E_{cell} for the above reaction is _____ V.

(Nearest integer)

[Given : log 2.5 = 0.3979, T = 298K]

[22 Jul 2021 Shift 2]

Answer: 3

Solution:

Solution:

$$E = E^\circ - \frac{0.059}{2}\log\frac{[Cu^{+2}]}{[Ag^{+}]^2}$$
$$= 2.97 - \frac{0.059}{2}\log\frac{0.25}{(10^{-3})^2} = 2.81V$$

Question93

The conductivity of a weak acid HA of concentration 0.001mol L^{-1} is $2.0 \times 10^{-5} Scm^{-1}$. If Λ_m^o(H A) = $190 Scm^2 mol^{-1}$, the ionization constant (K_a) of H A is equal to _____$\times 10^{-6}$.
(Round off to the Nearest Integer)
[27 Jul 2021 Shift 1]

Answer: 12

Solution:

Solution:

$$\Lambda_m = 1000 \times \frac{K}{M}$$

$$= 1000 \times \frac{2 \times 10^{-5}}{0.001} = 20 Scm^2 mol^{-1}$$

$$\Rightarrow \alpha = \frac{\Lambda_m}{\Lambda_m^{\infty}} = \frac{20}{190} = \left(\frac{2}{19}\right)$$

$$HA \rightleftharpoons H^+ + A^-$$

$$0.001(1-\alpha) \quad 0.001\alpha \quad 0.001\alpha$$

$$\Rightarrow k_a = 0.001\left(\frac{\alpha^2}{1-\alpha}\right) = \frac{0.001 \times \left(\frac{2}{19}\right)^2}{1-\left(\frac{2}{19}\right)}$$

$$= 12.3 \times 10^{-6}$$

Question94

Given below are two statements.
Statement I The limiting molar conductivity of KCl (strong electrolyte) is higher compared to that of CH_3 COOH (weak electrolyte).

Statement II Molar conductivity decreases with decrease in concentration of electrolyte. In the light of the above statements, choose the most appropriate answer from the options given below
[26 Aug 2021 Shift 1]

Options:

A. Statement I is true but statement II is false.

B. Statement I is false but statement II is true.

C. Both statement I and statement II are true.

D. Both statement I and statement II are false.

Answer: D

Solution:

Solution:
Limiting molar conductivity of KCl

$\Lambda_m^\circ KCl = \Lambda_m^\infty(K^+) + \Lambda_m^\infty(Cl^-)$

$= 73.55 S\ cm^2/mol + 76.3 S\ cm^2/mol$

$= 149.3 S\ cm^2/mol$

Limiting molar conductivity of CH_3COOH

$\Lambda_m^\circ(CH_3COOH) = \Lambda_m^\infty(CH_3COO^-) + \Lambda_m^\infty(H^+)$

$= 349.8 S\ cm^2/mol + 40.9 S\ cm^2/mol = 390.7 S\ cm^2/mol$

$\therefore$ Limiting molar conductivity of CH_3COOH is more than KCl.

Hence, statement I is false.
Also, as the concentration decreases, dilution of the electrolyte increases which will increase the degree of dissociation of weak electrolyte. Thus, the number of ions in the solution increases and hence, the molar conductance of electrolyte increases. Therefore, we can see that with decrease in concentration of electrolyte, molar conductance increases.
Hence, statement II is also false.

Question95

These are physical properties of an element.
A. Sublimation enthalpy
B. Ionisation enthalpy
C. Hydration enthalpy
D. Electron gain enthalpy
The total number of above properties that affect the reduction potential is..... (Integer answer)
[26 Aug 2021 Shift 1]

Answer: 3

Solution:

Solution:
The physical properties of an element is affected by sublimation enthalpy, ionisation enthalpy and hydration enthalpy as these 3 enthalpies will affect the reduction potential.

Question96

For the galvanic cell,
$Zn(s) + Cu^{2+}(0.02M) \rightarrow Zn^{2+}(0.04M) + Cu(s)$,
$E_{cell} = \ldots\ldots\ldots \times 10^{-2}V$.(Nearest integer)
[Use $E°_{Cu/cu^{2+}} = -0.34V$, $E°_{Zn/Zn^{2+}} = +0.76V$,
$\frac{2.303RT}{F} = 0.059V$]
[26 Aug 2021 Shift 2]

Answer: 109

Solution:

Solution:
Cell reaction,

$$\underset{0.02M}{Cu^{2+}(aq)} + Zn(s) \rightleftharpoons Cu(s) + \underset{0.04M}{Zn^{2+}(aq)}$$

$$E°_{cell} = E°_{Cu^{2+}/Cu} - E°_{Zn^{2+}/Zn}$$
$$= 0.34 - (-0.75) = 1.1V$$
$$E_{cell} = E°_{cell} - \frac{0.059}{2}\log\frac{[Zn^{2+}]}{[Cu^{2+}]}$$
$$= 1.1 - \frac{0.059}{2}\log\frac{[0.04]}{[0.02]}$$
$$= 1.1 - 0.03\log 2 = 1.1 - 0.03 \times 0.30$$
$$= 1.09V = 109 \times 10^{-2}V$$

∴ Answer is 109.

Question97

The resistance of a conductivity cell with cell constant $1.14cm^{-1}$, containing 0.001 M KCl at 298K is 1500Ω. The molar conductivity of 0.001 M KCl solution at 298K in Scm^2mol^{-1} is ..... .(Integer answer)
[27 Aug 2021 Shift 2]

Answer: 760

Solution:

Solution:
Given, $C^* = 1.14 cm^{-1} = 1.14 \times 10^3 m^{-1}$
R = 1500 Ω
C = 0.001 M
$G^* = R \times \kappa$
[where, R = resistance
G^* = cell constant
κ = conductivity
C = concentration]

$$\therefore \kappa = \frac{G^*}{R} = \frac{1.14 \times 10^3 m^{-1}}{1500}$$

$$\Lambda m = \frac{\kappa}{C}$$

[where , Λ_m = molar conductivity]

$$= \frac{1.14 \times 10^3}{1500 \times 0.001} = 760 Scm^2 mol^{-1}.$$

Question98

Section B : Numerical Type Questions

Consider the following cell reaction,

$Cd(s) + Hg_2SO_4(s) + \frac{9}{5}H_2O(l) \rightleftharpoons CdSO_4 . \frac{9}{5}H_2O(s) + 2\,Hg(l)$

The value of E_{cell}° is 4.315V at 25°C. If $\Delta H^\circ = -825.2\ kJ\ mol^{-1}$, the standard entropy change ΔS° in JK^{-1} is…. (Nearest integer)

[Given, Faraday constant $= 96487 Cmol^{-1}$]

[31 Aug 2021 Shift 1]

Answer: 25

Solution:

Solution:
$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ ...(i)
where, ΔG° = standard Gibb's free energy change
ΔH° = standard change in enthalpy
ΔS° = standard entropy change
Also, $\Delta G^\circ = -nFE^\circ$
where, n = number of electrons
F = Faraday constant
E° = standard cell potential
∴ Equation (i) is,
$-nFE^\circ = \Delta H^\circ - T\Delta S^\circ$
$-nFE^\circ - \Delta H^\circ = T\Delta S^\circ$

$$\Rightarrow \Delta S^\circ = \frac{nFE^\circ + \Delta H^\circ}{T} \quad (2 \times 96487 Cmol^{-1} \times 4.315V)$$

$$= \frac{+(-825.2\,\text{kJ/mol})}{298\text{K}}$$
$$= \frac{832.682 \times 10^3\text{J mol} - 825.2 \times 10^3\text{J/mol}}{298\text{K}}$$
$$= 25.11\text{J/Kmol}$$

Question99

Match List-I with List-II.

List-I (Parameter)	**List-II** (Unit)
A. Cell constant	1. $\text{Scm}^2\text{mol}^{-1}$
B. Molar conductivity	2. Dimensionless
C. Conductivity	3. m^{-1}
D. Degree of dissociation of electrolyte	4. $\Omega^{-1}\text{m}^{-1}$

Choose the most appropriate answer from the options given below [31 Aug 2021 Shift 2]

Options:

A. A-3 B-1 C-4 D-2

B. A-3 B-1 C-2 D-4

C. A-1 B-4 C-3 D-2

D. A-2 B-1 C-3 D-4

Answer: A

Solution:

Solution:
(A) Cell constant $= \frac{\text{I}}{\text{A}}\text{m}^{-1}$
(B) Molar conductivity $(\lambda_m) = \frac{\kappa \times 1000}{\text{molarity}}\text{Scm}^2\text{mol}^{-1}$
(C) Conductivity $(\kappa) = \frac{1}{\rho} = \frac{1}{\text{RA}}\Omega^{-1}\text{m}^{-1}$
(D) Degree of dissociation of electrolyte = Number of moles dissociated out of one mole.
It is a ratio. Hence, it is dimensionless.
Thus, the correct match is
A → 3, B → 1, C → 4, D → 2.

Question100

If the conductivity of mercury at 0°C is $1.07 \times 10^6 Sm^{-1}$ and the resistance of a cell containing mercury is 0.243Ω, then the cell constant of the cell is $x \times 10^4 m^{-1}$. The value of x is..... .(Nearest integer)
[1 Sep 2021 Shift 2]

Answer: 26

Solution:

Solution:
Conductance (G) is reciprocal of resistance (R).

$R = \frac{1}{G}$ or $G = \frac{1}{R} = \frac{1}{0.243\Omega} = 4.115\Omega^{-1}$

Relation between conductance (G),

conductivity (kappa) and cell constant $\left(\frac{I}{A}\right)$ is given as

$k = \frac{Gl}{A}$

$\Rightarrow \frac{l}{A} = \frac{k}{G} = \frac{1.07 \times 10^6 Sm^{-1}}{4.115\Omega^{-1}} = 26 \times 10^4 m^{-1}$

$\Rightarrow x = 26$

$\therefore$ Answer is 26.

Question101

The equation that is incorrect is:
[Jan. 07, 2020 (II)]

Options:

A. $(\Lambda_m^0)_{NaBr} - (\Lambda_m^0)_{NaCl} = (\Lambda_m^0)_{KBr} - (\Lambda_m^0)_{KCl}$

B. $(\Lambda_m^0)_{KCl} - (\Lambda_m^0)_{NaCl} = (\Lambda_m^0)_{KBr} - (\Lambda_m^0)_{NaBr}$

C. $(\Lambda_m^0)_{H_2O} = (\Lambda_m^0)_{HCl} + (\Lambda_m^0)_{NaOH} - (\Lambda_m^0)_{NaCl}$

D. $(\Lambda_m^0)_{NaBr} - (\Lambda_m^0)_{Nal} = (\Lambda_m^0)_{KBr} - (\Lambda_m^0)_{NaBr}$

Answer: D

Solution:

Solution:

$(\Lambda_m^0)_{NaBr} - (\Lambda_{mNal}^0)_{Nal}$

$= \Lambda_m^0 Na^+ + \Lambda_m^0 Br^- - (\Lambda_m^0 N^+ + \Lambda_m^0 I^-)$

$= \Lambda_m^0 N_a^+ + \Lambda_m^0 Br^- - \Lambda_m^0 N^+ - \Lambda_m^0 I$

$= \Lambda_m^0 Br^- + \Lambda_m^0 I^-$

$(\Lambda_m^0)_{KBr} - (\Lambda_m^0)_{NaBr}$

$= \Lambda_m^0 K^+ + \Lambda_m^0 Br^- - (\Lambda_m^0 Na^+ + \Lambda_m^0 Br^-)$

$= \Lambda_m^0 K^+ + \Lambda_m^0 Br^- - \Lambda_m^0 N^+ - \Lambda_m^0 Br^-$

$= \Lambda_m^0 K^+ + \Lambda_m^0 Na^+$

$\therefore (\Lambda_m^0)_{NaBr} - (\Lambda_m^0)_{Nal} \neq (\Lambda_m^0)_{KBr} - (\Lambda_m^0)_{NaBr}$

Question102

108g of silver (molar mass $108 g mol^{-1}$) is deposited at cathode from $AgNO_3(aq)$ solution by a certain quantity of electricity. The volume (in L) of oxygen gas produced at 273K and 1 bar pressure from water by the same quantity of electricity is
[NV, Jan. 09, 2020(I)]

Answer: 5.68

Solution:

Solution:

No. of moles of silver deposited.

$= \frac{108}{108} = 1 mol$

$Ag^+ + e^- \rightarrow Ag$

$1F$ charge is required to deposit $1 mol\ e$ of Ag

$H_2O \rightarrow \frac{1}{2}O_2 + 2H^+ + 2e^-$

$2F$ charge deposit $\rightarrow \frac{1}{2}$ moles of oxygen

$1F$ charge will deposit $\rightarrow \frac{1}{4}$ moles of oxygen

$V_{O_2} = \frac{nRT}{P}$

$= \frac{1}{4} \times \frac{0.08314 L bar K^{-1} mol^{-1} \times 273K}{1 bar}$

$= \frac{1}{4} \times 22.7$

$V_{O_2} = 5.675L$

Question103

What would be the electrode potential for the given half cell reaction at pH = 5 ?

_____ .

$2H_2O \rightarrow O_2 + 4H^{\oplus} + 4c^{-}$; $E_{red}^{0} = 1.23V$

(R = 8.314J mol $^{-1}$K $^{-1}$; T emp = 298K ; oxygen under std. atm. pressureof 1 bar)
[NV, Jan. 08, 2020 (I)]

Answer: 1.52

Solution:

Solution:

$E = 1.23 - \frac{0.0591}{4} \log[H^+]^4$

$= 1.23 + 0.0591 \times pH$

$= 1.23 + 0.0591 \times 5$

$= 1.23 + 0.2955$

$= 1.52V$

Question104

For an electrochemical cell
Sn(s) | Sn^{2+}(aq, 1M) | Pb^{2+}(aq, 1M) | Pb(s) the ratio $\frac{[Sn^{2+}]}{[Pb^{2+}]}$
when this cell attains equilibrium is
[NV, Jan. 08, 2020 (II)]

Answer: 2.15

Solution:

Solution:
At equilibrium state $E_{cell} = 0$; $E_{cell}^{0} = 0.01V$

$Sn + Pb^{2+} \longrightarrow Sn^{2+} + Pb$

$E = E_{cell}^{0} - \frac{0.06}{n} \log \frac{[P]}{[R]}$

$0 = 0.01 - \frac{0.06}{2} \log \frac{[Sn^{2+}]}{[Pb^{2+}]}$

$-0.01 = -\frac{0.06}{2} \log \frac{[Sn^{2+}]}{[Pb^{2+}]}$

$\frac{1}{3} = \log \frac{[Sn^{2+}]}{[Pb^{2+}]}$

$\frac{[Sb^{2+}]}{[Pb^{2+}]} = 10^{1/3} = 2.15$

Question105

Given that the standard potentials (E) of Cu^{2+}/Cu and Cu^{+}/Cu are 0.34V and 0.522V respectively, the E^{0} of Cu^{2+}/Cu is: [Jan. 07,2020(I)]

Options:

A. +0.182V

B. +0.158V

C. −0.182V

D. −0.158V

Answer: B

Solution:

Solution:
$Cu^{2+} + 2e^{-} \longrightarrow Cu, \Delta G_1^{\circ} = -2F(0.34)$ . . . (i)

$Cu^{+} + e^{-} \longrightarrow Cu, \Delta G_2^{\circ} = -F(0.522)$ . . . (ii)

Subtract (ii) from (i)

$Cu^{2+} + e^{-} \longrightarrow Cu^{+}, \Delta G_3^{\circ} = -F(E^{0})$

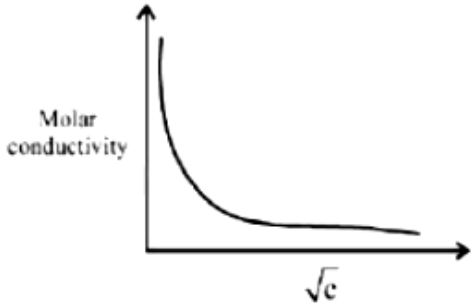
$\therefore \Delta G_1^{\circ} - \Delta G_2^{\circ} = G_3^{\circ}$

$\Rightarrow -F E^{\circ} = -2F(0.34) + F(0.522)$

$\Rightarrow E^{\circ} = 0.68 - 0.522 = 0.158V$

Question106

The variation of molar conductivity with concentration of an electrolyte (X) in aqueous solution is shown in the given figure.



The electrolyte X is :
[Sep. 05, 2020 (II)]

Options:

A. HCl

B. $NaCl$

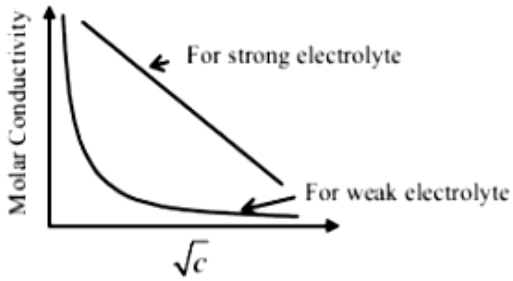
C. KNO_3

D. CH_3COOH

Answer: D

Solution:

Solution:



Among given electrolytes, CH_3COOH is weak electrolyte.

Question107

Let C_{NaCl} and C_{BaSO_4} be the conductances (in S) measured for saturated aqueous solutions of $NaCl$ and $BaSO_4$, respectively, at a temperature T .

Which of the following is false?
[Sep. 03, 2020 (I)]

Options:

A. Ionic mobilities of ions from both salts increase with T

B. $C_{BaSO_4}(T_2) > C_{BaSO_4}(T_1)$ for $T_2 > T_1$

C. $C_{NaCl}(T_2) > C_{NaCl}(T_1)$ for $T_2 > T_1$

D. $C_{NaCl} \gg C_{BaSO_4}$ at a given T

E. None of above

Answer: E

Solution:

Solution:
(N) Ionic mobility increases with increase in temperature, which increases the conductance of the solution, while conductance of $NaCl$ solution is independant of temperature above $400^{\circ}C$. Because the temperature is not given, so none of the option is correct.

Question108

Potassium chlorate is prepared by the electrolysis of KCl in basic solution $6OH^- + Cl^- \rightarrow ClO_3^- + 3H_2O + 6e^-$ If only 60% of the current is utilized in the reaction, the time (rounded to the nearest hour) required to produce 10 g of $KClO_3$ using a current of 2A is _____ .

(Given : $F = 96,500 C mol^{-1}$; molar mass of $KClO_3 = 122 g\ mol^{-1}$)

[NV, Sep. 06, 2020 (I)]

Answer: 11

Solution:

Solution:
$6OH^- + Cl^- \rightarrow ClO_3^- + 3H_2O + 6e^-$
For synthesis of 1 mole of ClO_3^-, $6F$ of charge is required.

$\because$ Current efficiency $= 60\%$

$\therefore$ To synthesis 1 mole of ClO_3^-, $10F$ of charge is required.

Tosynthesis $\frac{10}{122}$ moles of $KClO_3$, charge $= \frac{10 \times 10}{122}F$ $Q = I \cdot t$

$t = \frac{100 \times 96500}{122 \times 2} = 39549.18s$

$= \frac{79098.365}{3600s} = 10.99h$

$\therefore t = 11h.$

Question109

250mL of a waste solution obtained from the workshop of a goldsmith contains 0.1M $AgNO_3$ and 0.1M $AuCl$. The solution was electrolyzed at 2V by passing a current of 1 A for 15 minutes. The metal\/metals electrodeposited will be:
[Sep. 04, 2020(II)]

$$\left(E_{Ag^+/Ag}^0 = 0.80V, E_{Au^+/Au}^0 = 1.69V\right)$$

Options:

A. only gold

B. silver and gold in proportion to their atomic weights

C. only silver

D. silver and gold in equal mass proportion

Answer: A

Solution:

Solution:

Millimoles of $Au^+ = 0.1 \times 250 = 25$

Mole of $Au^+ = \frac{25}{1000} = \frac{1}{40} = 0.025$

Similarly, moles of $Ag^+ = 0.025$

Charge passed $= I \times t = 1 \times 15 \times 60 = 900C$

Moles of e- passed $= \frac{900}{96500} = 0.0093mol.$

Species withhigher value of SRP will get deposited first at cathode.

Question110

An acidic solution of dichromate is electrolyzed for 8 minutes using 2 A current. As per the following equation

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$$

The amount of Cr^{3+} obtained was 0.104g. The efficiency of the process (in %) is (Take : F = 96000C, At. mass of chromium = 52) _____.
[NV, Sep. 03,2020 (II)

Answer: 60

Solution:

Solution:

Charge $(Q) = I\,t = 2 \times 8 \times 60 = 960C$

$$\Rightarrow \frac{960}{96000} = 0.01F$$

$$Cr_2O_7^{2-} + \underset{0.01F}{14H^+} + 6e^- \longrightarrow \underset{\frac{1}{3} \times 0.01 \text{ mole}}{2Cr^{3+}} + 7H_2O$$

Theoritical mass of $Cr^{3+} = \frac{1}{3} \times \frac{960}{96000} \times 52 = 0.173g$

So, efficiency $= \frac{W_{actual}}{W_{Theoritical}} \times 100 = \frac{0.104}{0.173} \times 100 = 60\%$

Question111

For the given cell;

$Cu(s) \mid Cu^{2+}(C_1M) \mid Cu^{2+}(C_2M) \mid Cu(s)$

change in Gibbs energy (ΔG) is negative, if:
[Sep. 06, 2020 (II)]

Options:

A. $C_1 = C_2$

B. $C_2 = \frac{C_1}{\sqrt{2}}$

C. $C_1 = 2C_2$

D. $C_2 = \sqrt{2}C_1$

Answer: D

Solution:

Solution:
For the concentration cell, $E_{cell}^{0} = 0$

Anode : $Cu(s) \longrightarrow Cu^{2+}(aq)_A$

$$\text{Cathode} \quad : Cu^{2+}(aq)_e \longrightarrow Cu(s)$$

$$\text{Overall} \quad \underset{(C_2M)}{Cu^{2+}(aq)_C} \longrightarrow \underset{(C_1M)}{Cu^{2+}(aq)_A}$$

As $\Delta G = -nF\,E$

If $\Delta G = -ve$, then E_{cell} is $+v_e$.

$$E_{cell} = E_{cell}^{0} - \frac{RT}{2F} \ln \frac{C_1}{C_2}$$

$$E_{cell} = 0 - \frac{RT}{2F} \ln \frac{C_1}{C_2}$$

$$E_{cell} = \frac{RT}{2F} \ln \frac{C_2}{C_1}$$

So, $C_2 > C_1$.

Thus, $C_2 = \sqrt{2}C_1$ relation is correct.

Question112

An oxidation-reduction reaction in which 3 electrons are transferred has a ΔG^0 of 17.37kJ mol^{-1} at 25°C. The value of E_{cell}^{0}(inV) is _____ $\times 10^{-2}$

(1F = 96, 500Cmol^{-1})
[NV, Sep. 05, 2020(I)]

Answer: -6

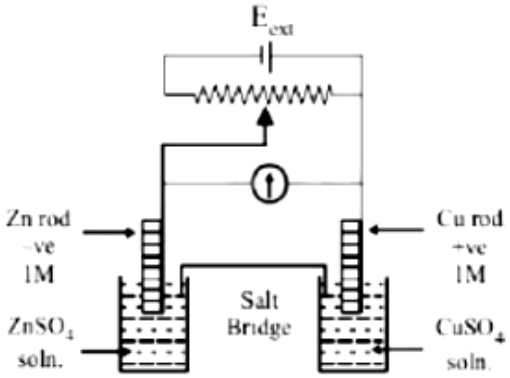
Solution:

Solution:
$\Delta G^0 = -nF\,E_{cell}^{0}$

$17.37 \times 10^3 = -3 \times 96500 \times E_{cell}^{o}$

$E_{cell}^{o} = -0.06V \sim eq - 6.0 \times 10^{-2}V$

Question113



$E_{Cu^{2+}|Cu}^{o} = +0.34V$; $E_{Zn^{2+}|Zn}^{o} = -0.76V$

Identify the incorrect statement from the options below for the above cell : [Sep. 04, 2020(I)]

Options:

A. If $E_{ext} > 1.1V$, e^- flows from Zn to Zn

B. If $E_{cxt} > 1.1V$, Zn dissolves at Zn electrode and Cu deposits at Cu electrode

C. If $E_{ext} < 1.1V$, Zn dissolves at anode and Cu deposits at cathode

D. If $E_{ext} = 1.1V$, no flow of e^- or current occurs

Answer: B

Solution:

Solution:

$E_{cell}^{\circ} = E_{Cu^{2+}|Cu}^{o} - E_{Zn^{2+}|Zn}^{\circ} = 1.1V$

So, if $E_{ext.} = 1.1V$ no electron will flow

At, $E_{ext.} > 1.1V$ cell act as electrolytic cell and electron will flowfrom Cu to Zn.

At, $E_{ext.} < 1.1V$ cell act as electrochemical cells so Zn dissolve and Cu deposit.

Question114

The photoelectric current from Na (work function, $w_0 = 2.3eV$) is stopped by the output voltage of the cell
Pt(s) | H_2(g, 1bar) | HCl (aq., pH = 1) | AgCl (s) | Ag(s) The pH of aq. HCl required to stop the photoelectric current from K ($w_0 = 2.25cV$), all other conditions remaining the same, is _____ $\times 10^{-2}$ (to the nearest integer).

Given,

$2.303\ \frac{RT}{F} = 0.06V$; $E^{0}_{AgCl\ |\ Ag\ |\ C^{-}} = 0.22V$

[NV, Sep. 03 , 2020(I)]

Answer: 142

Solution:

Solution:
Sodium metal :

$E = E_0 + (KE)_{max}$; $E_{cell}^{0} = 0.22V$

Cell reaction

Cathode: $AgCl(s) + e^{-} \longrightarrow Ag(s) + Cl^{-}(aq)$

Anode : $\frac{1}{2}H_2(g) \longrightarrow H^{+}(aq) + e^{-}$

Overall : $AgCl(s) + \frac{1}{2}H_2(g) \longrightarrow Ag(s) + H^{+}(aq) + Cl^{-}(aq)$

$E_{cell} = E_{cell}^{0} - \frac{0.06}{1}\log[H^{+}][Cl^{-}]$

$E_{cell} = 0.22 - \frac{0.06}{1}\log[10^{-1}][10^{-1}]$

$= 0.22 + 0.12 = 0.34V$

$(KE)_{max} = E_{cell} = 0.34eV$

So, $E = 2.3 + 0.34 = 2.64eV$ = Energy of photon incident

For potassium metal :

$E = E_0 + (KE)_{max}$

$2.64 = 2.25 + (KE)_{max}$

$(KE)_{max} = 0.39 = E_{cell}$

Cell reaction

Cathode: $AgCl(s) + e^{-} \longrightarrow Ag(s) + Cl^{-}(aq)$

Anode : $\frac{1}{2}H_2(g) \longrightarrow H^{+}(aq) + e^{-}$

Overall : $AgCl(s) + \frac{1}{2}H_2(g) \longrightarrow Ag(s) + H^{+}(aq) + Cl^{-}(aq)$

$E_{cell} = E_{cell}^{0} - \frac{0.06}{1}\log[H^{+}][Cl^{-}]$

$0.39 = 0.22 - 0.06\log[H^{+}]^2$

$0.39 = 0.22 - 0.12\log[H^{+}]$

$0.17 = 0.12 \times pH$

$pH = 17/12 = 1.4166 \sim eq1.42 = 142 \times 10^{-2}$

Question115

The Gibbs energy change (in J) for the given reaction at $[Cu^{2+}] = [Sn^{2+}] = 1M$ and 298K is :

$Cu(s) + Sn^{2+}(\ aq.\ ) \longrightarrow Cu^{2+}(\ aq.\ ) + Sn(s)$

$\left(E^{0}_{Sn^{2+}\,|\,Sn} = -0.16V\,,\ E^{0}_{Cu^{2+}\,|\,Cu} = 0.34V\,,\ \text{Take}\ F = 96500 cmol^{-1}\right)$

[NV, Sep. 02, 2020(I)]

Answer: 96500

Solution:

Solution:

Sol. $E_{cell}^{0} = E_{Sn^{2+}/Sn}^{0} - E_{Cu^{2+}/Cu}^{0}$

$= -0.16 - 0.34$

$= -0.50V$

$\Delta G^{0} = -nF\,E_{cell}^{0}$

$= -2 \times 96500 \times (-0.5)$

$= 96500J$

$= 96.5K\,J = 96500J$

Question116

For the disproportionation reaction $2Cu^{+}(aq) \rightleftharpoons Cu(s) + Cu^{2+}(aq)$ at 298K, ln K

(where K is the equilibrium constant) is $\times 10^{-1}$.

Given

[NV, Sep. 02,2020(II)]

$\left(E_{Cu^{2+}/Cu^{+}}^{0} = 0.16V\ ;\ E_{Cu^{+}/Cu}^{0} = 0.52V\ ;\ \frac{RT}{F} = 0.025\right)$

Answer: 144

Solution:

Solution:

$2Cu^{+}(aq) \rightleftharpoons Cu(s) + Cu^{2+}(aq)$

$E_{cell}^{0} = E_{Cu^{+}/Cu}^{o} - E_{Cu^{2+}/Cu^{+}}^{o} = 0.52 - 0.16 = 0.36V$

$E_{cell}^{0} = \frac{RT}{nF} \ln K_{eq}$

$\Rightarrow 0.36 = \frac{0.025}{1} \ln K$

$\Rightarrow \ln K = 14.4 = 144 \times 10^{-1}$

Question117

Λ_m° for NaCl, HCl and NaA are 126.4, 425.9 and 100.5Scm^2mol^{-1}, respectively. If the conductivity of 0.001 M HA is $5 \times 10^{-5} Scm^{-1}$, degree of dissociation of HA is : [Jan. 12, 2019 (II)]

Options:

A. 0.50

B. 0.25

C. 0.125

D. 0.75

Answer: C

Solution:

Solution:

Given: $\Lambda_m^{\circ}(NaCl) = 126.4 Scm^2mol^{-1}$, $\Lambda_m^{\circ}(HCl) = 425.9 Scm^2mol^{-1}$ and $\Lambda_m^{\circ}(NaA) = 100.5 Scm^2mol^{-1}$

$\Lambda_m^{\circ}(HA) = \lambda_{H}^{\circ} + \lambda_{A^-}^{\circ}$

$= \lambda_{H^+}^{\circ} + \lambda_{Cl}^{\circ} + \lambda_{A^+}^{\circ} + \lambda_{Na^+}^{\circ} - \lambda_{Cl}^{\circ} - \lambda_{Na^+}^{\circ}$

$= \Lambda_m^{\circ}(HCl) + \Lambda_m^{\circ}(NaA) - \Lambda_m^{\circ}(NaCl)$

$\Lambda_m^{\circ}(HA) = 429.5 - 126.4 + 100.5 = 400 Scm^2mol^{-1}$

$\kappa(HA) = 5 \times 10^{-5} Scm^{-1}$

$(HA) = \frac{\kappa(HA) \times 1000}{\text{Molality of HA}} = \frac{5 \times 10^{-5} \times 1000}{0.001} = 50 Scm^2mol^{-1}$

$\alpha = \frac{\Lambda_m(HA)}{\Lambda_m^{\circ}(HA)} = \frac{50}{400} = 0.125$

Question118

The anodic half-cell of lead-acid battery is recharged using electricity of 0.05 Faraday. The amount of $PbSO_4$ electrolyzed in g during the process is :

(Molar mass of $PbSO_4 = 303 g mol^{-1}$)

[Jan. 9,2019(I)]

Options:

A. 22.8

B. 15.2

C. 7.6

D. 11.4

Answer: A

Solution:

Solution:
According to the Faraday's law of electrolysis, nF of current is required for the deposition of $1 mol$
According to the reaction,
$Ni(NO_3)_2 \longrightarrow Ni^{2+} + 2NO_3^-$
$2F$ of current deposits $= 1 mol$
$\therefore 0.1F$ of currentdeposits $= \frac{0.1}{2} = 0.05 mol$

Question119

The standard electrode potential E^0 and its temperature coefficient $\left(\frac{dE^0}{dT}\right)$ for a cell are $2V$ and $-5 \times 10^{-4} V K^{-1}$ at $300K$ respectively. The cell reaction is:

$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu$

The standard reaction enthalpy ($\Delta_r H^0$) at $300K$ in $kJ\ mol^{-1}$ is,

[Use $R = 8 J K^{-1} mol^{-1}$ and $F = 96,000 C mol^{-1}$]

[Jan. 12, 2019 (I)]

Options:

A. −412.8

B. −384.0

C. 192.0

D. 206.4

Answer: A

Solution:

Solution:

$\Delta G^\circ = \Delta_r H^\circ - T\ \Delta S^\circ$

$\Delta_r H^\circ = \Delta G^\circ + T\ \Delta S^\circ$

$\Delta_r H^\circ = -nF\ E^\circ + T\ nF\ \frac{dE}{dT}$

$\Delta_r H^\circ = -nF\ E^\circ + nF\ T\ \frac{dE}{dT}$

Cell reaction: $Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$

$\Delta_r H^\circ = -nF\left(E^\circ - \frac{T\,dE^\circ}{dT}\right)$

$\Delta_r H^\circ = -2 \times 96000(2 - 300 \times -5 \times 10^{-4})$

$\Delta_r H^\circ = -2 \times 96000(2 + 300 \times 5 \times 10^{-4})$

$= -2 \times 96000(2 + 0.15)$

$= -412.8 \times 10^3 J/mol\ ;\ = -412.8 kJ/mol$

Question120

For the cell $Zn(s) \mid Zn^{2+}(aq) \mid M^{x+}(aq) \mid M(s)$, different half cells and their standard electrode potentials are given below:

$M^{x+}(aq)/$	$Au^{3+}(aq)$	$Ag^{+}(aq)/$	$Fe^{3+}(aq)$	$Fe^{2+}(aq)$
$M(s)$	$Au(s)$	$Ag(s)$	$Fe^{2+}(aq)$	$Fe(s)$
$E^\circ M^{x+/M}/(V)$	1.40	0.80	0.77	0.44

If $E^{\circ}_{Zn^{2+}/Zn}{}^{2} = -0.76V$, which cathode will give a maximum value of E°_{ecll} per electron transferred?

[Jan. 11, 2019 (I)]

Options:

A. Ag^{+}/Ag

B. Fe^{3+}/Fe^{2+}

C. Au^{3+}/Au

D. Fe^{2+}/Fe

Answer: A

Solution:

Solution:

$Zn + 2Ag^{+} \longrightarrow 2Ag + Zn^{2+}(n = 2e^{-})$

$E_{cell}^{\circ} = (E_{R.P}^{\circ})_{cathode} - (E_{R.P}^{\circ})_{anode}$

$= 0.80 - (-0.76) = 1.56V$ for 2e

$\therefore E_{cell}^{\circ}$ for $1e^{-} = \frac{1.56}{2} = 0.78V$

Question121

Given the equilibrium constant: K_C of the reaction: $Cu(s) + 2Ag^{+}(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$ is 10×10^{15} calculate the E_{cell}^{0} of this reaction at 298K

$\left[2.303 \frac{RT}{F} \text{at } 298K = 0.059V\right]$

[Jan. 11, 2019 (II)]

Options:

A. 0.04736mV

B. 0.4736mV

C. 0.4736V

D. 0.04736V

Answer: C

Solution:

Solution:

$E_{cell}^{0} = \frac{2.303RT}{nF} \log K_C$ or $E_{cell}^{0} = \frac{0.059V}{n} \log K_C$

$= \frac{0.059V}{2} \log 10^{16} = 0.4736V$

Question122

In the cell
$Pt(s) \mid H_2(g, 1\ bar) / HCl(aq) \mid AgCl(s) / Ag(s) \mid Pt(s)$, the cell potential is 0.92V when a 10^{-6} molal HCl solution is used. The standard electrode potential of $(AgCl / Ag, Cl^-)$ electrode is:

{ Given : $\frac{2.303RT}{F} = 0.06V$ at 298K }
[Jan. 10, 2019 (II)]

Options:

A. 0.94V

B. 0.76V

C. 0.40V

D. 0.20V

Answer: D

Solution:

Solution:
Given that:
$Pt(s) \mid H_2(g, 1bar) / HCl(aq) \mid AgCl(s) / Ag(s) \mid Pt(s)$
$E_{cell} = 0.92V$
Now, $E_{cell} = E^\circ_{H_2(g)/H^+(aq)} + E^\circ_{AgCl(s)Ag(s)Cl^-} - \frac{0.06}{n} \log Q$
Cell reaction: $\frac{1}{2}H_2(g) \longrightarrow H^+(aq) + e^-$
$AgCl(s) + e^- \longrightarrow Ag(s) + Cl^-(aq)$
Net cell reaction:
$\frac{1}{2}H_2(g) + AgCl(s) \longrightarrow Ag(s) + H^+ + Cl^-(aq)$
$\therefore Q = \frac{[H^+][Cl^-]}{(P_{H_2})^{1/2}}$
Here, 10^{-6} molal HCl solution is used
So $Q = \frac{10^{-6} \times 10^{-6}}{1} = 10^{-12}$
(assuming molality = molarity)
Now, $0.92 = E_{AgCl(s)Ag(s), Cl^-} - \frac{0.06}{1} \log 10^{-12}$
$E^\circ_{AgCl(s)Ag(s), Cl^-} = 0.92 + [0.06 \times (-12)]$
$= 0.92 - 0.72 = 0.20V$

Question123

If the standard electrode potential for a cell is 2V at 300K , the equilibrium constant (K) for the reaction
$Zn(s) + Cu^{2+}(aq) \rightleftharpoons Zn^{2+}(aq) + Cu(s)$
at 300K is approximately ($R = 8J\,K^{-1}mol^{-1}$, $F = 96000Cmol^{-1}$)
[Jan. 9, 2019 (II)]

Options:

A. e^{-80}

B. e^{-160}

C. e^{320}

D. c^{160}

Answer: D

Solution:

Solution:
We know that,

$$\ln K = \frac{nF E^{\circ}_{cell}}{RT}$$

$$\Delta G = -nF E^{\circ}_{cell}$$

$$\Delta G = -RT \ln k$$

Now,

After putting the given values, we get $\ln K = \frac{2 \times 96000 \times 2}{8 \times 300} = 160$

$\therefore\ K = e^{160}$

Question124

The decreasing order of electrical conductivity of the following aqueous solutions is:

0.1 M Formic acid (A),
0.1 MAcetic acid (B),
0.1 M Benzoic acid (C).
[April 12, 2019 (II)]

Options:

A. A > C > B

B. C > B > A

C. A > B > C

D. C > A > B

Answer: A

Solution:

Solution:
Order of acidic strength is $HCOOH > C_6H_5COOH > CH_3COOH$
More the acidic strength more will be the dissociation of acid into ions and more will be the conductivity. Thus, order of conductivity will be,
$\underset{(A)}{HCOOH} > \underset{(C)}{C_6H_5COOH} > \underset{(B)}{CH_3COOH}$

Question125

Consider the statements S1 and S2 : S1: Conductivity always increases with decrease in the concentration of electrolyte. S2: Molar conductivity always increases with decrease in the concentration of electrolyte. The correct option among the following is: [April 10,2019 (I)]

Options:

A. Both S1 and S2 are wrong

B. S1 is wrong and S2 is correct

C. Both S1 and S2 are correct

D. S1 is correct and S2 is wrong

Answer: B

Solution:

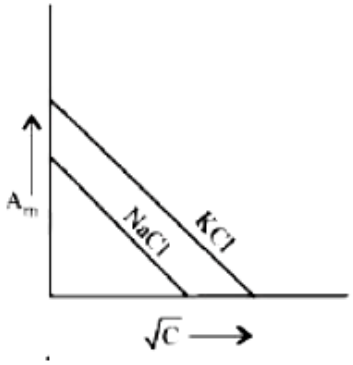
Solution:
Conductivity of an electrolyte is the conductance of $1cm^3$ of the given electrolyte. It increases with the increase in concentration of electrolyte due to increase in the number of ions per unit volume. Molar conductivity (λ_m) is theconductance of a solution containing 1 mole of the electrolyte. It increases with the decrease of concentration due to increase in the total volume having one mole of electrolyte. Thus, interionic attraction increses and degree of ionisation decreases.Therefore, (S_1) is wrong and (S_2) is correct.

Question126

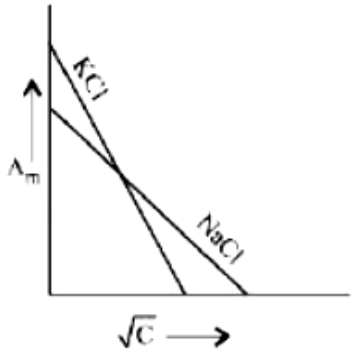
Which one of the following graphs between molar conductivity (A_m) versus $\sqrt{C}$ is correct?
[April 10, 2019 (II)]

Options:

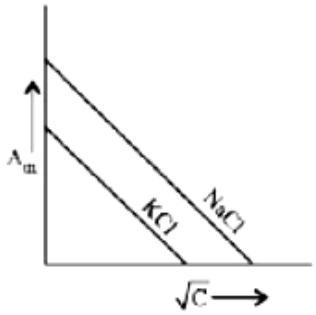
A.



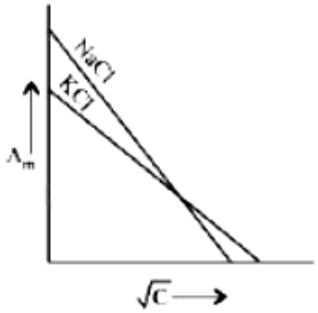
B.



C.



D.



Answer: A

Solution:

Solution:
Since, KCl is more conducting than $NaCl$, therefore, graph (a) is correct.

Question127

A solution of $Ni(NO_3)_2$ is electrolysed between platinum electrodes using 0.1 Faraday electricity. How many mole of Ni will be deposited at the cathode?
[April 9, 2019 (II)]

Options:

A. 0.05

B. 0.20

C. 0.15

D. 0.10

Answer: C

Solution:

Solution:
The reaction involved in the process is given below
$PbSO_4 \rightarrow Pb^{2+} + SO_4^{2-}$
$Pb^{2+} + H_2O \rightarrow PbO_2 + 4H^+ + 2e^-$
So, half cell reaction is
$PbSO_4 \rightarrow Pb^{4+} + 2e^-$
According tothe reaction:
$PbSO_4 \rightarrow Pb^{4+} + 2e^-$
We require $2F$ for the electrolysis of $1mol$ or $303g$ of $PbSO_4$

$\therefore$ Amount of $PbSO_4$ electrolysed by $0.05F$

$= \frac{303}{2} \times 0.05 = 7.575g \approx 7.6g$

Question128

The standard Gibbs energy for the given cell reaction in $kJ\ mol^{-1}$ at 298K is:

$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

$E^{\circ} = 2V$ at 298K

(Faraday's constant, $F = 96000Cmol^{-1}$)

[April 9, 2019(I)]

Options:

A. -384

B. 384

C. 192

D. -192

Answer: A

Solution:

Solution:

$\Delta G^{\circ} = -nF E_{cell}^{\circ}$

$= -2 \times (96000) \times 2V = -384000J/mol = -384kJ/mol$

Question129

Calculate the standard cell potential (in V) of the cell in which following reaction takes place:

$Fe^{2+}(aq) + Ag^{+}(aq) \rightarrow Fe^{3+}(aq) + Ag(s)$

Given that

$E_{Ag^{+}/Ag}^{0} = xV$; $E_{Fe^{2+}/Fe}^{0} = yV$; $E_{Fe^{3+}/Fe}^{0} = zV$

[April 8, 2019 (II)]

Options:

A. $x - z$

B. $x - y$

C. $x + 2y - 3z$

D. $x + y - z$

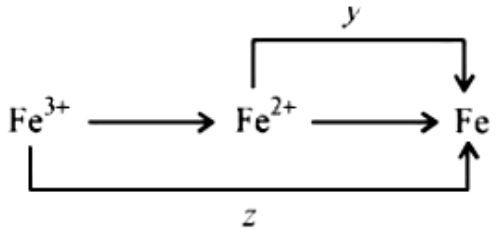
Answer: C

Solution:

Solution:

$Ag^{+}(aq) + Fe^{2+}(aq) \longrightarrow Fe^{3+}(aq) + Ag(s)$

$E_{cell}^{0} = E_{Ag^{+}/Ag}^{0} - E_{Fe^{3+}/Fe^{2+}}^{0}$



$E_{Fe^{3+}/Fe^{2+}}^{0} = 3z - 2y$

$E_{Ag^{+}/Ag}^{0} = x$ (given)

$\therefore E_{cell}^{0} = x - 3z + 2y$

Question130

How long (approximate) should water be electrolysed by passing through 100 amperes current so that the oxygen released can completely burn 27.66g of diborane? (Atomic weight of B = 10.8u) [2018]

Options:

A. 6.4 hours

B. 0.8 hours

C. 3.2 hours

D. 1.6 hours

Answer: C

Solution:

Solution:

$B_2H_6 + 3O_2 \longrightarrow B_2O_3 + 3H_2O$

$27.66g$ of $B_2H_6(1mol\ e)$ requires 3 moles of oxygen (O_2) for complete burning.

Now the oxygen is produced by the electrolysis of H_2O.

On electrolysis: $2H_2O \xrightarrow{4F} 2H_2 + O_2$

1 mole O_2 is produced by $4F$ charge

$\therefore 3mol\ eO_2$ will be produced by $12F$ charge.

$\because Q = I\,t$

$12 \times 96500C = I \times t$

$12 \times 96500C = 100 \times t$

$t = \frac{12 \times 96500}{100} second$

$= \frac{12 \times 96500}{100 \times 3600}$ hour

$= 3.2$ hours

Question131

When 9.65 ampere current was passed for 1.0 hour into nitrobenzene in acidic medium, the amount of p -aminophenol produced is [Online April 16, 2018]

Options:

A. 109.0g

B. 98.1g

C. 9.81g

D. 10.9g

Answer: C

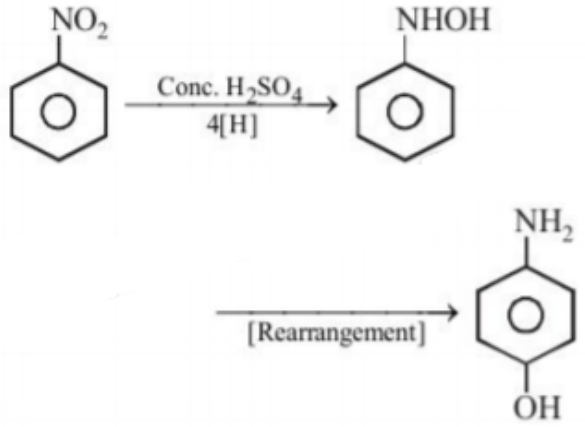
Solution:

Solution:

9.65 ampere current was passed for 1.0 hour (3600 seconds)

Number of moles of electrons passed

$= \frac{I(A) \times t(s)}{96500} = \frac{9.65A \times 3600s}{96500} = 0.36mol\ es$



$C_6H_5NO_2 + 4e^- + 4H^+ \longrightarrow$ p -Aminophenol $+H_2O$

$\because 4$ moles of electrons reduces 1 mole of nitrobenzene to p -aminophenol. $\therefore 0.36$ moles of electrons will reduce $\frac{0.36}{4} = 0.09$ moles of nitrobenzene to p -aminophenol.

p -aminophenol molar mass $= 109.14 g/mol$

Mass of p -aminophenol obtained $= 109.14 g/mol \times 0.09 mol = 9.81 g$

Question132

When an electric current is passes through acidified water, 112mL of hydrogen gas at N.T.P. was collected at the cathode in 965 seconds. The current passed, in ampere, is : [Online April 15,2018 (I)]

Options:

A. 2.0

B. 0.1

C. 0.5

D. 1.0

Answer: D

Solution:

Solution:
Reduction at cathode:

$2e^- + 2H_2O \rightarrow H_2 + 2OH^-$ (valence factor) $H_2 = 2$

At NTP $22400 mL$ of $H_2 = 1 mol\ e$ of H_2

$112 mL$ of $H_2 = \frac{1}{22400} \times 112 = 0.005 mol\ e$ of H_2

Moles of H_2 produced $= \frac{I \times t}{96500} \times$ molar ratio

$0.005 = \frac{I \times 965}{96500} \times \frac{1 mol\ e\ of\ H_2}{2 mol\ e\ of\ e^-}\ I = 1.0A$

Question133

Given

$E_{Cl_2/Cl^-}^{0} = 1.36V$, $E_{Cr^{3+}/Cr}^{\circ} = -0.74V$

$E_{Cr_2/O_7^{2-}/Cr^{3+}} = 1.33V$, $E_{MnO\bar{4}^-Mn^{2+}}^{0} = 1.51V$

Among the following, the strongest reducing agent is [2017]

Options:

A. Cr

B. Mn^{2+}

C. Cr^{3+}

D. Cl

Answer: A

Solution:

Solution:

$E_{MnO_4^-/Mn^{2+}}^{\circ} = 1.51V$

$E_{Cl_2/Cl^-}^{\circ} = 1.36V$

$E_{C_2O_7^{2-}/Cr^{3+}} = 1.33V$

$E_{Cr^{3+}/Cr}^{\circ} = -0.74$

Since Cr^{3+} is having least reduction potential, so Cris the best reducing agent.

Question134

Consider the following standard electrode potentials (E° in volts) in aqueous solution:

Element	M^{3+}/M	M^{+}/M
Al	-1.66	+0.55
Tl	+1.26	-0.34

Based on these data, which of the following statements is correct? [Online April 8, 2017]

Options:

A. Tl^{+} is more stable than Al^{3+}

B. Al^{+} is more stable than Al^{3+}

C. Tl^{+} is more stable than Al^{+}

D. Tl^{3+} is more stable than Al^{3+}

Answer: C

Solution:

Solution:

(i) $\underset{\text{more stable}}{Al^{3+}} \xrightarrow{E^{\circ}=-1.66} Al \xleftarrow{E^{\circ}=+0.55} \underset{\text{Less stable}}{Al^{+}}$

(ii) $\underset{\text{Less stable}}{Tl^{3+}} \xrightarrow{E^{\circ}=+1.26} Tl \xleftarrow{E^{\circ}=-0.34} \underset{\text{more stable}}{Tl^{+}}$

Tl^{+} has negative electrode potential ($E^{\circ}=-0.34$) means, it does not prefer to convert into Tl but reverse must be preferred that's why it is more stable than Tl^{3+} ($E^{\circ}=+1.26$). In Al, Al^{3+} is more stable ($E^{\circ}=-1.66$) than Al^{+} ($E^{\circ}=+0.55$) and also from Tl^{+} due to more negative value of E°. Therefore, by comparison it confirms that Tl^{+} is more stable than Al^{+}.

Question135

What is the standard reduction potential (E°) for $Fe^{3+} \rightarrow Fe$?
Given that :
$Fe^{2+} + 2e^{-} \rightarrow Fe$; $E^{\circ}_{Fe^{2+}/Fe} = -0.47V$
$Fe^{3+} + e^{-} \rightarrow Fe^{2+}$; $E^{\circ}_{Fe^{3+}/Fe^{2+}} = +0.77V$
[Online April 8, 2017]

Options:

A. $-0.057V$

B. $+0.057V$

C. $+0.30V$

D. $-0.30V$

Answer: A

Solution:

Solution:
$\Delta G^{\circ} = -nF\,E^{\circ}$
(i) $F\,e^{2+} + 2e^{-} \longrightarrow F\,e$; $E^{0} = -0.47V$;
(ii) $F\,e^{3+} + e^{-} \longrightarrow F\,e^{2+}$; $E^{0} = +0.77V$:
(iii) $F\,e^{3+} + 3e^{-} \longrightarrow F\,e$
(i) $\Delta G^{0} = -nF\,E^{0} = -2(-0.47)F = 0.94F$
(ii) $\Delta G^{0} = -nF\,E^{0} = -1(+0.77)F = -0.77F$
(iii) On adding : $\Delta G^{0} = +0.17F$
$\Delta G^{0} = -nF\,E^{0}$
E^{0} for $(F\,e^{3+} \longrightarrow F\,e) = \dfrac{\Delta G^{0}}{-nF} = \dfrac{0.17F}{-3F} = -0.057V$

Question136

To find the standard potential of M^{3+}/M electrode, the following cell is constituted : $Pt/M/M^{3+}(0.001 mol\,L^{-1}) \vee Ag^{+}(0.01 mol\,L^{-1})/Ag$

The emf of the cell is found to be 0.421 volt at 298K . The standard potential of half reaction $M^{3+} + 3e \longrightarrow M$ at 298K will be:
(Given $E^{\circ}_{Ag^{+}/Ag}$ at 298K = 0.80V ol t)
[Online April 9, 2017]

Options:

A. 0.38 Volt

B. 0.32 Volt

C. 1.28 Volt

D. 0.66 Volt

Answer: B

Solution:

Solution:
Cell reaction:
$M + 3Ag^{+}(aq) \longrightarrow M^{3+} + 3Ag(s)$
$E = E^{0} - \dfrac{0.0591}{n} \log \dfrac{[\text{ Reduced state }]}{[\text{ Oxidised state }]}$
$0.421 = E^{0} - \dfrac{0.0591}{3} \log \dfrac{0.001}{(0.01)^{3}}$
$E^{0} = 0.48$

$E^0 = E_{Ag^+/Ag}{}^0 - E_{M^{3+}/M}{}^0$

$E_{M^{3+}/M}{}^0 = 0.8V - 0.48V = 0.32volt$

Question137

What will occur if a block of copper metal is dropped into a beaker containing a solution of 1M $ZnSO_4$?

[Online April 9, 2016]

Options:

A. The copper metal will dissolve with evolution of oxygen gas

B. The copper metal will dissolve with evolution of hydrogen gas

C. Noreaction will occur

D. The copper metal will dissolve and zinc metal will be deposited.

Answer: C

Solution:

Solution:
No reaction will occur as the Zn is placed above Cu in electrochemical series. Hence there will be no $_2$ displacement reaction.

Question138

Galvanization is applying a coating of:
[2016]

Options:

A. Cu

B. Zn

C. Pb

D. Cr

Answer: B

Solution:

Solution:
Galvanization is the process by which zinc is coated over corrosive (easily rusted) metals to prevent them from corrosion.

Question139

Identify the correct statement:
[Online April 10,2016]

Options:

A. Corrosion of iron can be minimized by forming a contact with another metal with a higher reduction potential

B. Iron corrodes in oxygen free water

C. Corrosion of iron can be minimized by forming an impermeable barrier at its surface

D. Iron corrodes more rapidly in salt water because its electrochemical potential is higher

Answer: C

Solution:

Solution:
Corrosion of iron can be minimized by forming an impermeable barrier at its surface.

Question140

Two Faraday of electricity is passed through a solution of $CuSO_4$. The mass of copper deposited at the cathode is
(at. mass of Cu = 63.5amu)
[2015]

Options:

A. 2g

B. 127g

C. 0g

D. 63.5g

Answer: D

Solution:

Solution:
$Cu^{2+} + 2e^- \longrightarrow Cu$
2F i.e. $2 \times 96500C$ deposit $Cu = 1mol$
$= 63.5g$

Question141

A variable, opposite external potential (E_{ext}) is applied to the cell $Zn \mid Zn^{2+}(1M) \mid Cu^{2+}(1M) \mid Cu$, of potential 1.1V When $E_{cxt} < 1.1V$ and $E_{cxt} > 1.1V$, respectively electrons flow from:
[Online April 10, 2015]

Options:

A. anode to cathode in both cases

B. cathode to anode and anode to cathode

C. anode to cathode and cathode to anode

D. cathode to anode in both cases

Answer: C

Solution:

Solution:
EMF of galvanic cell $= 1.1$ volt
If $E_{ext} < EMF$ then electrons flow steadily from anode to cathode, while if $E_{ext} > EMF$ then electrons flow from cathode to anode as polarity is changed.

Question142

At 298K, the standard reduction potentials are 1.51V for $MnO_4^- \mid Mn^{2+}$, 1.36V for $Cl_2 \mid Cl^-$, 1.07V for $Br_2 \mid Br^-$ and 0.54V for $I_2 \mid I$. At pH = 3, permanganate is expected to oxidize : $\left(\frac{RT}{F} = 0.059V \right)$
[Online April 11, 2015]

Options:

A. Cl , Br and I

B. Br and I^-

C. Cl^- and Br^-

D. I^- only

Answer: B

Solution:

Solution:

$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$

$E = 1.51 - \frac{0.059}{5} \log \frac{[Mn^{2+}]}{[MnO_4^-][H^+]^8}$

Taking Mn^{2+} and MnO_4^- in standard state i.e. $1M$,

$E = 1.51 - \frac{0.059}{5} \times 8 \log \frac{1}{[H^+]}$

$= 1.51 - \frac{0.059}{5} \times 8 \times 3 = 1.2268V$

Hence at this pH , MnO_4^- will oxidise only Br^- and I^- as SRP of Cl_2/Cl^- is $1.36V$ which is greater than that for MnO_4^-/Mn^{2+}

Question143

Resistance of 0.2M solution of an electrolyte is 50Ω. The specific conductance of the solution is $1.4Sm^{-1}$. The resistance of 0.5M solution of the same electrolyte is 280 Ω. The molar conductivity of 0.5M solution of theelectrolyte in Sm^2mol^{-1} is:
[2014]

Options:

A. 5×10^{-4}

B. 5×10^{-3}

C. 5×10^3

D. 5×10^2

Answer: A

Solution:

Solution:
Given for $0.2M$ solution
$R = 50\Omega$
$\kappa = 1.4Sm^{-1} = 1.4 \times 10^{-2}Scm^{-1}$
Now, $R = \rho \frac{l}{a} = \frac{1}{\kappa} \times \frac{l}{a}$
$\Rightarrow \frac{l}{a} = R \times \kappa = 50 \times 1.4 \times 10^{-2}cm^{-1}$
For 0.5 M solution
$R = 280\Omega$
$\kappa = ?$
$\frac{l}{a} = 50 \times 1.4 \times 10^{-2}cm^{-1}$
$\Rightarrow R = \rho \frac{l}{a} = \frac{1}{\kappa} \times \frac{l}{a}$
$\Rightarrow \kappa = \frac{1}{280} \times 50 \times 1.4 \times 10^{-2}$
$= \frac{1}{280} \times 70 \times 10^{-2}$
$= 2.5 \times 10^{-3}Scm^{-1}$
Now, $\Lambda_m = \frac{\kappa \times 1000}{M}$
$= \frac{2.5 \times 10^{-3}Scm^{-1} \times 1000cm^3/L}{0.5mol/L}$
$= 5Scm^2mol^{-1} = 5 \times 10^{-4}Sm^2mol^{-1}$

Question144

The equivalent conductance of N aCl at concentration C and at infinite dilution are λ_C and λ_∞, respectively. The correct relationship between λ_C and λ_∞ is given as:

(Where the constant B is positive)
[2014]

Options:

A. $\lambda_C = \lambda_\infty + (B)C$

B. $\lambda_C = \lambda_\infty - (B)C$

C. $\lambda_C = \lambda_\infty - (B)\sqrt{C}$

D. $\lambda_C = \lambda_\infty + (B)\sqrt{C}$

Answer: C

Solution:

Solution:

According to Debye Huckle onsager equation,
$\lambda_C = \lambda_\infty - B\sqrt{C}$

Question145

The standard electrode potentials $\left(E_{M^+/M}^\circ\right)$ of four metals A, B, C and D are $-1.2V$, $0.6V$, $0.85V$ and $-0.76V$, respectively. The sequence of deposition of metals on applying potential is: [Online April 9, 2014]

Options:

A. A, C, B, D

B. B, D, C, A

C. C, B, D, A

D. D, A, B, C

Answer: C

Solution:

Solution:
As the value of reduction potential decreases the reducing power increases i.e.
$C < B < D < A$
$(0.85) \cdot (0.6) \cdot (-0.76)(-1.2)$

Question146

A current of 10.0A flows for 2.00h through an electrolytic cell containing a molten salt of metal X . This results in the decomposition of 0.250 mol of metal X at the cathode. The oxidation state of X in the molten salt is: (F = 96, 500C) [Online April 9, 2014]

Options:

A. 1+

B. 2+

C. 3+

D. 4

Answer: C

Solution:

Solution:
According to Faraday's first law of electrolysis
$W = \frac{E \times i \times t}{96500}$
Where $E =$ equivalent weight
$= \frac{\text{mol. mass of metal } (M)}{\text{oxidation state of metal } (x)}$
Substituting the value in the formula
$W = \frac{M}{x} \times \frac{i \times t}{96500}$
or $x = \frac{M}{W} \times \frac{i \times t}{96500} = \frac{10 \times 2 \times 60 \times 60}{96500 \times 0.250} = 3$
[Given : no. of moles $= \frac{W}{M} = 0.250$]
Hence oxidation state of metal is $(+3)$

Question147

Given

$Fe^{3+}(aq) + c^{-} \rightarrow Fe^{2+}(aq); E^{\circ} = +0.77V$

$Al^{3+}(aq) + 3e^{-} \rightarrow Al(s); E^{\circ} = -1.66V$

$Br_2(aq) + 2e \rightarrow 2Br; E^{\circ} = +1.09V$

Considering the electrode potentials, which of the following represents the correct order of reducing power?
[Online April 11, 2014]

Options:

A. $Fe^{2+} < Al < Br^{-}$

B. $Br^{-} < Fe^{2+} < Al$

C. $Al < Br^{-} < Fe^{2+}$

D. $Al < Fe^{2+} < Br$

Answer: D

Solution:

Solution:

Reducing character decreases down the series. Hence the correct order is
$Al < Fe^{2+} < Br^{-}$

Question148

How many electrons would be required to deposit 6.35g of copper at the cathode during the electrolysis of an aqueous solution of copper sulphate? (Atomic mass of copper = 63.5u, N_A = Avogadro's constant)

[Online April 12, 2014]

Options:

A. $\frac{N_A}{20}$

B. $\frac{N_A}{10}$

C. $\frac{N_A}{5}$

D. $\frac{N_A}{2}$

Answer: C

Solution:

Solution:
$Cu \longrightarrow Cu^{2+} + 2e^{-}$
i.e, to deposit 1 mole of Cu at cathode from $Cu^{2+}SO_4^{2-}$ solution $= 2$ moles of electrons are required
i.e, To deposit $6.35g = \frac{6.35}{63.5} \times 2 = \frac{2}{10} = \frac{1}{5}mol$
Thus total no. of electronsrequired $= \frac{N_A}{5}$

Question149

Given below are the half-cell reactions:
$Mn^{2+} + 2e^{-} \rightarrow Mn; E^{\circ} = -1.18V$
$2(Mn^{3+} + e^{-} \rightarrow Mn^{2+}); E^{\circ} = +1.51V$
The E for $3Mn^{2+} \rightarrow Mn + 2Mn^{3+}$ will be:
[2014]

Options:

A. −2.69V ; the reaction will not occur

B. −2.69V ; the reaction will occur

C. −0.33V ; the reaction will not occur

D. −0.33V ; the reaction will occur

Answer: A

Solution:

Solution:
(a) $Mn^{2+} + 2e^- \rightarrow Mn$; $E^\circ = -1.18V$;...(i)
(b) $Mn^{3+} + e^- \rightarrow Mn^{2+}$; $E^0 = -1.51V$; ... (ii)
Now multiplying equation (ii) by two and subtracting from equation (i)
$3Mn^{2+} \rightarrow Mn^+ + 2Mn^{3+}$
$E^0 = E_{Ox.} + E_{Red}$
$= -1.18 + (-1.51) = -2.69V$
(-ve value of EMF (i.e. $\Delta G = +ve$) shows that the reaction is non-spontaneous)

Question150

Electrode potentials (E°) are given below:

$Cu^+/Cu = +0.52V$

$Fe^{3+}/Fe^{2+} = +0.77V$

$\frac{1}{2}I_2(s)/I^- = +0.54V$

$Ag^+/Ag = +0.88V$

Based on the above potentials, strongest oxidizing agent will be:
[Online April 9, 2013]

Options:

A. Cu^+

B. Fe^{3+}

C. Ag^+

D. I_2

Answer: C

Solution:

Solution:
Higher the value of reduction potential stronger will be the oxidising agent, hence based on the given values Ag^{+} will be strongest oxidizing agent.

Question151

Given:

$E_{\frac{1}{2}Cl_2/Cl^-} = 1.36V$, $E_{Cr^{3+}/Cr}^{\circ} = -0.74V$

$E^{\circ}Cr_2O_7^{2-}/Cr^{3+} = 1.33V$, $E^{\circ}MnO_4^{-}/Mn^{2+} = 1.51V$

The correct order of reducing power of the species(Cr, Cr^{3+}, Mn^{2+} and Cl^{-}) will be:
[Online April 23, 2013]

Options:

A. $Mn^{2+} < Cl^{-} < Cr^{3+} < Cr$

B. $Mn^{2+} < Cl^{3+} < Cl^{-} < Cr$

C. $Cr^{3+} < Cl^{-} < Mn^{2+} < Cr$

D. $Cr^{3+} < Cl^{-} < Cr < Mn^{2+}$

Answer: A

Solution:

Solution:
Lower the value of reduction potential higher will be reducing power hence the correct order will be
$Mn^{2+} < Cl^{-} < Cr^{3+} < Cr$

Question152

A solution of copper sulphate ($CuSO_4$) is electrolysed for 10 minutes with a current of 1.5 amperes. The mass of copper deposited at the cathode (at. mass of Cu = 63u) is:
[Online April 25, 2013]

Options:

A. 0.3892g

B. 0.2938g

C. 0.2398g

D. 0.3928g

Answer: B

Solution:

Solution:
$W = Z\,it$
where Z = Electrochemical equivalent
Eq. wt. of copper $= \frac{63}{2} = 31.5$
$Z = \frac{31.5}{96500}$
$W = Z\,it = \frac{31.5}{96500} \times 1.5 \times 10 \times 60 = 0.2938g$

Question153

Given:

$\mathbf{E_{Cr^{3+}/Cr}^{\circ} = -0.74V\ ;\ E_{MnO_4^-/Mn^{2+}}^{0} = 1.51V}$

$\mathbf{E_{Cr_2O_7^{2-}/Cr^{3+}} = 1.33V\ ;\ E_{Cl/Cl^-}^{\circ} = 1.36V}$

Based on the data given above, strongest oxidising agentwill be: [2013]

Options:

A. Cl

B. Cr^{3+}

C. Mn^{2+}

D. MnO_4^-

Answer: D

Solution:

Solution:
Higher the value of standard reduction potential, stronger is the oxidising agent, hence MnO_4^- is the strongest oxidising agent.

Question154

The standard reduction potentials for Zn^{2+}/Zn, Ni^{2+}/Ni and Fe^{2+}/Fe are −0.76, −0.23 and −0.44V respectively.
The reaction $X + Y^{2+} \longrightarrow X^{2+} + Y$ will be spontaneous when :
[2012]

Options:

A. $X = Ni, Y = Fe$

B. $X = Ni, Y = Zn$

C. $X = Fe, Y = Zn$

D. $X = Zn, Y = Ni$

Answer: D

Solution:

Solution:
For a spontaneous reaction ΔG° must be -ve
Since $\Delta G^{\circ} = -nFE^{\circ}$
Hence for ΔG° to be − ve, ΔE° has to be positive. Which is possible when $X = Zn, Y = Ni$
$Zn + Ni^{2+} \longrightarrow Zn^{2+} + Ni$
$E_{cell}^{\circ} = E_{Ni^{2+}/Ni}^{\circ} - E_{Zn^{2+}/Zn}^{\circ}$
$= -0.23 - (-0.76) = +0.53$ (positive)

Question155

A battery is constructed of Cr and $Na_2Cr_2O_7$. The unbalanced chemical equation when such a battery discharges is following:
$Na_2Cr_2O_7 + Cr + H^{+} \rightarrow Cr^{3+} + H_2O + Na^{+}$
If one Faraday of electricityis passed through the batteryduring the charging, the number of moles of Cr^{3+} removed from the solution is
[Online May 7, 2012]

Options:

A. $\frac{4}{3}$

B. $\frac{1}{3}$

C. $\frac{3}{3}$

D. $\frac{2}{3}$

Answer: C

Solution:

Solution:
Reduction half reaction:
$Cr_2O_7^{2-} + 6e^- + 14H^+ \longrightarrow 2Cr^{3+} + 7H_2O$
Oxidation half reaction:
$Cr \longrightarrow Cr^{3+} + 3e^-$
Overall reaction :
$Cr_2O_7^{2-} + Cr + 14H^+ + 3e^- \longrightarrow 3Cr^{3+} + 7H_2O$
$3F$ of electricity $= 3$mol es of Cr^{3+}
$1F$ of electricity $= \frac{3}{3}$ moles of Cr^{3+}

Question156

The standard potentials of Ag^+/Ag, $Hg_2^{2+}/2Hg$, Cu^{2+}/Cu and Mg^{2+}/Mg electrodes are 0.80, 0.79, 0.34 and $-2.37V$, respectively. An aqueous solution which contains one mole per litre of the salts of each of the four metals is electrolyzed. With increasing voltage, the correct sequence of deposition of the metals at the cathode is [Online May 19, 2012]

Options:

A. Ag, Hg, Cu, Mg

B. Cu, Hg, Ag only

C. Ag, Hg, Cu only

D. Mg, Cu, Hg, Ag

Answer: C

Solution:

Solution:

More the reduction potential, more is the deposition of metals at cathode. Cation having E° value less than $-0.83V$ (reduction potential of H_2O) will not deposit from aqueous solution. Hence correct order of deposition of themetal at the cathode is
$Ag > Hg > Cu$

Question157

Given

$E_{Cu^{2+}/Cu}^\circ = 0.34V$, $E_{Cu^{2+}/Cu^+}^\circ = 0.15V$

Standard electrode potential for the half cell Cu^+/Cu is [Online May 26,2012]

Options:

A. 0.38V

B. 0.53V

C. 0.19V

D. 0.49V

Answer: B

Solution:

Solution:
$Cu^{2+} + e^- \longrightarrow Cu^+; E_1^\circ = 0.15V ; \Delta G_1^\circ = -n_1 E_1^\circ F$
$Cu^{2+} + 2e \longrightarrow Cu; E_2^\circ = 0.34V ; \Delta G_2^\circ = -n_2 E_2^\circ F$
On subracting eq.(i) from eq. (ii) we get
$Cu^+ + e^- \longrightarrow Cu; \quad \Delta G^\circ = \Delta G_2^\circ - \Delta G_1^\circ$
$-nE^\circ F = -(n_2 E_2^\circ F - n_1 E_1^0 F)$
$E^\circ = \frac{n_2 E_2^\circ F - n_1 E_1^\circ F}{nF}$
$= \frac{2 \times 0.34 - 0.15}{1} = 0.53V$

Question158

Resistance of 0.2M solution of an electrolyte is 50Ω. The specific conductance of the solution is $1.3Sm^{-1}$. If resistance of the 0.4M solution of the same electrolyte is 260 Ω, its molar conductivity is : [2011RS]

Options:

A. $6.25 \times 10^{-4} Sm^2 mol^-$

B. $625 \times 10^{-4} Sm^2 mol^{-1}$

C. $62.5 Sm^2 mol^{-1}$

D. $6250 Sm^2 mol$

Answer: A

Solution:

Solution:

$k = \frac{1}{R} \times \frac{l}{a}$

$1.3 = \frac{1}{50} \times \frac{l}{a}$

$\frac{l}{a} = 65 m^{-1}$

$\Lambda_m = \frac{k \times 1000}{molarity}$

[molarity is in moles/litre but 1000 is used to convert litre into cm^3]

$= \frac{\left(\frac{1}{260} \times 65 m^{-1}\right) \times 1000 cm^3}{0.4 mol}$

$= \frac{65 m^{-1}}{260 \times 0.4 mol} \times \frac{1}{1000} m^3$

$= 6.25 \times 10^{-4} Sm^2 mol^{-1}$

Question159

The Gibbs energy for the decomposition of Al_2O_3 at 500 $^\circ$C is as follows:

$\frac{2}{3} Al_2O_3 \rightarrow \frac{4}{3} Al + O_2, \Delta_r G = +966 kJ\ mol^{-1}$

The potential difference needed for electrolytic reduction of Al_2O_3 at 500°C is at least

[2010]

Options:

A. 4.5V

B. 3.0V

C. 2.5V

D. 5.0V

Answer: C

Solution:

Solution:
$\frac{2}{3}Al_2O_3 \rightarrow \frac{4}{3}Al + O_2$
The ionic reactions are
$\frac{2}{3} \times (2Al^{3+}) + 4e^- \rightarrow \frac{4}{3}Al$
$\frac{2}{3} \times (3O^{2-}) \rightarrow O_2 + 4e^-$
Thus, no. of electron transferred $\Rightarrow n = 4$
$\Delta G = -nF\,E$
or $E = \frac{\Delta G}{-nF} = \frac{966 \times 10^3}{4 \times 96500} = -2.5V$
$\therefore$ The potential difference needed for the reduction $= 2.5V$.

Question160

Given:

$E^\circ_{Fe^{3+}/Fe} = -0.036V$

$E^\circ Fe^{2+}/Fe = -0.439V$

The value of standard electrode potential for the change, $Fe^{3+}(aq) + e^- \longrightarrow Fe^{2+}(aq)$ will be:

[2009]

Options:

A. 0.385V

B. 0.770V

C. −0.270V

D. −0.072V

Answer: B

Solution:

Solution:
Given
$Fe^{3+} + 3e^- \rightarrow Fe$
$E^\circ_{Fe^{3+}/Fe} = -0.036V$ ...(i)

$Fe^{2+} + 2e^- \rightarrow Fe$

$E^\circ_{Fe^{2+}/Fe} = -0.439V$ ...(ii)

we have to calculate

$Fe^{3+} + e^- \rightarrow Fe^{2+}, \Delta G^\circ = ?$

To obtain this equationsubtract equ. (ii) from (i) we get

$Fe^{3+} + e^- \rightarrow Fe^{2+}$ ...(iii)

As we know that $\Delta G^\circ = -nFE^\circ$

Thus for reaction (iii)

$\Delta G_3 = \Delta G_1 - \Delta G_2$

$-nFE_3^\circ = -nFE_1^\circ - (-nFE_2^\circ)$

$-nFE_3^\circ = nFE_2^\circ - nFE_1^\circ$

$-1FE_3^\circ = 2 \times 0.439F - 3 \times 0.036F$

$-1FE_3^\circ = 0.770F$

$\therefore E_3^\circ = -0.770V$

Question161

In a fuel cell methanol is used as fuel and oxygen gas is used as an oxidizer. The reaction is

$CH_3OH(l) + 3/2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

At 298K standard Gibb's energies of formation for $CH_3OH(l)$, $H_2O(l)$ and and $CO_2(g)$ are $-166.2 - 237.2$ and $-394.4 kJ\ mol^{-1}$ respectively. If standard enthalpy of combustion of methonal is $-726 kJ\ mol^{-1}$, efficiency of the fuel cell will be:

[2009]

Options:

A. 87%

B. 90%

C. 97%

D. 80%

Answer: C

Solution:

Solution:

$CH_3OH(l) + \frac{3}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

$\Delta G_r = \Delta G_f(CO_2, g) + 2\Delta G_f(H_2O, 1) -$

$\Delta G_f(CH_3OH, 1) - \frac{3}{2}\Delta G_f(O_2, g)$

$= -394.4 + 2(-237.2) - (-166.2) - 0$

$= -394.4 - 474.4 + 166.2 = -702.6kJ$

$\%$ efficiency $= \frac{702.6}{726} \times 100 = 97\%$

Question162

Given $E^\circ Cr^{3+}/Cr = -0.72V$
$E^\circ Fe^{2+}/Fe = -0.42V$.
The potential for the cell
$Cr \mid Cr^{3+}(0.1M) \mid Fe^{2+}(0.01M) \mid Fe$ is
[2008]

Options:

A. 0.26V

B. 0.336V

C. −0.339

D. 0.26V

Answer: D

Solution:

Solution:
From the given representation of the cell, E_{cell} can be found as follows.

$E_{cell} = E_{Fe^{2+}/Fe}^{o} - E_{Cr^{3+}/Cr}^{o} - \frac{0.059}{6}\log\frac{[Cr^{3+}]^2}{[Fe^{2+}]^3}$

[Nernst -Equ.]

$= -0.42 - (-0.72) - \frac{0.059}{6}\log\frac{(0.1)^2}{(0.01)^3}$

$= -0.42 + 0.72 - \frac{0.059}{6}\log\frac{0.1 \times 0.1}{0.01 \times 0.01 \times 0.01}$

$= 0.3 - \frac{0.059}{6}\log\frac{10^{-2}}{10^{-6}} = 0.3 - \frac{0.059}{6} \times 4$

$= 0.30 - 0.0393 = 0.26V$

Question163

The equivalent conductances of two strong electrolytes at infinite dilution in H_2O (where ions move freely through a solution) at 25°C are given below:

$\Lambda_{CH_3COONa} = 91.0 Scm^2/$ equiv.

$\Lambda_{HCl}^{\circ} = 426.2 Scm^2/$ equiv.

What additional information\/ quantity one needs to calculate Λ° of an aqueous solution of acetic acid?
[2007]

Options:

A. Λ of chloroacetic acid ($ClCH_2COOH$)

B. Λ of $NaCl$

C. Λ of CH_3COOK

D. the limiting equivalent coductance of $H^+ \left(\lambda_{H^+}^{\circ}\right)$.

Answer: B

Solution:

Solution:
Note: According to Kohlrausch's law, equivalent conductivity of weak electrolyte, acetic acid (CH_3COOH) can be calculated as follows:
$\Lambda_{CH_3COOH}^{\circ} = \left(\Lambda_{CH_3COONa}^{\circ} + \Lambda_{HCl}^{\circ}\right) - \Lambda_{NaCl}^{\circ}$
$\therefore$ Value of Λ_{NaCl}° should also be known for calculating value of $\Lambda_{CH_3COOH}^{\circ}$

Question164

The cell, $Zn | Zn^{2+}(1M) | Cu^{2+}(1M) | Cu (E_{cell}^{0} = 1.10V)$ was allowed to be completely discharged at 298K.

The relative concentration of Zn^{2+} to $Cu^{2+} \left(\frac{[Zn^{2+}]}{[Cu^{2+}]}\right)$ is

[2007]

Options:

A. 9.65×10^4

B. antilog (24.08)

C. 37.3

D. 10^{373}.

Answer: D

Solution:

Solution:
$E_{cell} = 0$; when cell is completely discharged.

$$E_{cell} = E_{cell}^{\circ} - \frac{0.059}{2}\log\left(\frac{[Zn^{2+}]}{[Cu^{2+}]}\right)$$

$$\text{or } 0 = 1.1 - \frac{0.059}{2}\log\left(\frac{[Zn^{2+}]}{[Cu^{2+}]}\right)$$

$$\log\left(\frac{[Zn^{2+}]}{[Cu^{2+}]}\right) = \frac{2 \times 1.1}{0.059} = 37.3$$

$$\therefore \left(\frac{[Zn^{2+}]}{[Cu^{2+}]}\right) = 10^{37.3}$$

Question165

The molar conductivities Λ_{NaOAc}^{o} and Λ_{HCl}^{o} at infinite dilution in water at 25°C are 91.0 and 426.2Scm^2/mol respectively. To calculate Λ_{HOAc}°, the additional value required is
[2006]

Options:

A. Λ_{NaOH}^{0}

B. Λ_{NaCl}^{o}

C. $\Lambda_{H_2O}^{O}$

D. Λ_{KCl}^{0}

Answer: B

Solution:

Solution:

$\Lambda_{CH_3COOH}^{\circ}$ is given by the following equation

$\Lambda_{CH_3COOH}^{\circ} = \left(\Lambda_{CH_3COONa}^{\circ} + \Lambda_{HCl}^{\circ}\right) - (\Lambda_{NaCl}^{\circ})$

Hence Λ_{NaCl}° is required.

Question166

Resistance of a conductivity cell filled with a solution of an electrolyte of concentration 0.1M is 100Ω. The conductivity of this solution is $1.29 Sm^{-1}$. Resistance of the same cell when filled with 0.2M of the same solution is 520Ω. The molar conductivity of 0.2M solution of electrolyte will be [2006]

Options:

A. $1.24 \times 10^{-4} Sm^2 mol^{-1}$

B. $12.4 \times 10^{-4} Sm^2 mol^{-1}$

C. $124 \times 10^{-4} Sm^2 mol^{-1}$

D. $1240 \times 10^{-4} Sm^2 mol^{-1}$

Answer: B

Solution:

Solution:

$R = 100\Omega$, $\kappa = \frac{1}{R}\left(\frac{l}{a}\right)$

$\frac{l}{a}\left(\text{ cell constant }\right) = 1.29 \times 100 m^{-1}$

Given, $R = 520\Omega$

$C = 0.2M = 0.2 \times 1000 mol\, m^{-3}$

Λ_m (molar conductivity) $= ?$

$\Lambda_m = K \times V$ (K can be calculated as cell constantis known.)

$K = \frac{1}{R}\left(\frac{l}{a}\right)$

Hence,

$K = \frac{129 m^{-1}}{520\Omega} = 0.248 Sm^{-1}$

$\Lambda_m = \frac{K}{C} = \frac{0.248 Sm^{-1}}{0.2 \times 1000 mol\, m^{-3}}$

$= 12.4 \times 10^{-4} Sm^2 mol^{-1}$

Question167

The highest electrical conductivity of the following aqueous solutions is of [2005]

Options:

A. 0.1M difluoroacetic acid

B. 0.1M fluoroacetic acid

C. 0.1M chloroacetic acid

D. 0.1M acetic acid

Answer: A

Solution:

Solution:
Difluoro acetic acid being strongest acid will furnish maximum number of ions showing highest electrical conductivity. The decreasing acidic strength of the carboxylic acids given is
difluoro acetic acid > fluoro acetic acid > chloro acetic acid> acetic acid.

Question168

Electrolyte:	KCl	KNO_3	HCl	$NaOAc$	$NaCl$
$\Lambda^{\infty}(Scm^2mol^{-1})$:	149.9	145	426.2	91	126.5

Calculate Λ_{HOAc}^{∞} using appropriate molar conductance's of the electrolytes listed above at infinite dilution in H_2O at 25°C [2005]

Options:

A. 217.5

B. 390.7

C. 552.7

D. 517.2

Answer: B

Solution:

Solution:
$\Lambda_{HCl}^{\infty} = 426.2$ . . . (i)
$\Lambda_{ACONa}^{\infty} = 91.0$ . . . (ii)
$\Lambda_{NaCl}^{\infty} = 126.5$ . . . (iii)
$\Lambda_{AcOH}^{\infty} = (i) + (ii) - (iii)$
$= [426.2 + 91.0 - 126.5] = 390.7$

Question169

Aluminium oxide may be electrolysed at 1000°C to furnish aluminium metal (At. Mass = 27 amu; 1 Faraday = 96, 500 Coulombs). The cathode reaction is- $Al^{3+} + 3e^{-} \rightarrow Al^{\circ}$ Toprepare 5.12kg of aluminium metal by this methodwe require
[2005]

Options:

A. 5.49×10^{1}C of electricity

B. 5.49×10^{4}C of electricity

C. 1.83×10^{7}C of electricity

D. 5.49×10^{7}C of electricity

Answer: D

Solution:

Solution:
1 mole of $e^{-} = 1F = 96500C$
27g of Al is deposited by $3 \times 96500C$
5120g of Al will be deposited by
$= \frac{3 \times 96500 \times 5120}{27} = 5.49 \times 10^{7}C$
We know, $Q = \frac{mFz}{M}$
$\therefore Q = \frac{5120 \times 96500 \times 3}{27} = 5.49 \times 10^{7}C$

Question170

For a spontancous reaction the ΔG, equilibrium constant (K) and E_{cell}° will be respectively
[2005]

Options:

A. $-ve, >1, -ve$

B. $-ve, <1, -ve$

C. $+ve, >1, -ve$

D. $-ve, >1, +ve$

Answer: D

Solution:

Solution:
Note: For spontaneous reaction ΔG should be negative. Equilibrium constant should be more than one ($\Delta G = -2.303RT \log K_c$, If $K_c = 1$ then $\Delta G = 0$; If $K_c > 1$ then $\Delta G = -ve$).
Again $\Delta G = -nF E_{cell}^{\circ}$
E_{cell}° must be + ve to have $\Delta G - ve$.

Question171

The limiting molar conductivities Λ° for NaCl, KBr and KCl are 126,152 and 150Scm^2mol^{-1} respectively. The Λ for NaBr is
[2004]

Options:

A. $278 Scm^2 mol^{-1}$

B. $176 Scm^2 mol^{-1}$

C. $128 Scm^2 mol^{-1}$

D. $302 Scm^2 mol^{-1}$

Answer: C

Solution:

Solution:
$\Lambda^{\circ} NaCl = \lambda^{\circ} Na^{+} + \lambda_{Cl}$ . . . (i)
$\Lambda_{KBr}^{\circ} = \lambda_{K^{+}}^{\circ} + \lambda_{Br^{-}}$ . . . (ii)
$\Lambda_{KCl}^{\circ} = \lambda_{K^{+}}^{\circ} + \lambda_{Cl^{-}}$ . . . (iii)
operating (i) + (ii) − (iii)
$\Lambda_{NaBr}^{\circ} = \lambda_{Na^{+}}^{\circ} + \lambda_{Br^{-}}$
$= 126 + 152 - 150 = 128 Scm^{2} mol^{-1}$

Question172

Consider the following
E° values $E_{Fe^{3+}/Fe^{2+}} = +0.77V$; $E_{Sn^{2+}/Sn}^{\circ} = -0.14V$
Under standard conditions the potential for the reaction $Sn(s) + 2Fe^{3+}(aq) \rightarrow 2Fe^{2+}(aq) + Sn^{2+}(aq)$is
[2004]

Options:

A. 0.91V

B. 1.40V

C. 1.68V

D. 0.63V

Answer: A

Solution:

Solution:
$Fe^{3+} + e^{-} \rightarrow Fe^{2+}$; $Sn^{2+} + 2e^{-} \rightarrow Sn(s)$;
for $Sn(s) + 2Fe^{3+}(aq) \rightarrow 2Fe^{2+}(aq) + Sn^{2+}(aq)$
∴ Standard potential for the given reaction or
$E_{cell}^{\circ} = E_{Sn/Sn^{2+}}^{\circ} + E_{Fe^{3+}/Fe^{2+}}$
$= 0.14 + 0.77 = 0.91V$

Question173

The $E^{0}M^{3+}/M^{2+}$ values for Cr, Mn, Fe and Coare −0.4l , + 1.57, +0.77 and +1.97V respectively. For which one of these metals the change in

oxidation state from +2 to +3 is easiest?
[2004]

Options:

A. F e

B. M n

C. G

D. Co

Answer: C

Solution:

Solution:
The given values show that Cr has maximum oxidation potental, therefore its oxidation will be easiest.
(Change the sign to get the oxidation values)

Question174

The standard e.m.f. of a cell involving one electron change is found to be 0.591V at 25°C. The equilibrium constant of the reaction is
($F = 96,500 C mol^{-1}$; $R = 8.314 J K^{-1} mol^{-1}$)
[2004]

Options:

A. 1.0×10^{10}

B. 1.0×10^{5}

C. 1.0×10^{1}

D. 1.0×10^{30}

Answer: A

Solution:

Solution:

(a) $E_{cell} = E_{cell}^{\circ} - \frac{0.059}{n} \log K_c$

or $0 = 0.591 - \frac{0.0591}{1} \log K_c$

$$\text{or } \log K_c = \frac{0.591}{0.0591} = 10 \text{ or } K_c = 1 \times 10^{10}$$

Question175

In a cell that utilises the reaction $Zn(s) + 2H^+(aq) \rightarrow Zn^{2+}(aq) + H_2(g)$ addition of H_2SO_4 to cathode compartment, will
[2004]

Options:

A. increase the E and shift equilibrium to the right

B. lower the E and shift equilibrium to the right

C. lower the E and shift equilibrium to the left

D. increase the E and shift equilibrium to the left

Answer: A

Solution:

Solution:

$Zn(s) + 2H^+(aq) \rightleftharpoons Zn^{2+}(aq) + H_2(g)$

$$E_{cell} = E_{cell}^{\circ} - \frac{0.059}{2} \log \frac{[Zn^{2+}][H_2]}{[H^+]^2}$$

Addition of H_2SO_4 will increase $[H^+]$ and E_{cell} will also increase and thus the equilibrium shifts towards RHS.

Question176

In a hydrogen-oxygen fuel cell, combustion of hydrogen occurs to
[2004]

Options:

A. produce high purity water

B. create potential difference between two electrodes

C. generate heat

D. remove adsorbed oxygen from elctrode surfaces

Answer: B

Solution:

Solution:
In $H_2 - O_2$ fuel cell, the combustion of H_2 occurs to create potential difference between the two electrodes

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Question177

Standard reduction electrode potentials of three metals A, B & C are respectively +0.5V , −3.0V − 1.2V .
The reducing powers of these metals are
[2003]

Options:

A. A > B > C

B. C > B > A

C. A > C > B

D. B > C > A

Answer: D

Solution:

Solution:
Note: The higher the negative value of reduction potential, the more is the reducing power.
Hence B > C > A.

Question178

When during electrolysis of a solution of $AgNO_3$ 9650 coulombs of charge pass through the electroplating bath, the mass of silver deposited on the cathode will be
[2003]

Options:

A. 10.8g

B. 21.6g

C. 108g

D. 1.08g

Answer: A

Solution:

Solution:
When 96500 coulomb of electricity is passed through the electroplating bath the amount of Ag deposited $= 108g$
$\therefore$ when 9650 coulomb of electricity is passed deposited Ag.
$= \frac{108}{96500} \times 9650 = 10.8g$

Question179

For a cell reaction involving a two-clectron change, the standard e.m.f. of the cell is found to be 0.295V at 25°C. The equilibrium constant of the reaction at 25°C will be
[2003]

Options:

A. 29.5×10^{-2}

B. 10

C. 1×10^{10}

D. 1×10^{-10}

Answer: C

Solution:

Solution:
The equilibrium constant is related to the standard emf of cell by the expression
$\log K = E_{cell}^{\circ} \times \frac{n}{0.059} = 0.295 \times \frac{2}{0.059}$
$\log K = \frac{590}{59} = 10$ or $K = 1 \times 10^{10}$

Question180

For the redox reaction :
[2003]

$$Zn(s) + Cu^{2+}(0.1M) \rightarrow Zn^{2+}(1M) + Cu(s)$$

taking place in a cell, E_{cell}° is 1.10 volt. E_{cell} for the cell will be

$$\left(2.303 \frac{RT}{F} = 0.0591\right)$$

Options:

A. 1.80 volt

B. 1.07 volt

C. 0.82 volt

D. 2.14 volt

Answer: B

Solution:

Solution:

$$E_{cell} = E_{cell}^{\circ} + \frac{0.059}{n} \log \frac{[Cu^{2+}]}{[Zn^{2+}]}$$

$$= 1.10 + \frac{0.059}{2} \log[0.1]$$

$$= 1.10 - 0.0295 = 1.07V$$

Question181

Conductivity (unit Siemen's S) is directly proportional to area of the vessel and the concentration of the solution in it and is inversely proportional to the length of the vessel then the unit of the constant of proportionality is
[2002]

Options:

A. $Smmol^{-1}$

B. Sm^2mol^{-1}

C. $S^{-2}m^2mol$

D. $S^2m^2mol^{-2}$.

Answer: B

Solution:

Solution:
Given conductivity $\propto \frac{\text{area} \times \text{conc.}}{\text{length}}$

$= \frac{K\,m^2mol}{m \times m^3}$

$\therefore\ K = Sm^2mol^{-1}$

Question182

Which of the following reaction is possible at anode? [2002]

Options:

A. $2Cr^{3+} + 7H_2O \rightarrow Cr_2O_7^{2-} + 14H^+$ [2002]

B. $F_2 \rightarrow 2F$

C. $(1/2)O_2 + 2H^+ \rightarrow H_2O$

D. none of these.

Answer: A

Solution:

Solution:
$2Cr^{3+} + 7H_2O \rightarrow Cr_2O_7^{2-} + 14H^+$
O.S. of Cr changes from $+3$ to $+6$ by loss of electrons. At anode oxidation takes place.

Question183

When the sample of copper with zinc impurity is to be purified by electrolysis, the appropriate electrodes are Cathode Anode [2002]

Options:

A. Cathode - pure zinc
Anode - pure copper

B. Cathode - impure sample
Anode - pure copper

C. Cathode - impure zinc
Anode - impure sample

D. Cathode - pure copper
Anode - impure sample.

Answer: D

Solution:

Solution:
Pure metal always deposits at cathode.

Question184

EMF of a cell in terms of reduction potential of its left and right electrodes is
[2002]

Options:

A. $E = E_{left} - E_{right}$

B. $E = E_{left} + E_{right}$

C. $E = E_{right} - E_{left}$

D. $E = -(E_{righ} + E_{lef})$.

Answer: C

Solution:

Solution:
E_{cell} = Reduction potential of cathode (right)
− reduction potential of anode (left)
= $E_{right} - E_{left}$

Question185

What will be the emf for the given cell

$Pt \mid H_2(P_1) \mid H^+(aq) \mid \mid H_2(P_2) \mid Pt$

[2002]

Options:

A. $\frac{RT}{F}\log_e \frac{P_1}{P_2}$

B. $\frac{RT}{2F}\log_e \frac{P_2}{P_1}$

C. $\frac{RT}{F}\log_e \frac{P_2}{P_1}$

D. none of these.

Answer: C

Solution:

Solution:
Oxidation half call:-
$H_2(g) \longrightarrow 2H^+(1M) + 2e^-$
P_1 Reduction half cell
$2H^+(1M) + 2e^- \longrightarrow H_2(g)$
P_2 The net cell reaction
$H_2(g) \longrightarrow H_2(g)$
P_t
$E_{cell}^\circ = 0.00V ; n = 2$
$\therefore E_{cell} = E_{cell}^\circ - \frac{RT}{nF}\log_e K$
$= 0 - \frac{RT}{nF}\log_e \frac{P_2}{P_1}$
or $E_{cell} = \frac{RT}{2F}\log_e \frac{P_1}{P_2}$
